
Clock and Data Recovery Architectures & Circuits

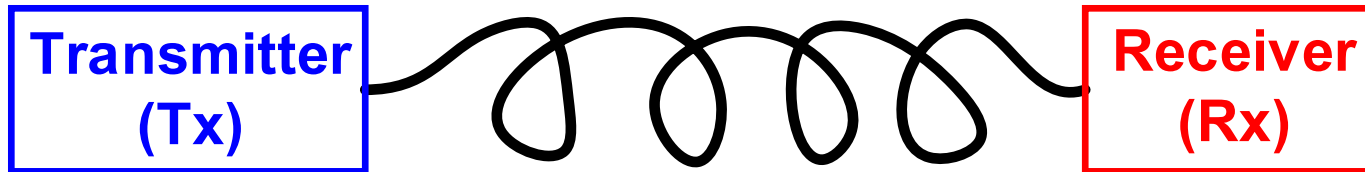
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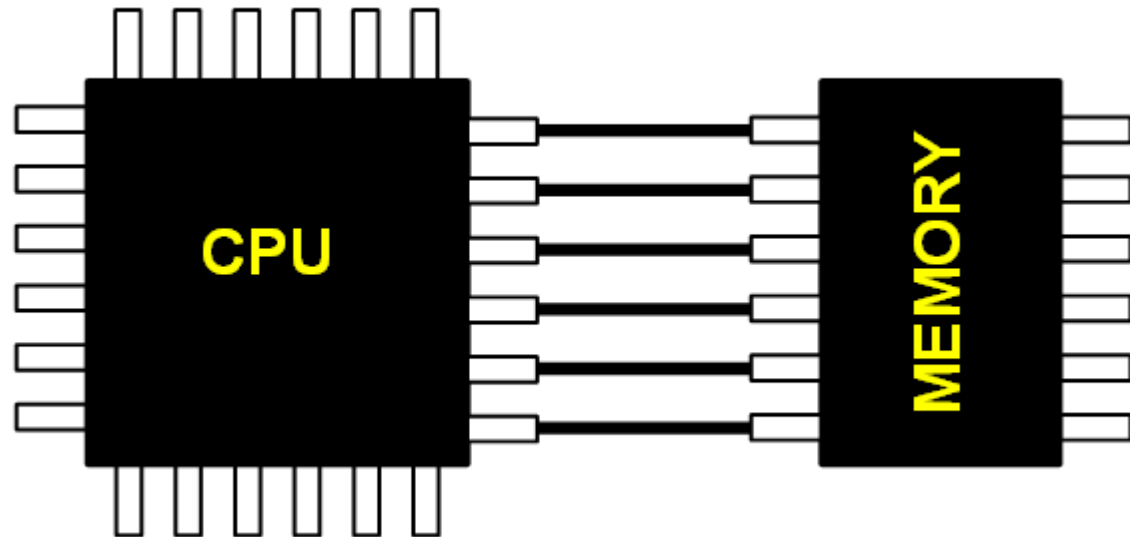
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Feb. 22, 2015

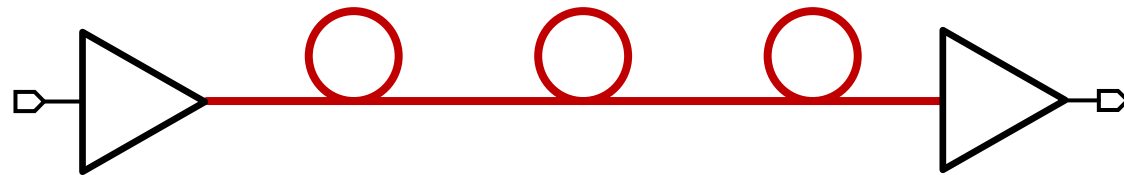
Serial Link Applications



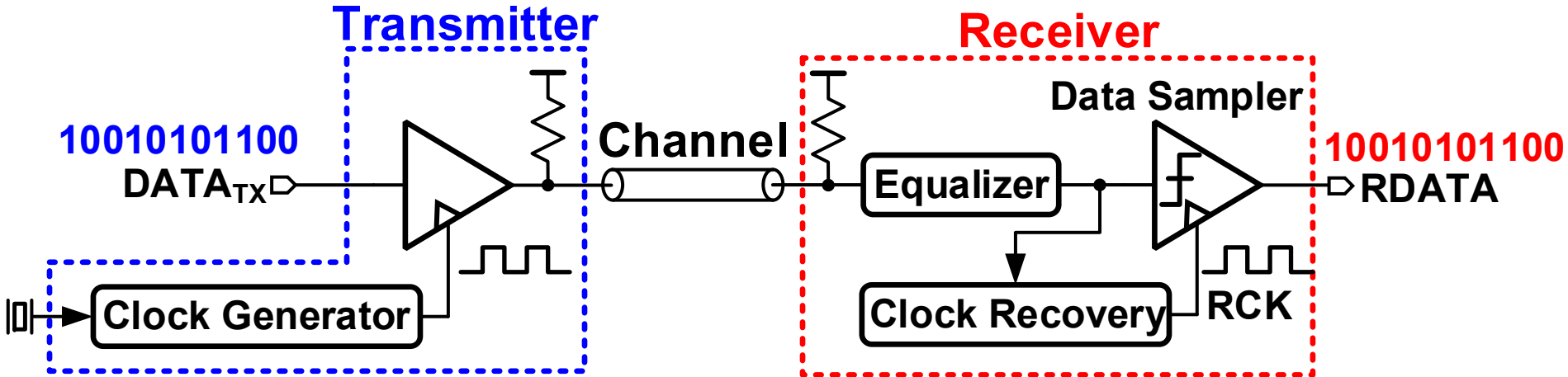
**Chip-to-chip
Electrical link**



Optical link



Serial Link Components



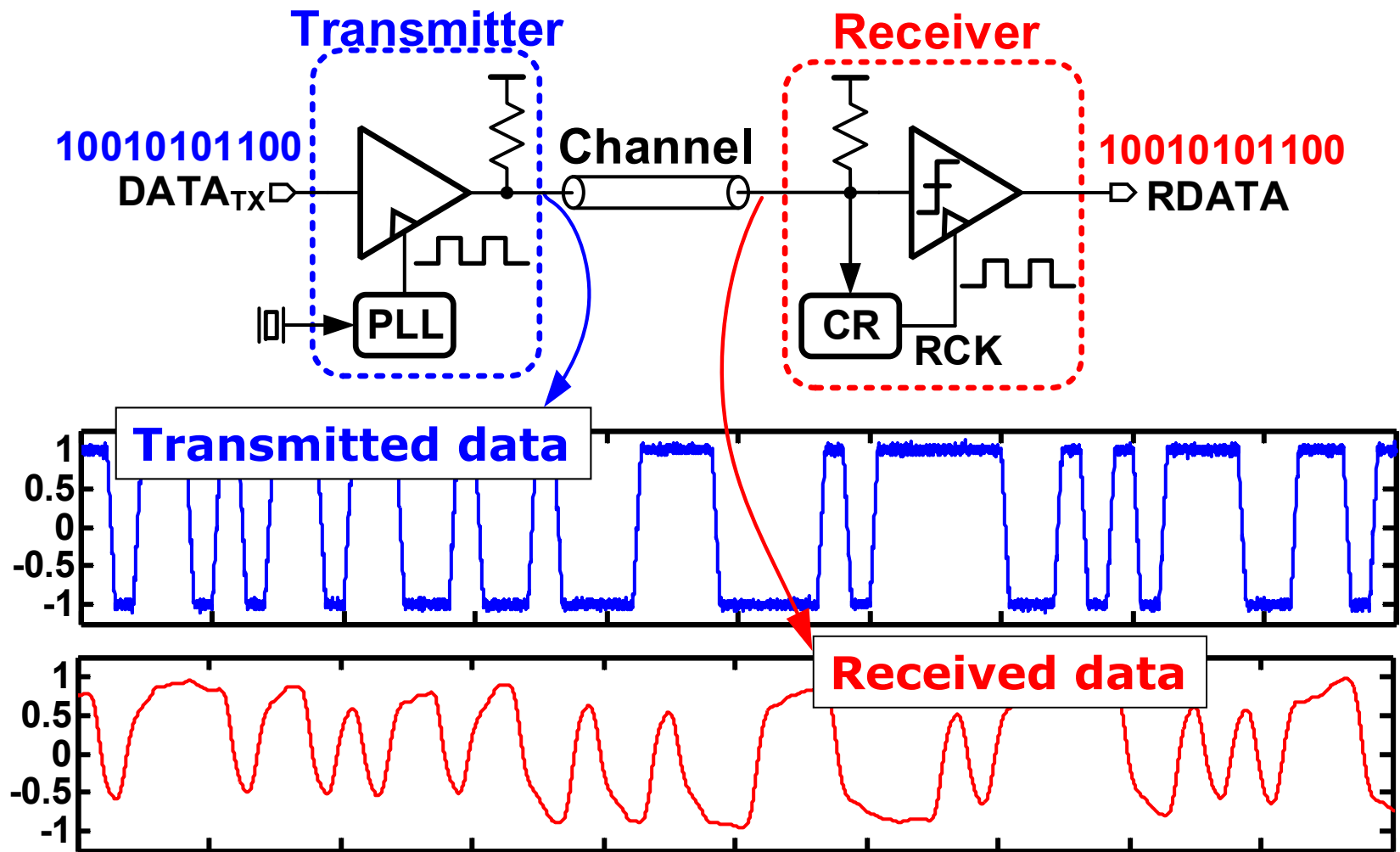
□ Transmitter

□ Channel

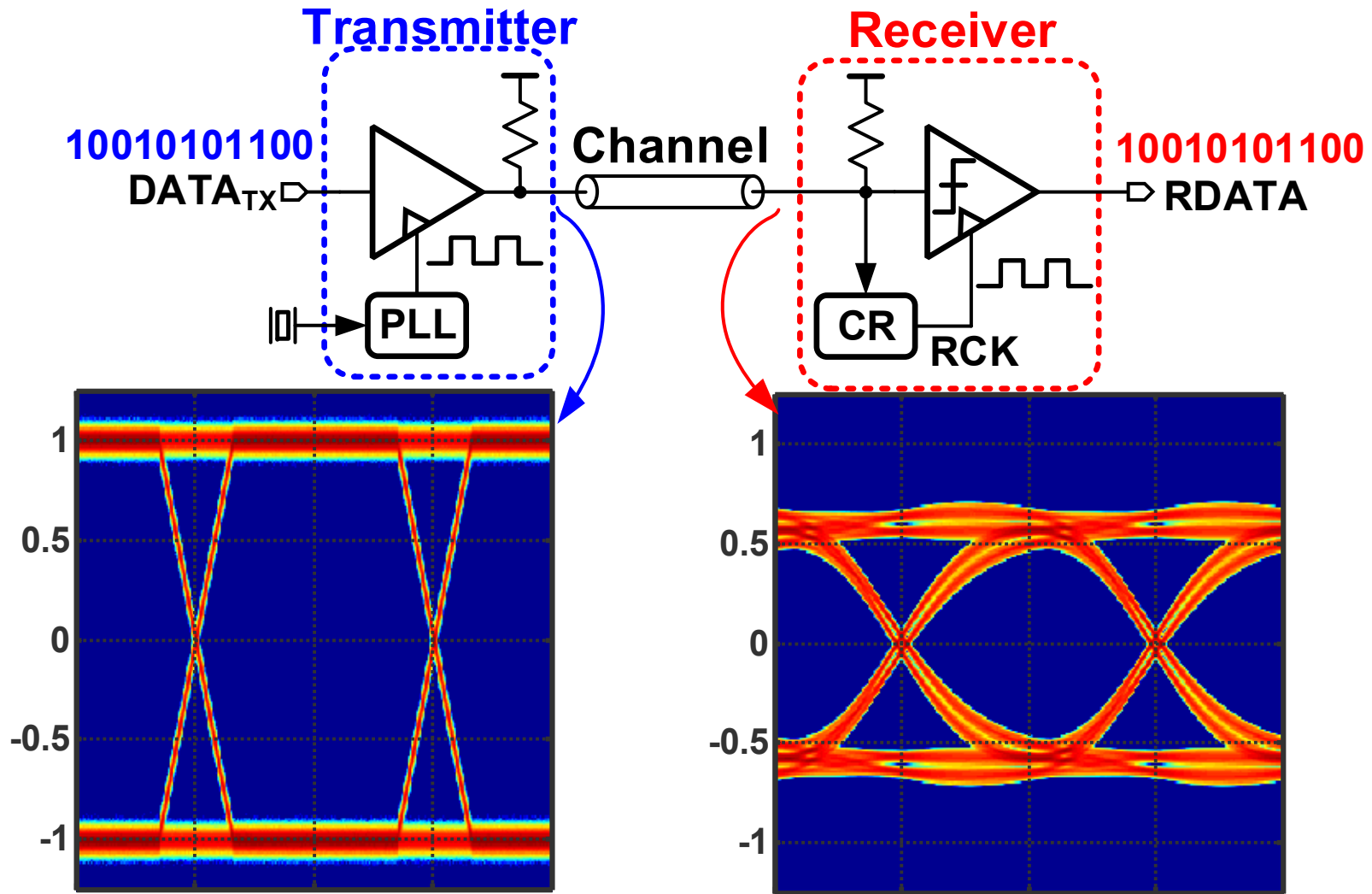
□ Receiver

□ Clock Recovery (CR) + Data sampler = CDR

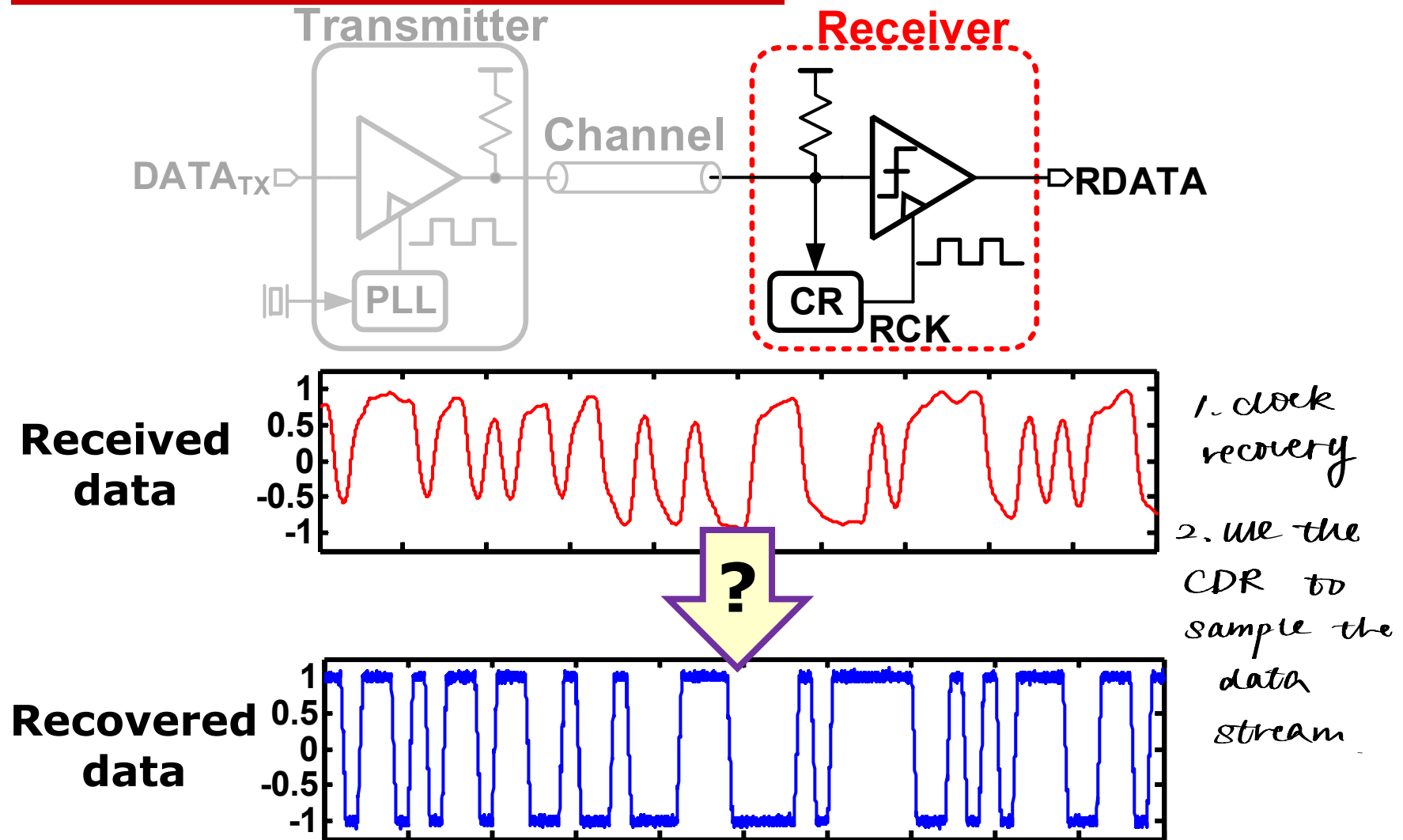
Serial Link Waveforms



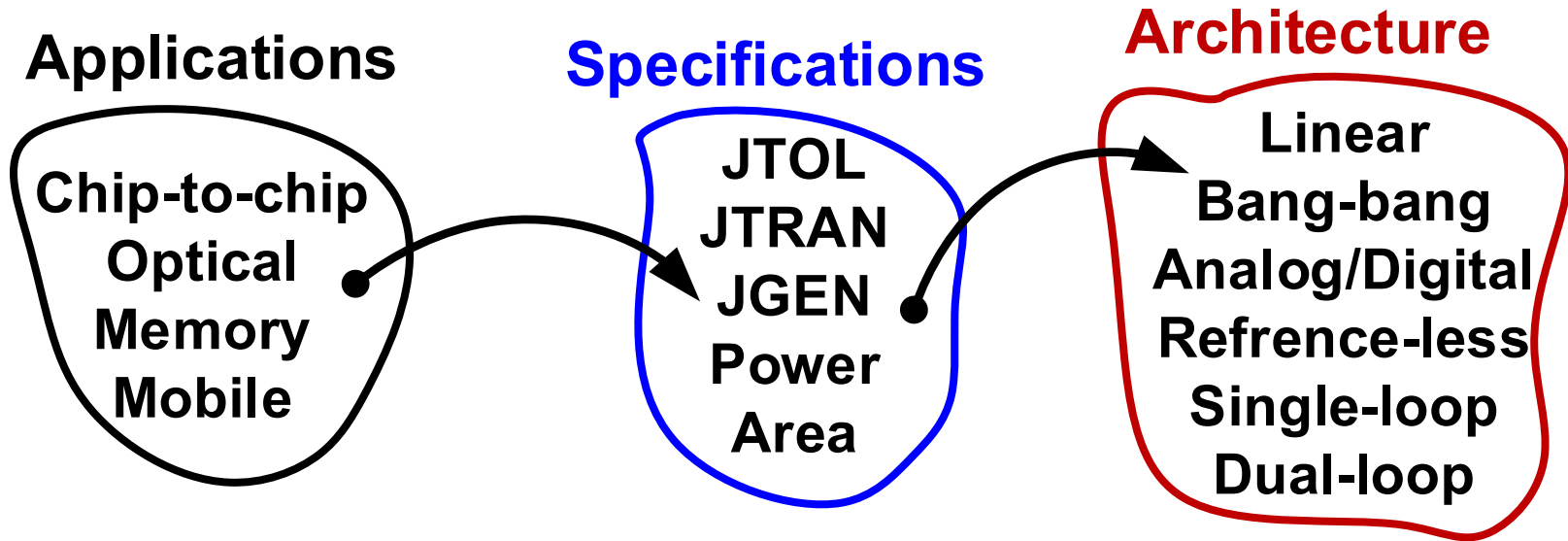
Serial Link Eye Diagrams



This Tutorial Focus: CDR



Tutorial Goals



- ❑ Map application requirements to CDR specifications
- ❑ Optimal architecture choice based on CDR specs.
 - Exposure to different CDR architectures
 - Develop intuition for design tradeoffs
 - Awareness to practical considerations

Tutorial Roadmap

- Performance metrics *unique to CDR*
 - Basic architectures
 - Linear/Bang-bang
 - Digital
 - Hybrid
 - Application-specific CDRs
 - Multi-lane chip-to-chip links
 - Repeaters for optical links and active cables
 - Summary
-

CDR Performance Metrics

- ❑ Jitter tolerance (JTOL)
- ❑ Jitter generation (JGEN)
- ❑ Jitter transfer (JTRAN)

Jitter metrics

Scalability

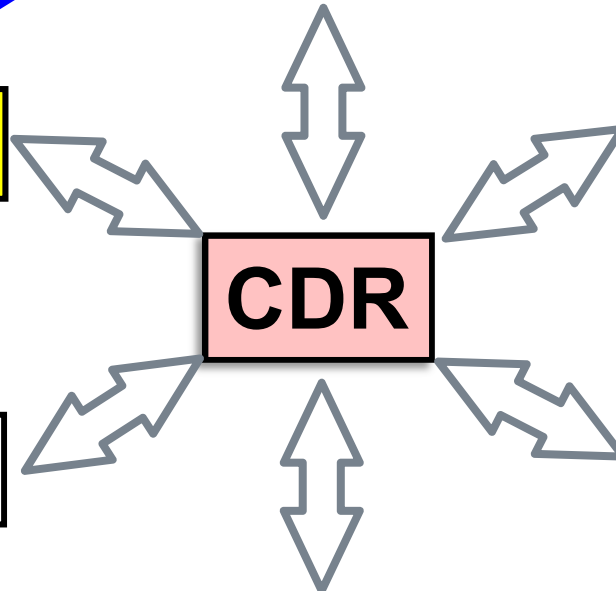
Power

CDR

Supply noise sensitivity

Area

Operating range

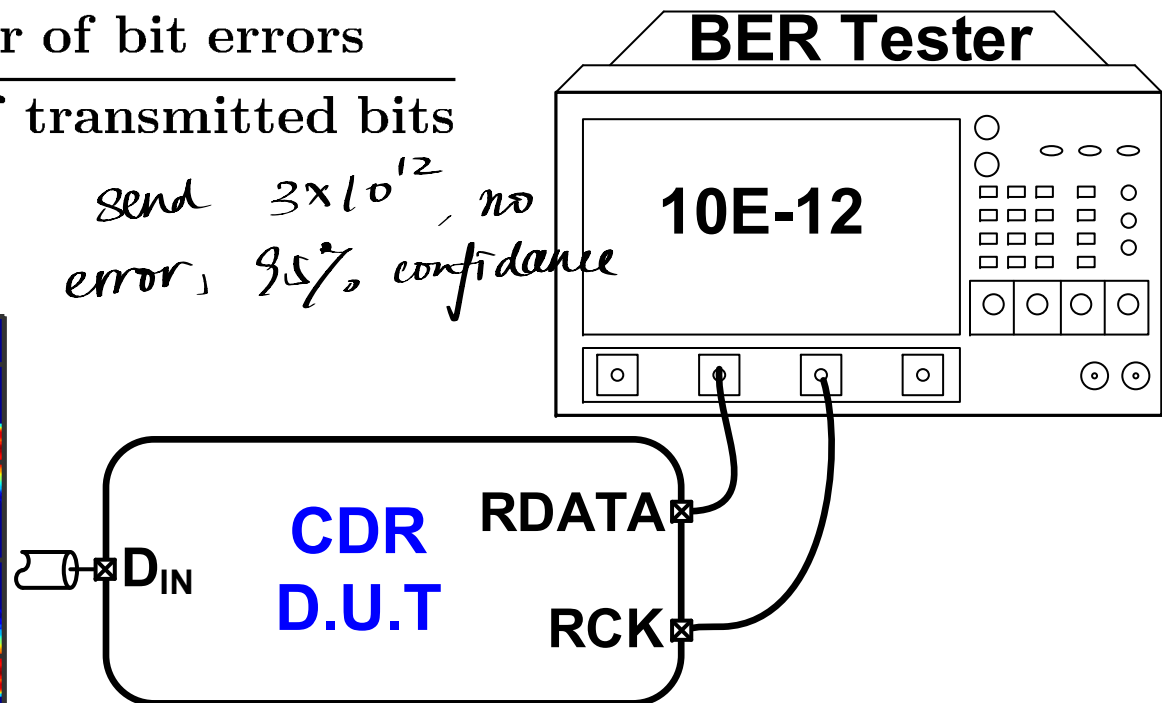
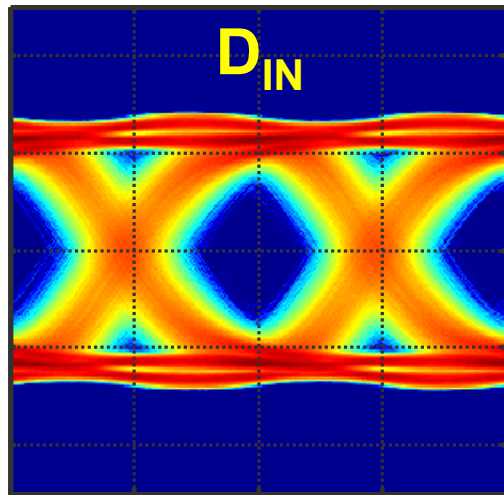


Jitter Tolerance (JTOL)

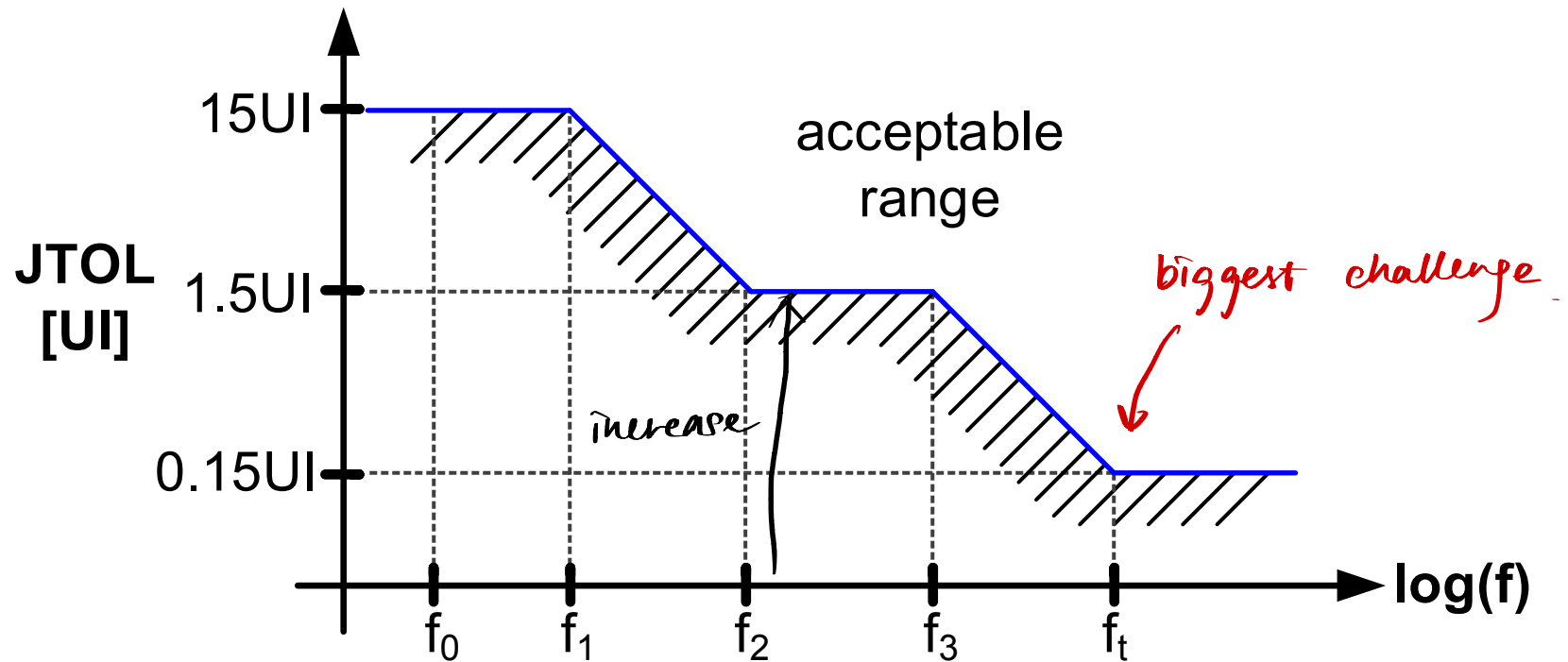
- Maximum tolerable input jitter for a given BER

$$\text{BER} = \frac{\text{Number of bit errors}}{\text{Number of transmitted bits}}$$

send 3×10^{12} , no error, 95% confidence



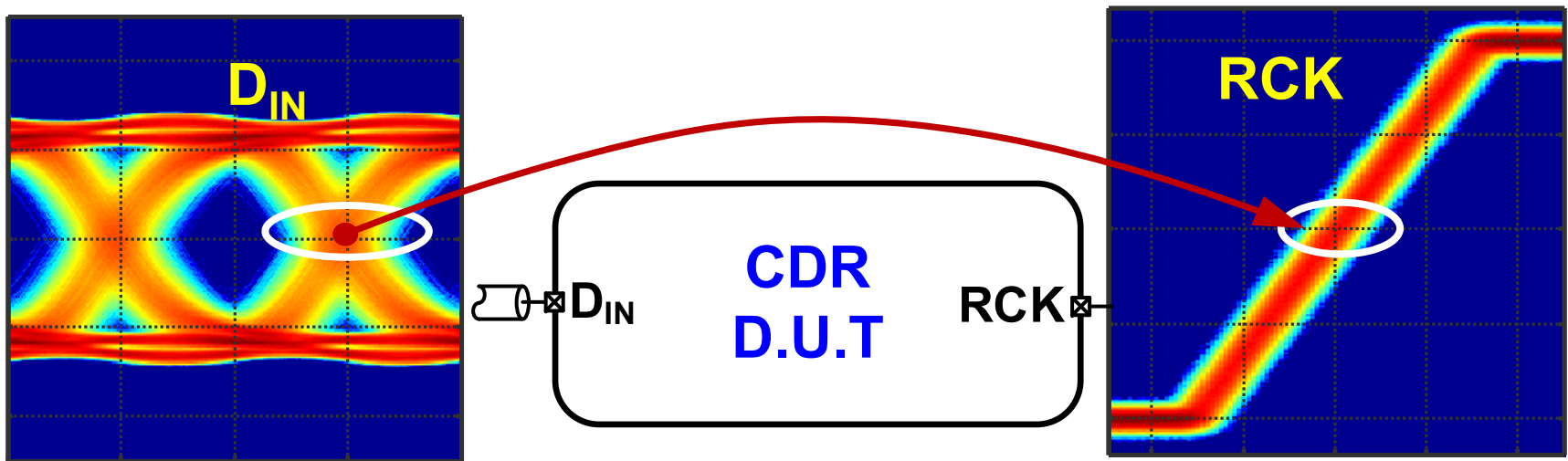
JTOL Mask



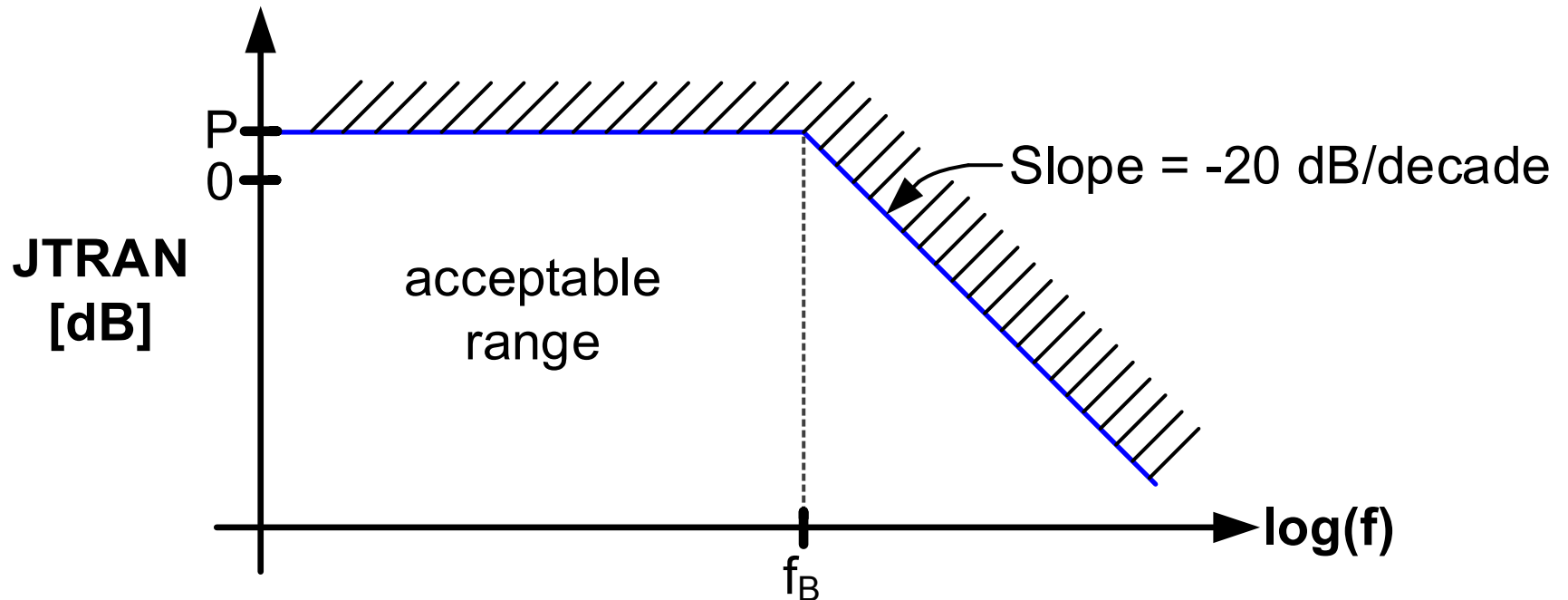
- Increase data input sinusoidal jitter until BER exceeds target

Jitter Transfer (JTRAN)

- Amount of jitter attenuation provided by CDR



JTRAN Mask

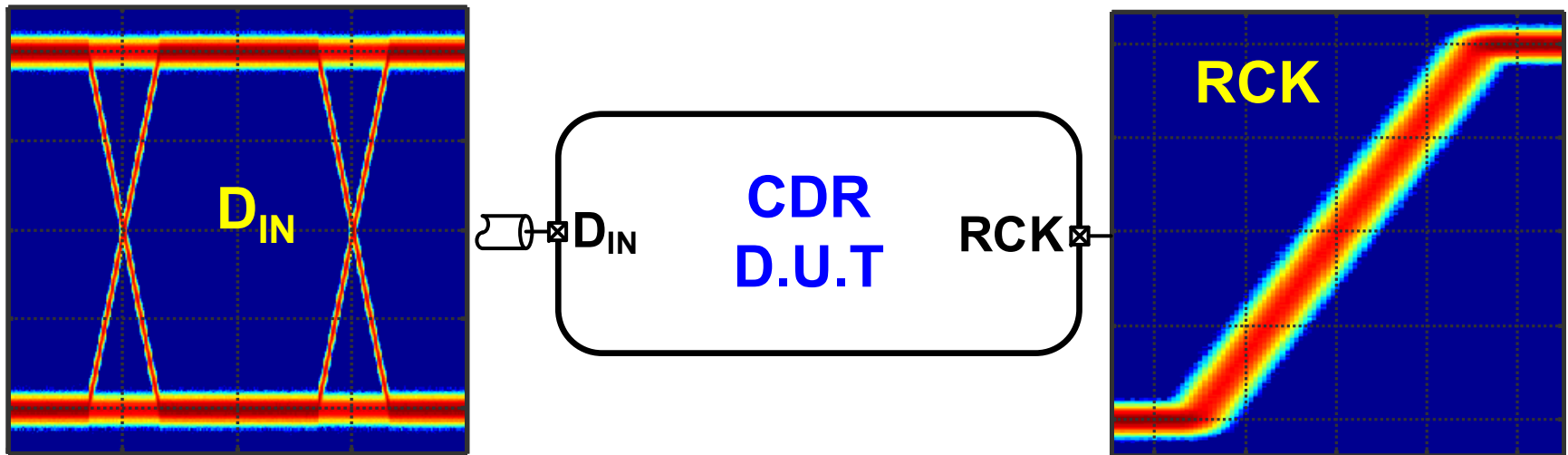


- Modulate data input with sinusoidal jitter and measure resulting output jitter

Jitter Generation (JGEN)

- Amount of output jitter when fed with clean data

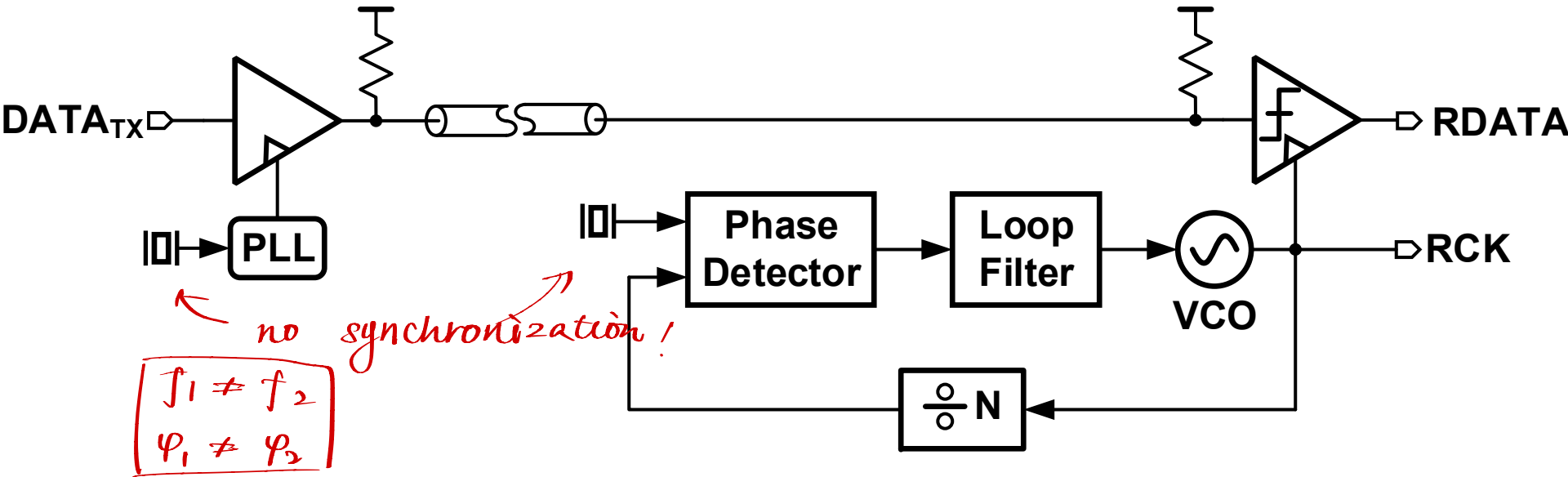
How does jitter getting generated by CDR w/ clean input



Tutorial Roadmap

- Performance metrics
 - **Basic architectures**
 - **Linear/Bang-bang**
 - **Digital**
 - **Hybrid**
 - Application-specific CDRs
 - Multi-lane chip-to-chip links
 - Repeaters for optical links and active cables
 - Summary
-

Phase-Locked Loop based CDR

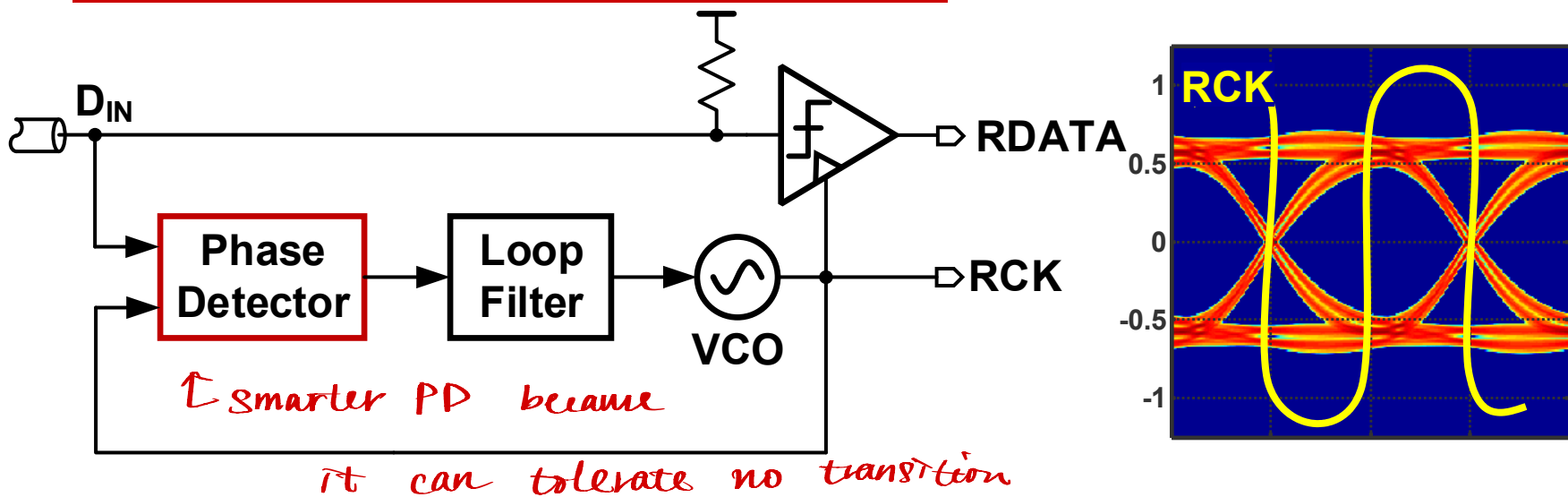


❑ This **WON'T work** because:

- Crystal oscillators at TX and RX do not match
- No phase relationship between received data and RCK

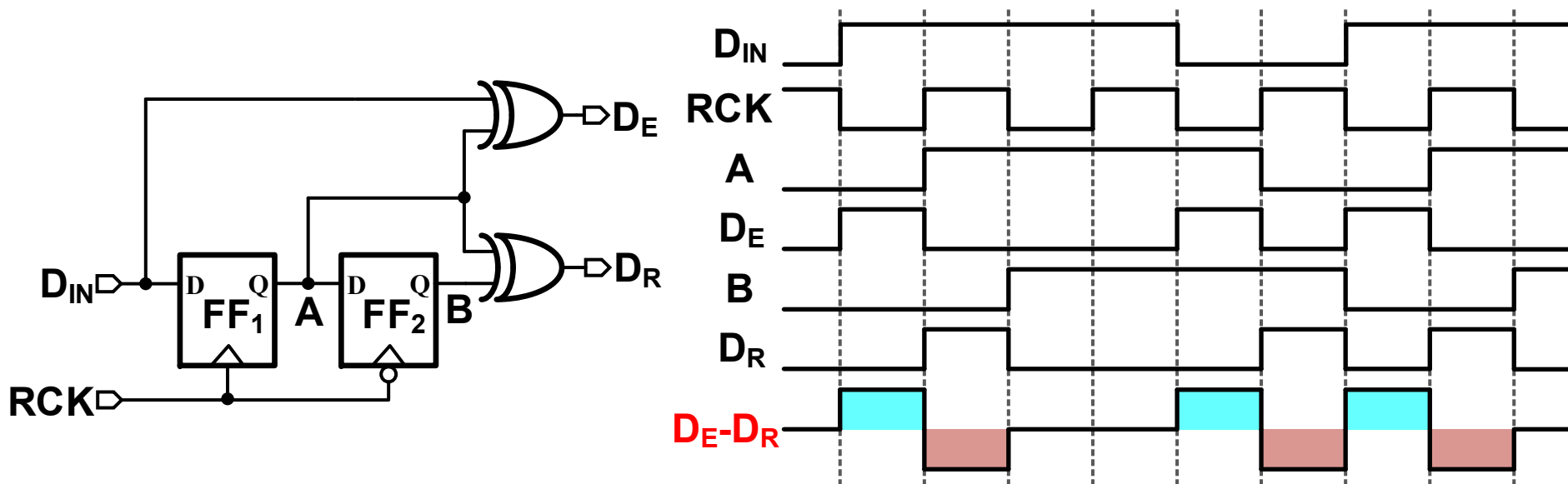
❑ **Need:** Acquire freq. & phase information from data

PLL-based CDR



- ❑ Phase detector should tolerate missing transitions
- ❑ Rest of the building blocks similar to a PLL
- ❑ Neg. edge of recovered clock locks to data edge
- ❑ Pos. edge samples data in the middle of the eye

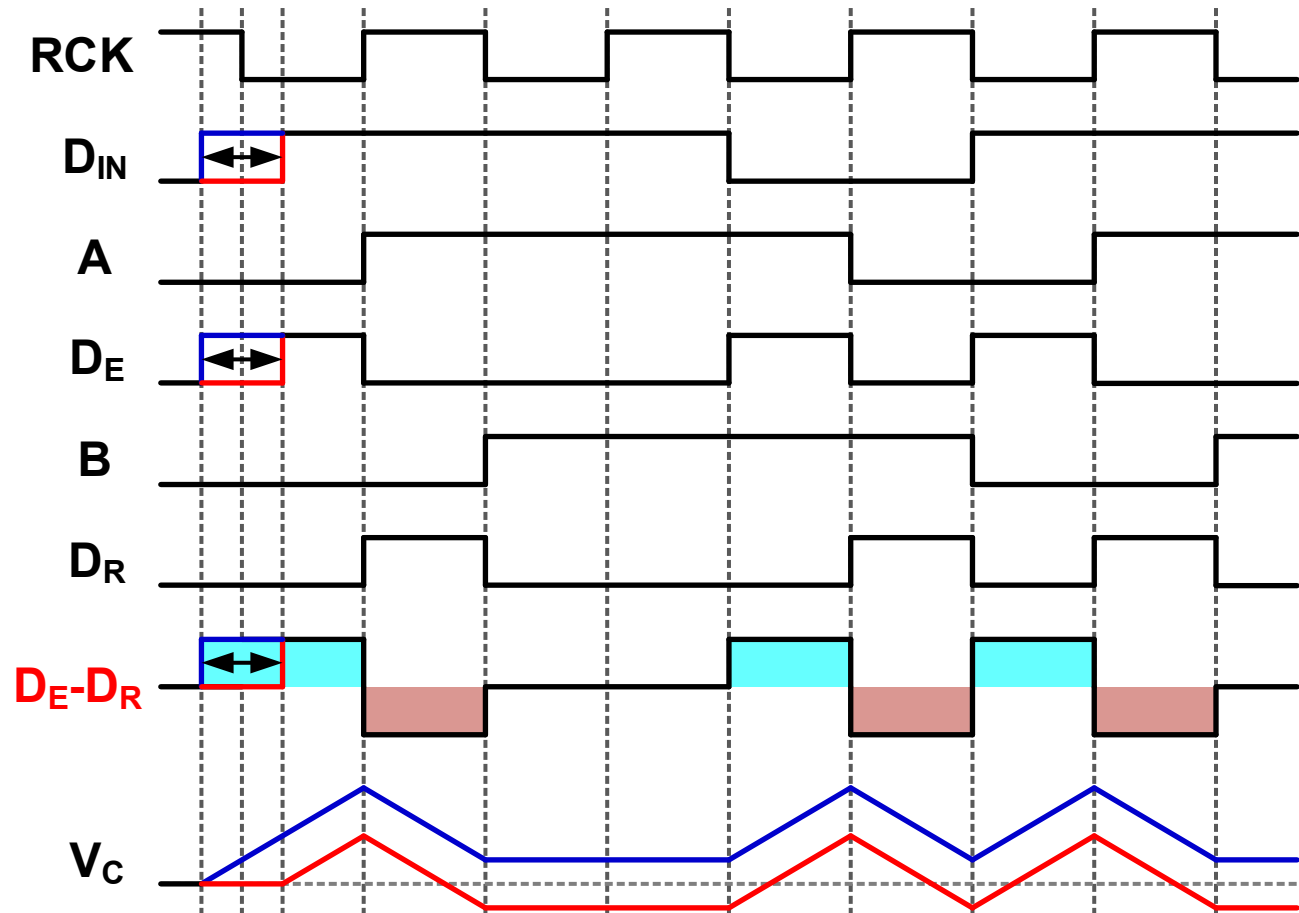
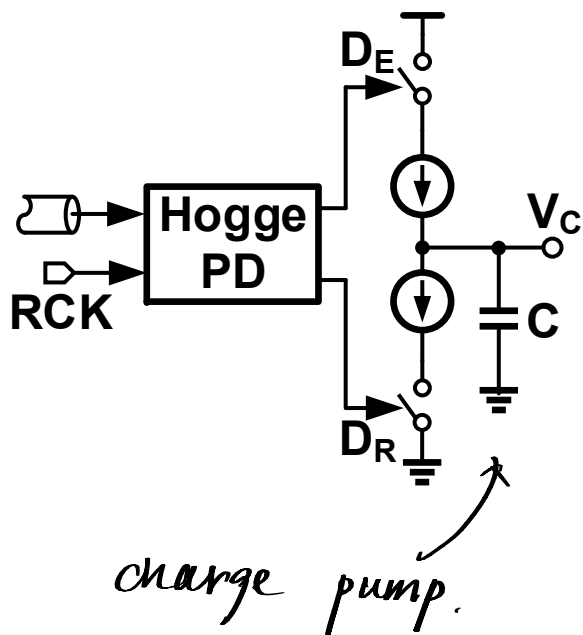
Linear (Hogge) Phase Detector^[1]



- Error output ($D_E - D_R$) is difference of 2 pulses
 - Pulse width of D_E is proportional to phase error
 - Pulse width of D_R is fixed and is equal to $T_{RCK}/2$
- Area under $D_E - D_R$ is proportional to phase error
 - Area is zero when RCK is aligned with D_{IN}

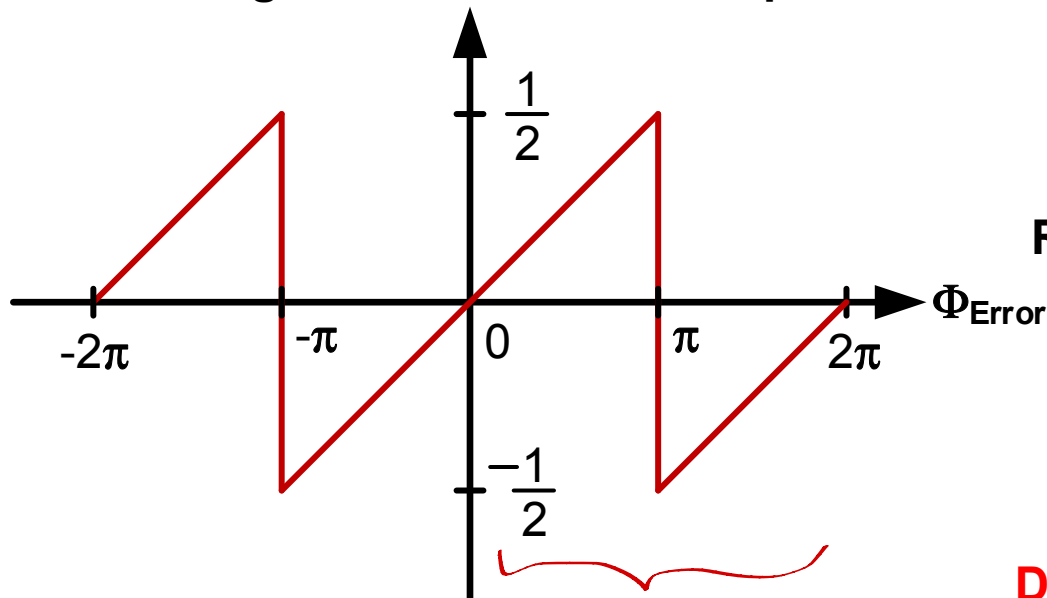
*when there is a transition, D_E pulses $\Delta\phi$, D_R pulses $\frac{T}{2}$.
They are quite if no transition presents. Compare D_E to D_R.*

Hogge PD Waveforms



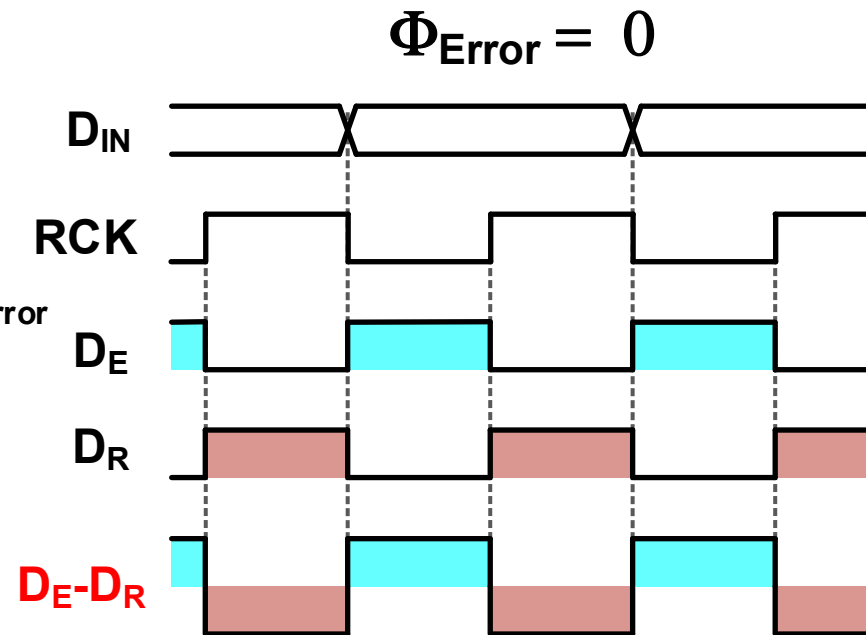
Hogge PD Transfer Function

Average Phase Detector Output

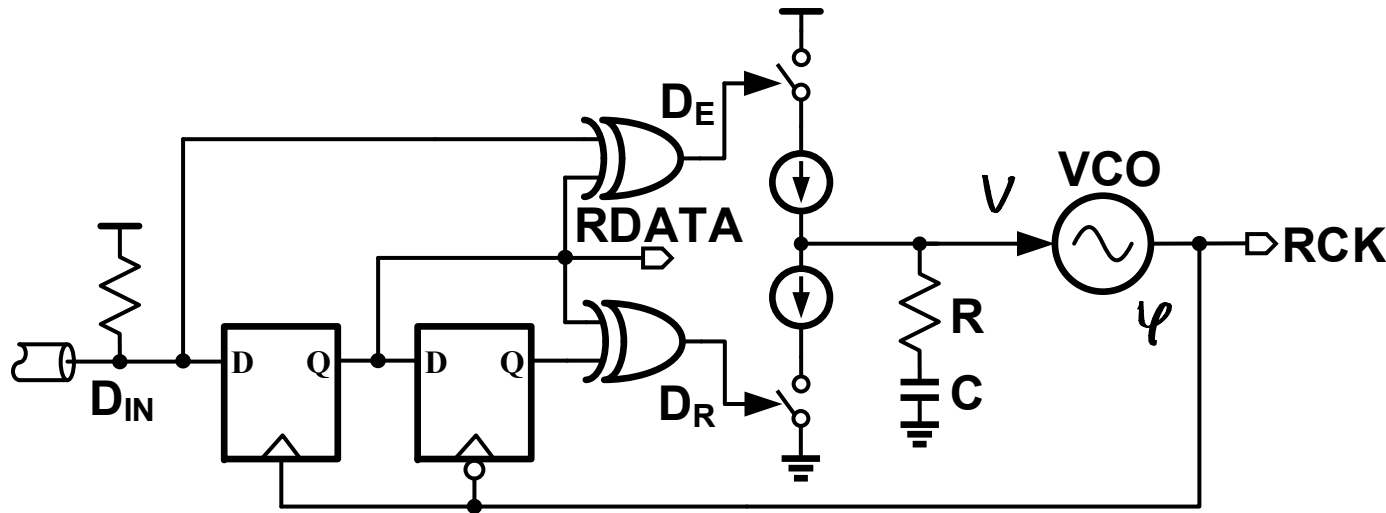


Linear range : $\pm\pi$

$$\text{Gain } K_{\text{pd}} = \frac{1}{2\pi}$$



when $\Phi_{\text{ERROR}} \neq 0$ because
 $\exists \Delta f$, Hogge PD is not a
 frequency detector.

[illegible]

□ Type-II response

- 2 integrators – one in the loop filter and the other is VCO

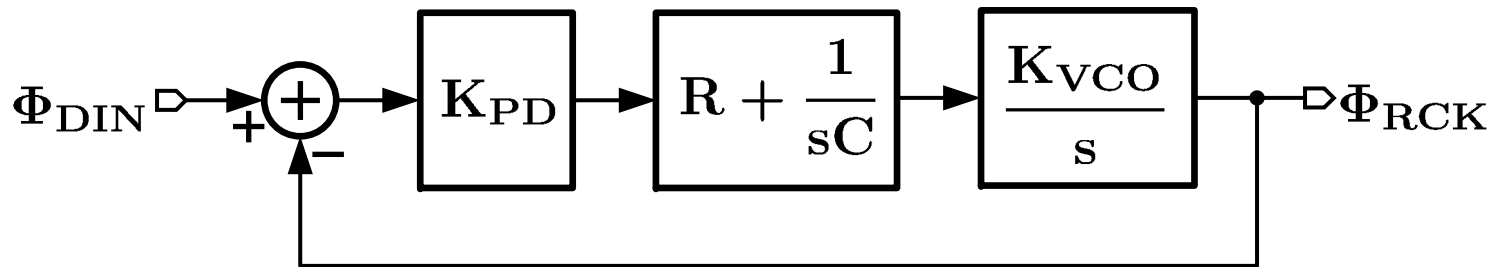
- ❑ Zero static phase offset (ideally)

- CP output should be zero in steady state
- Implies input phase error = 0

most CDR is type 2 because we don't tolerate $\Delta\varphi$

b/w data & clock $\leftarrow$ in the middle of the eye.

Choosing Loop Parameters

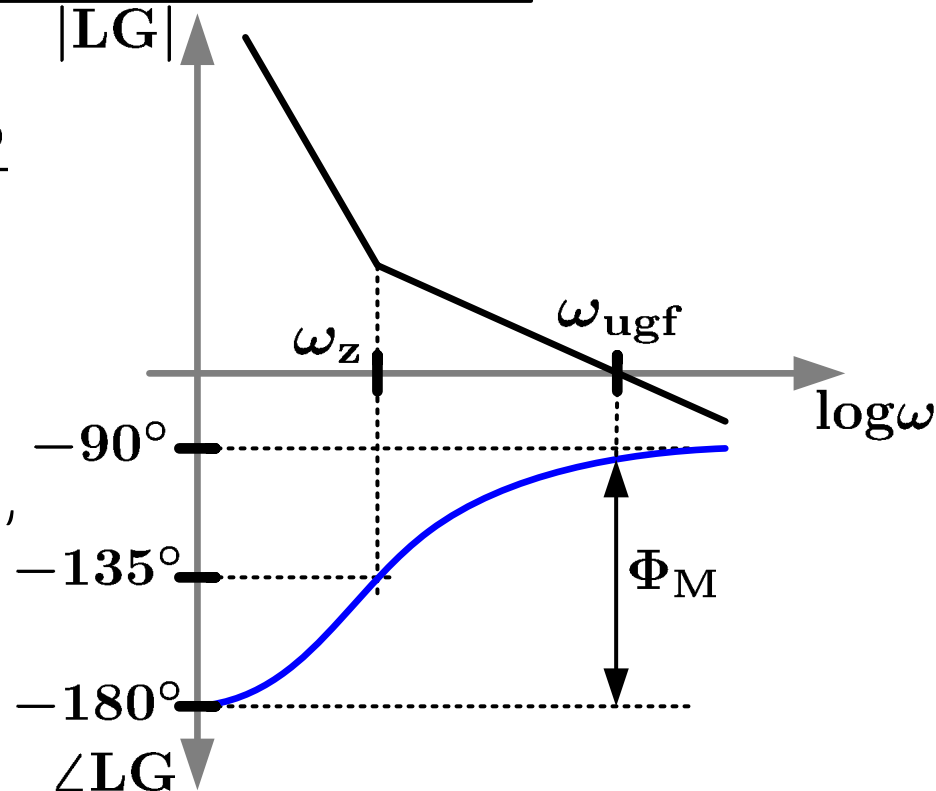


$$LG(s) = K_{PD} \cdot \left(R + \frac{1}{sC} \right) \cdot \frac{K_{VCO}}{s}$$

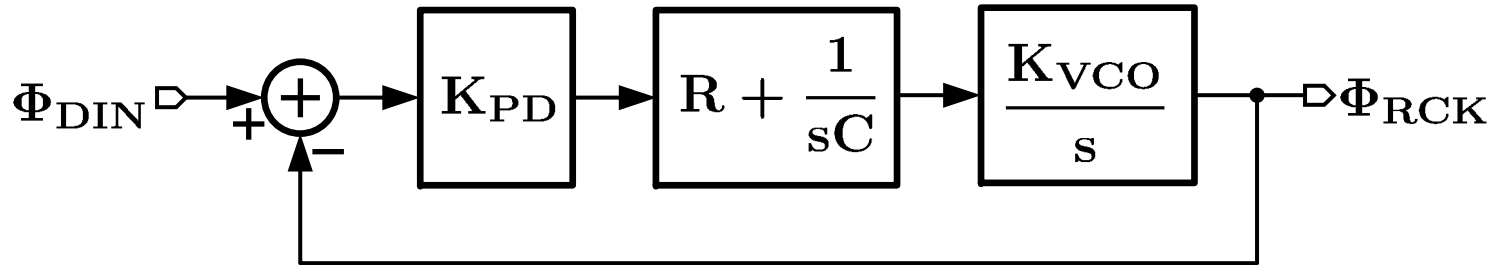
$$\omega_z = \frac{1}{RC}, \quad \omega_{p1} = 0, \quad \omega_{p2} = 0$$

$R \sim \text{few } k\Omega s$,
JTRAN BW $\approx \omega_{ugf}$ *unit gain freq*,

$$\Phi_M = \arctan \left(\frac{\omega_{ugf}}{\omega_z} \right)$$



Jitter Transfer Function *Mason's Rule*



$$\begin{aligned}
 H_{\text{JTRAN}}(s) &= \frac{\Phi_{\text{RCK}}(s)}{\Phi_{\text{DIN}}(s)} = \frac{\text{FPG}(s)}{1 + \text{LG}(s)} \\
 &= \frac{1 + sRC}{1 + sRC + s^2 \cdot \frac{C}{K_{\text{VCO}}K_{\text{PD}}}} \\
 &\equiv \frac{1 + s/\omega_z}{(1 + s/\omega_{\text{PL}})(1 + s/\omega_{\text{PH}})}
 \end{aligned}$$

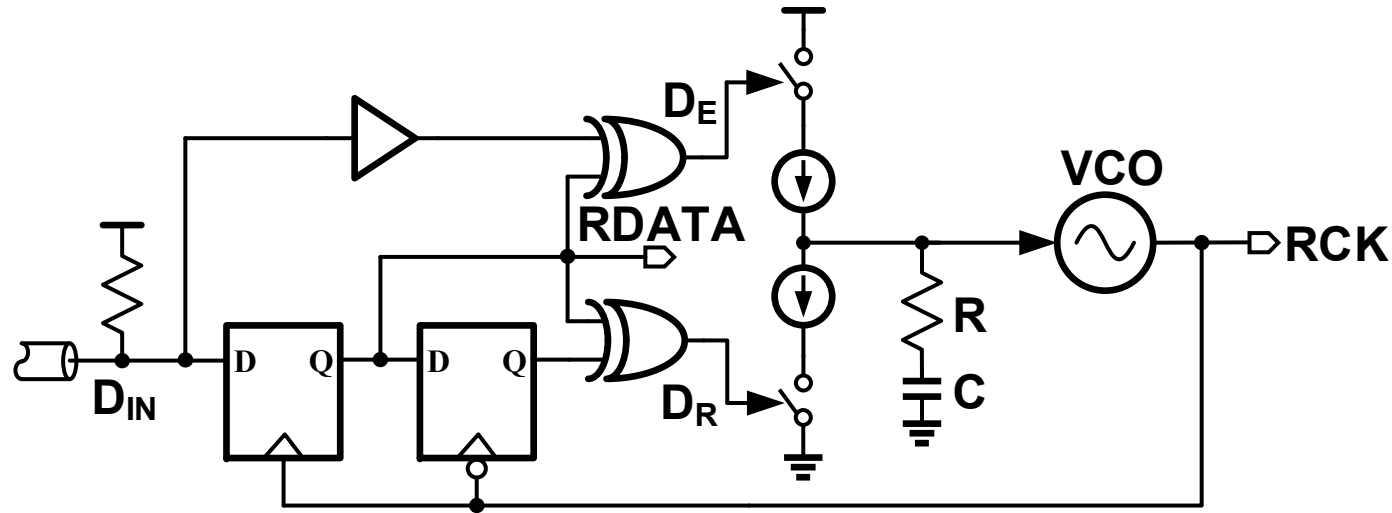
usually $\omega_z \sim \omega_{\text{PL}}$

$\omega_z = 1/RC$
 $\omega_{\text{PL}} \approx 1/RC$
 $\omega_{\text{PH}} \approx K_{\text{VCO}} \cdot K_{\text{PD}} \cdot R$

JTRAN BW = $\omega_{-3\text{dB}} \approx \omega_{\text{PH}} \approx K_{\text{VCO}} \cdot K_{\text{PD}} \cdot R$

• *usually overdamped.*

Linear CDR Drawbacks



- ❑ Jitter peaking (large loop filter area)
- ❑ Coupled JTRAN and JTOL
can't choose one over another
- ❑ Hogge PD non-idealities

Jitter Peaking (I)

□ Zero in feed-forward path → inevitable peaking

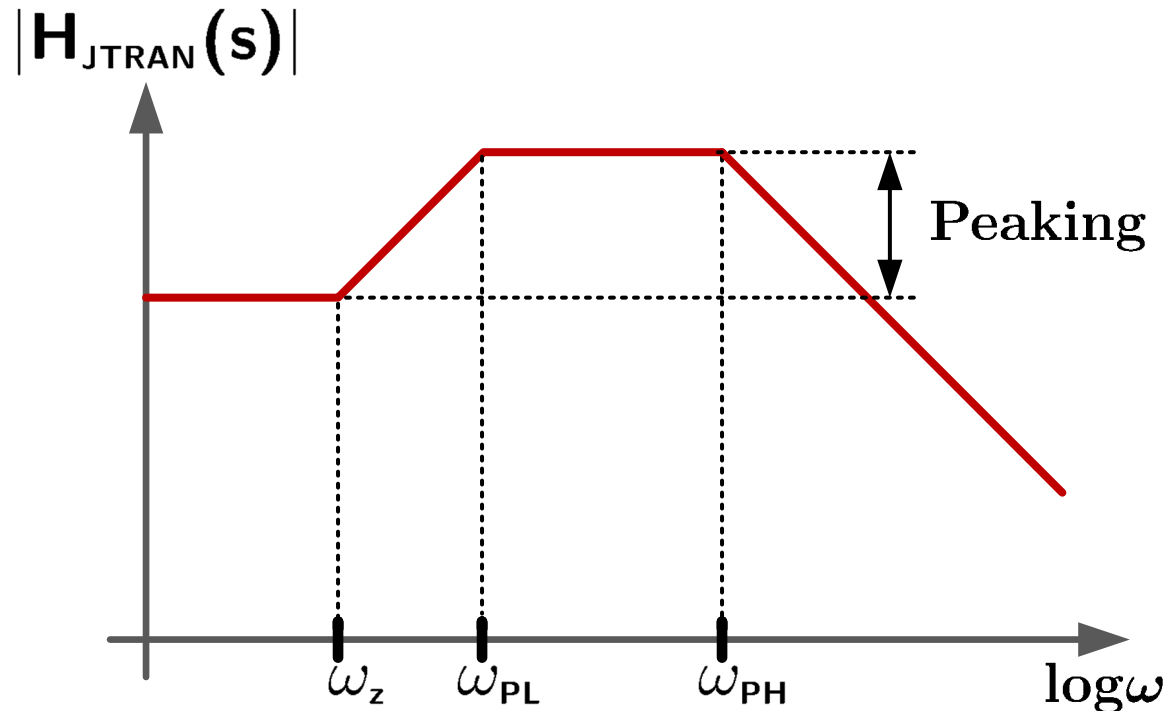
$$H_{JTRAN}(s) = \frac{1 + sRC}{1 + sRC + s^2 \cdot \frac{C}{K_{VCO}K_{PD}}}$$

ω_z has to be
less than $\omega_{p1,2}$.

$$\omega_z = 1/RC$$

$$\omega_{PL} \approx 1/RC$$

$$\omega_{PH} \approx K_{VCO} \cdot K_{PD} \cdot R$$



• Assume the system is like $H(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

Then the system has $\omega_z = -\frac{\omega_n}{2\zeta} = -\frac{1}{RC}$

and $\omega_{p_{1,2}} = (-\zeta \pm \sqrt{\zeta^2 - 1})\omega_n$

• If $\zeta = 1$, then $\omega_{p_1} = \omega_{p_2} = -\omega_n$ where $|\omega_{p_1}| > |\omega_z|$.

• If $\zeta \gg 1$, then $\sqrt{\zeta^2 - 1} \approx \zeta(1 - \frac{1}{2\zeta^2}) = \zeta - \frac{1}{2\zeta}$, and

• $\omega_{p_1} \approx (-\zeta + \zeta - \frac{1}{2\zeta})\omega_n = \frac{\omega_n}{2\zeta} \approx \omega_z$

• $\omega_{p_2} \approx (-\zeta - \zeta + \frac{1}{2\zeta})\omega_n \approx -2\zeta\omega_n$.

vanish

Razavi, "Design of CMOS
Phase-Locked Loops", Chapter
8.1.

8.1 More on PLL Transfer Functions

Our study of simple charge-pump PLLs has led to various closed-loop transfer functions from different points to the output. In summary, we have found that (a) the input phase fluctuations experience a second-order low-pass transfer with one zero, and (b) noise voltages in series with the control or supply of the VCO see a band-pass response. In this section, we further explore the PLL transfer functions.

The input/output transfer function expressed by (7.23) can be rewritten as

$$H(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (8.1)$$

The zero is located at $\omega_z = -\omega_n/(2\zeta)$ and the poles at

$$\omega_{p1,2} = (-\zeta \pm \sqrt{\zeta^2 - 1})\omega_n. \quad (8.2)$$

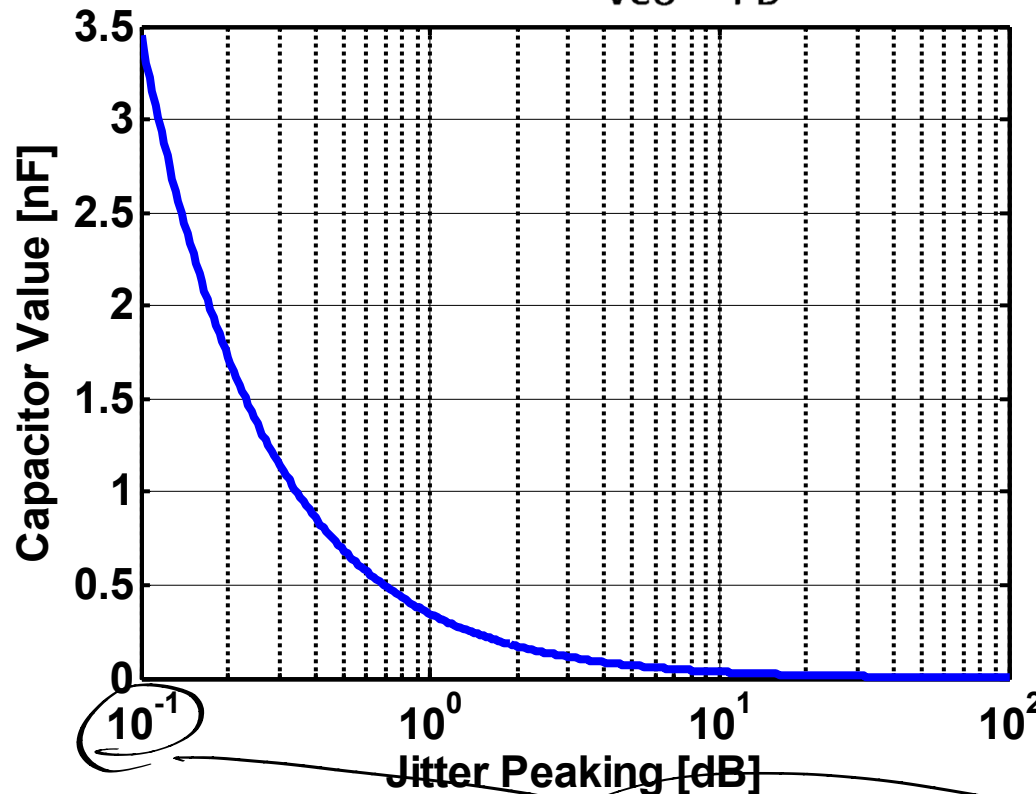
For $\zeta = 1$, we have $\omega_z = -\omega_n/2$ and $\omega_{p1} = \omega_{p2} = -\omega_n$. If, on the other hand, $\zeta^2 \gg 1$, then $\sqrt{\zeta^2 - 1} \approx \zeta[1 - 1/(2\zeta^2)] = \zeta - 1/(2\zeta)$, yielding

$$\omega_{p1} \approx -\frac{\omega_n}{2\zeta} = -\frac{1}{R_1 C_1} \quad (8.3)$$

$$\omega_{p2} \approx -2\zeta\omega_n = -\frac{R_1 I_p K_{VCO}}{2\pi}. \quad (8.4)$$

Jitter Peaking (II)^[2]

$$\text{Jitter Peaking [dB]} \approx \frac{8.686}{K_{\text{VCO}} K_{\text{PD}} C R^2} \approx \frac{8.686}{\omega_{-3\text{dB}} C R}$$

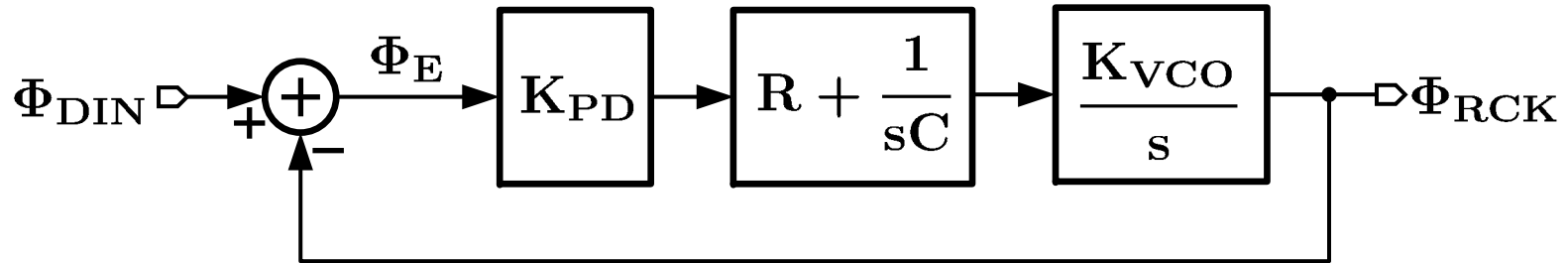


□ BW = 4MHz, R = 1kΩ & 0.1dB peaking, C ~ 3.5nF

repeaters : can't tolerate any jitter peaking.

expensive

Jitter Tracking

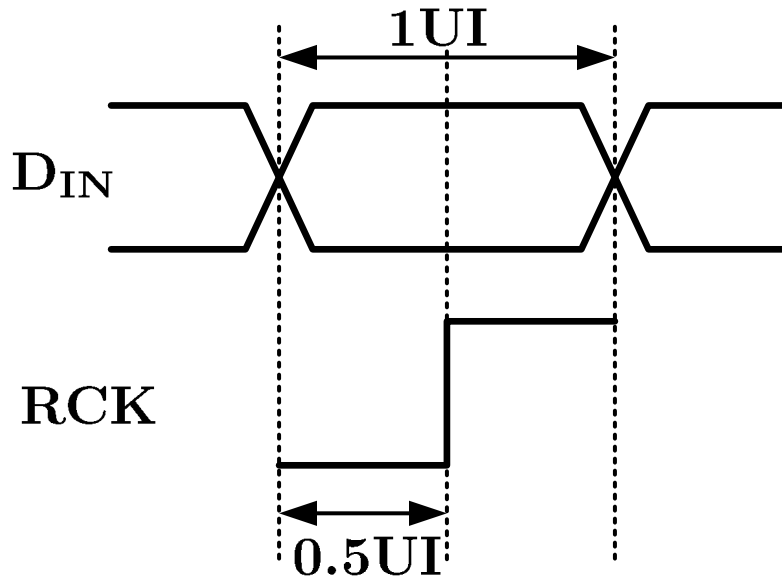


$$\begin{aligned}
 H_{\text{JTRACK}}(s) &= \frac{\Phi_E(s)}{\Phi_{\text{DIN}}(s)} = \frac{1}{1 + \mathbf{LG}(s)} \\
 &= \frac{s^2 \cdot \frac{C}{K_{\text{VCO}} K_{\text{PD}}}}{1 + sRC + s^2 \cdot \frac{C}{K_{\text{VCO}} K_{\text{PD}}}} \\
 &\equiv \frac{s^2 / \omega_{\text{PL}} \omega_{\text{PH}}}{(1 + s/\omega_{\text{PL}})(1 + s/\omega_{\text{PH}})}
 \end{aligned}$$

$$\text{JTRACK BW} = \underbrace{\omega_{-3\text{dB}}}_{\approx \omega_{\text{PH}}} \approx K_{\text{VCO}} \cdot K_{\text{PD}} \cdot R$$

When jitter $f \ll \text{BW}$, fully tracked.

Jitter Tolerance (JTOL) (I)



$$|\Phi_E| < 0.5 UI$$

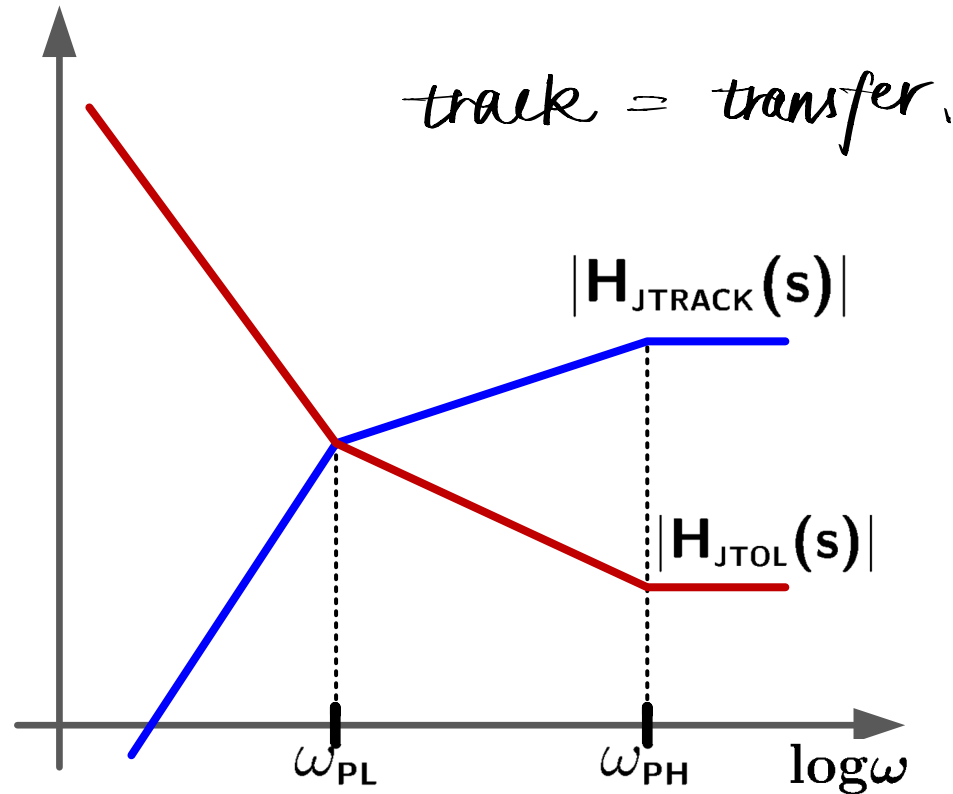
$$\Phi_{DIN} \times H_{JTRACK}(s) < 0.5 UI$$

$$\Rightarrow \Phi_{DIN} < \frac{0.5 UI}{H_{JTRACK}(s)}$$

$$\underline{\underline{H_{JTOL}(s) = \frac{0.5}{H_{JTRACK}(s)}}}$$

$$H_{JTOL}(s) = \frac{0.5 (1 + s/\omega_{PL}) (1 + s/\omega_{PH})}{s^2 / \omega_{PL} \omega_{PH}}$$

Jitter Tolerance (JTOL) (II)

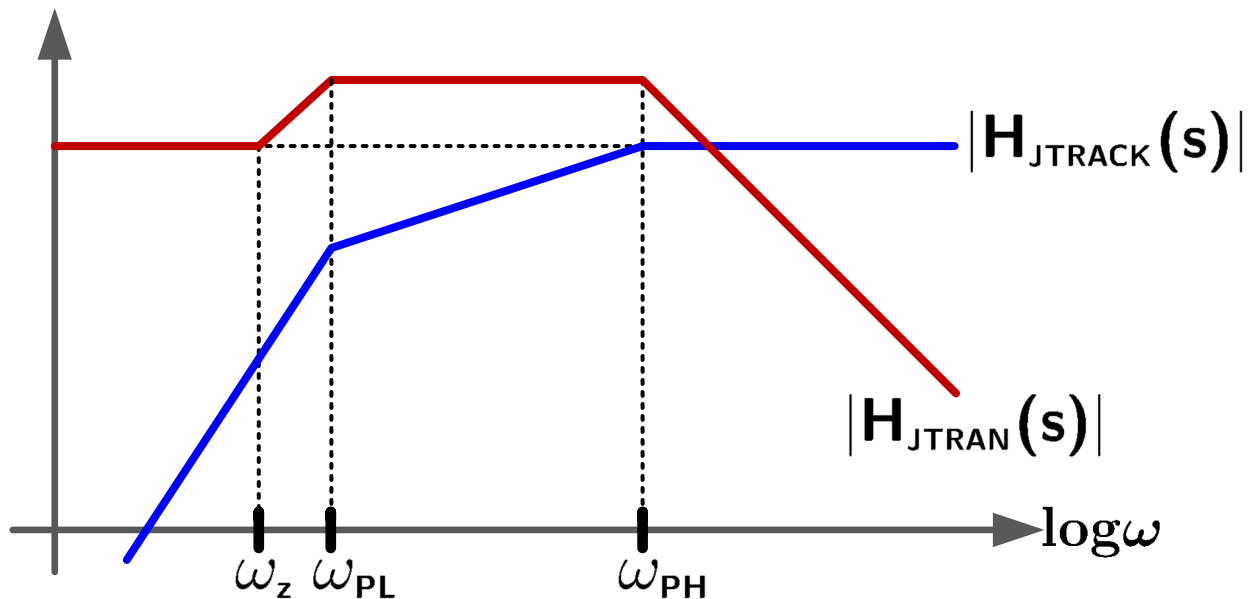


- JTOL improves w/ better jitter tracking
 - Better jitter tracking → wider JTRAN bandwidth!

Coupled JTRAN/JTOL Behavior

$$H_{JTRAN}(s) = \frac{1 + sRC}{1 + sRC + s^2 \cdot \frac{C}{K_{VCO}K_{PD}}}$$

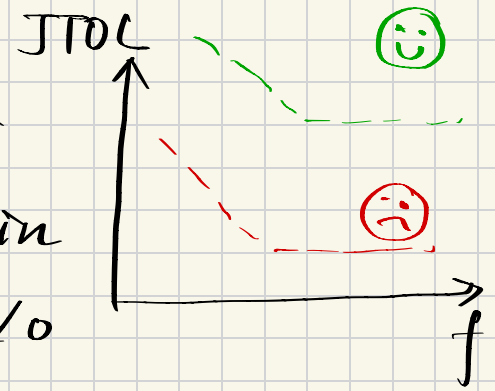
$$H_{JTRACK}(s) = \frac{s^2 \cdot \frac{C}{K_{VCO}K_{PD}}}{1 + sRC + s^2 \cdot \frac{C}{K_{VCO}K_{PD}}}$$



□ Both JTRAN and JTOL are governed by ω_{PH}

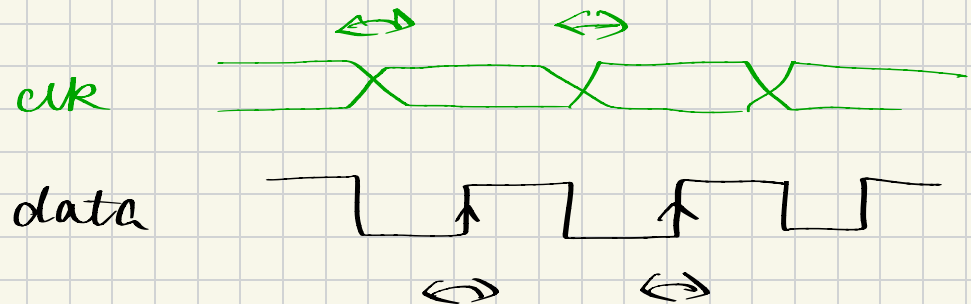
* JTOL

usually JTOL is the larger the better since it characterizes how much jitter in the data stream can CDR tolerate w/o causing a bit error.



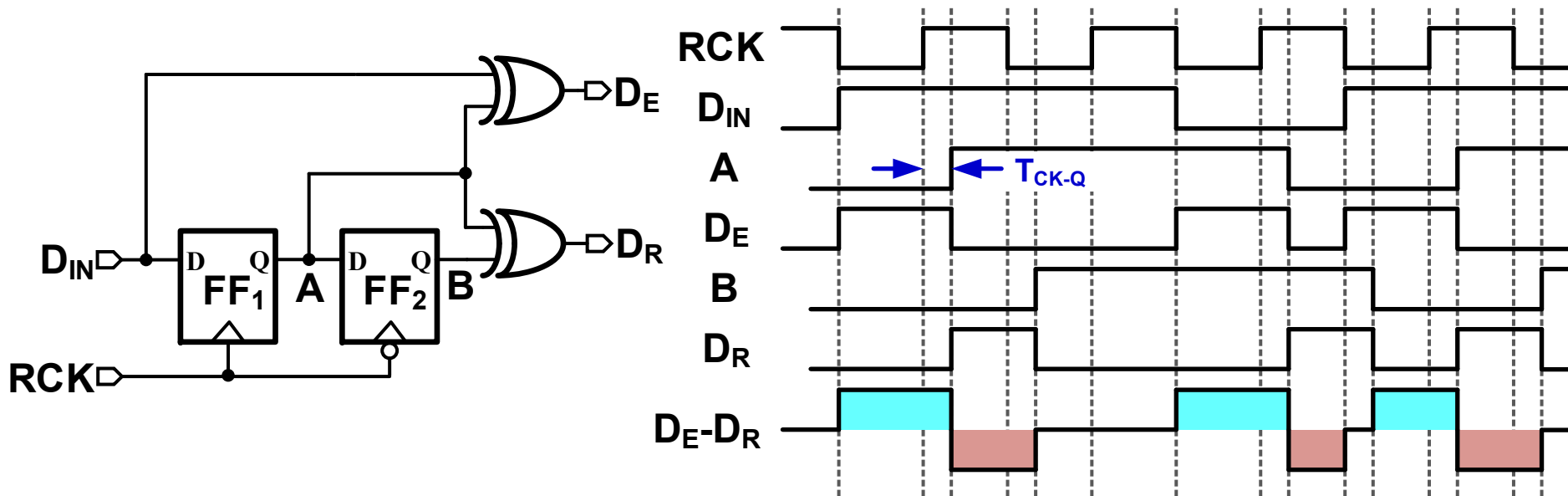
* Jtran

however, this does require the recovered clock to "track" the jittery data in order to track the behavior well.



Δ this can cause the recovered clock to be jittery, or even sometimes missing edges.

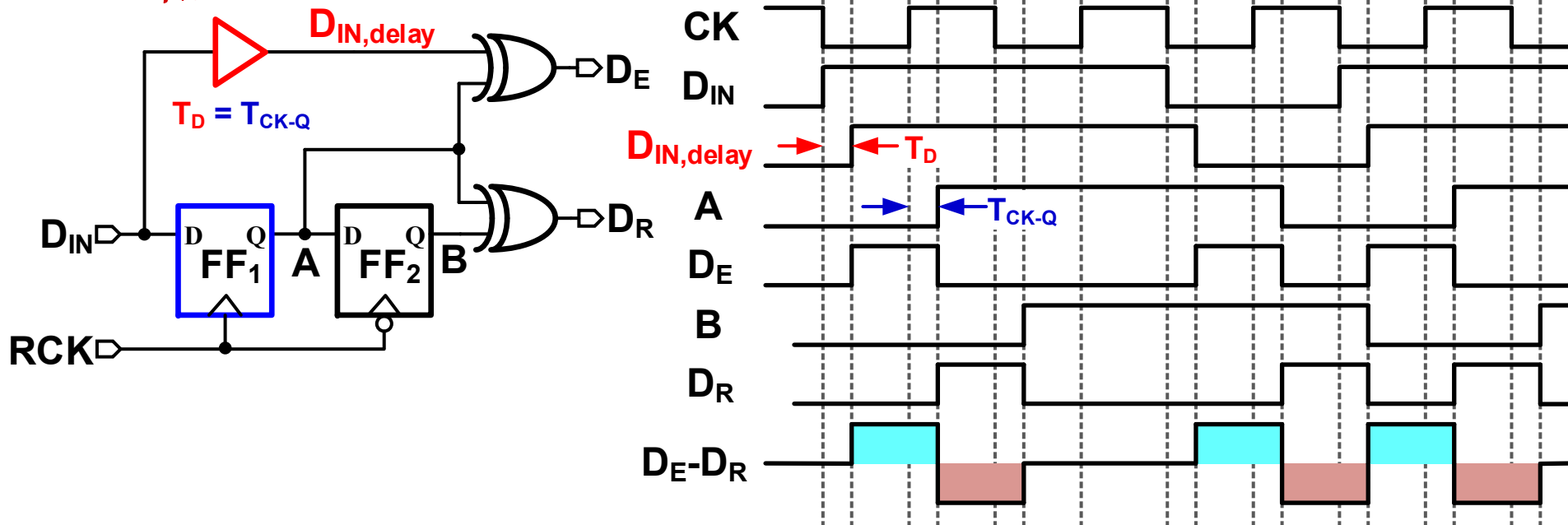
Hogge PD Non-idealities: Offset^[2]



- FF1 clock-to-Q delay introduces phase offset
causing locking errors.

Hogge PD Offset Mitigation^[2]

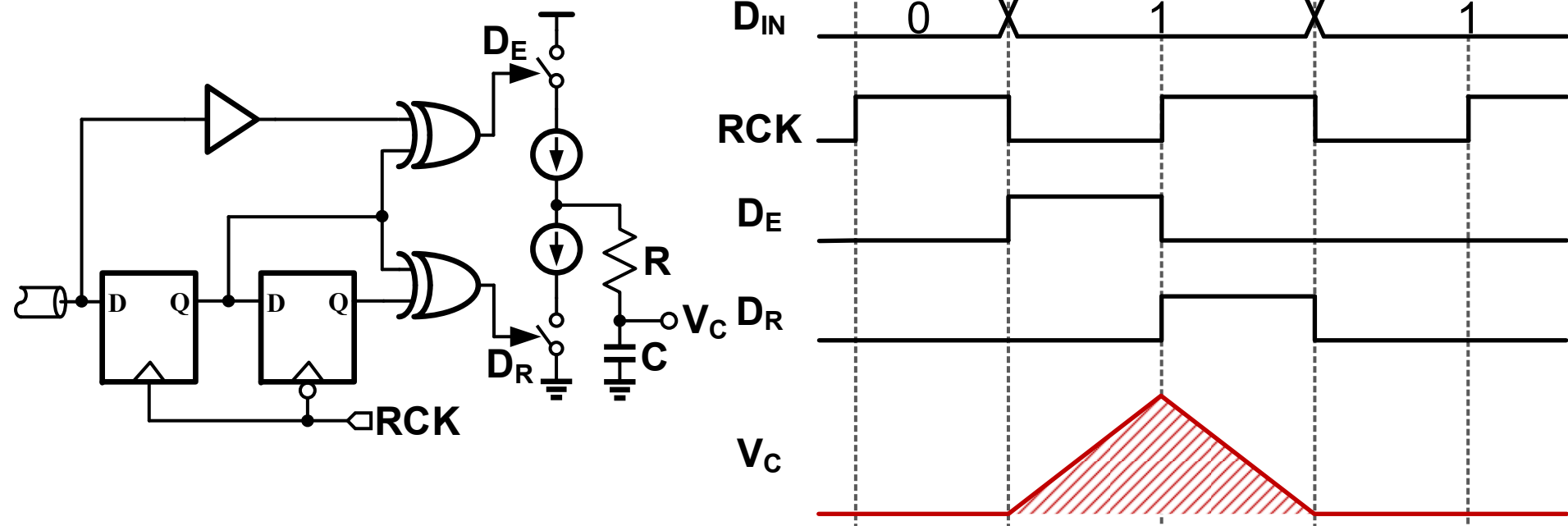
hard to track PVT



- ❑ FF1 clock-to-Q delay introduces phase offset
- ❑ FF1 delay compensated by inserting buffer
 - T_D may not track T_{CK-Q} across supply and temperature
 - Generating small well controlled T_D is also difficult

main reason Hogge PD not used in high f.

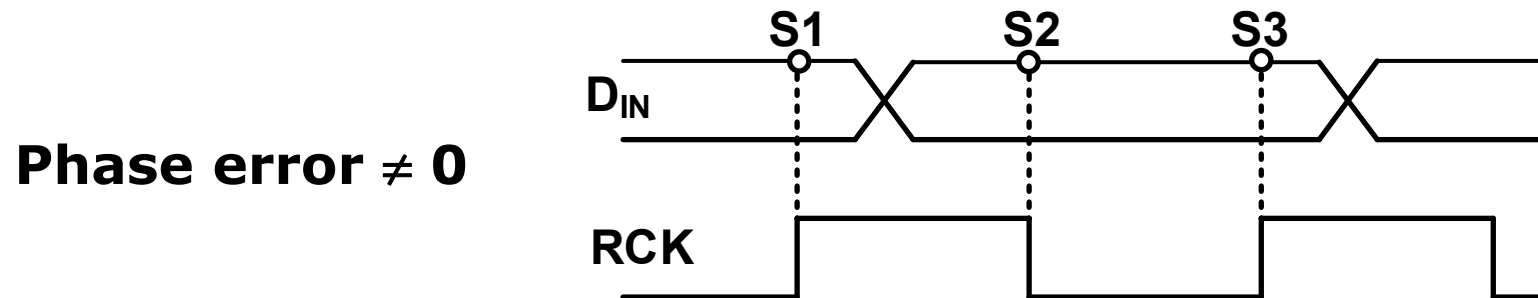
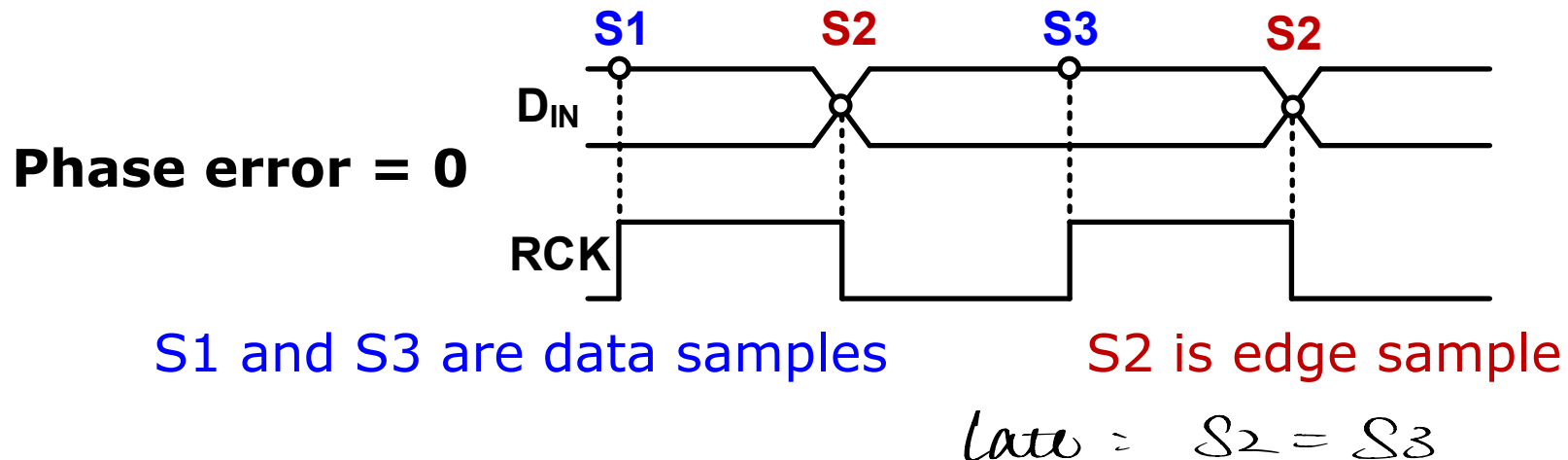
Hogge PD Non-idealities: DDJ^[2]



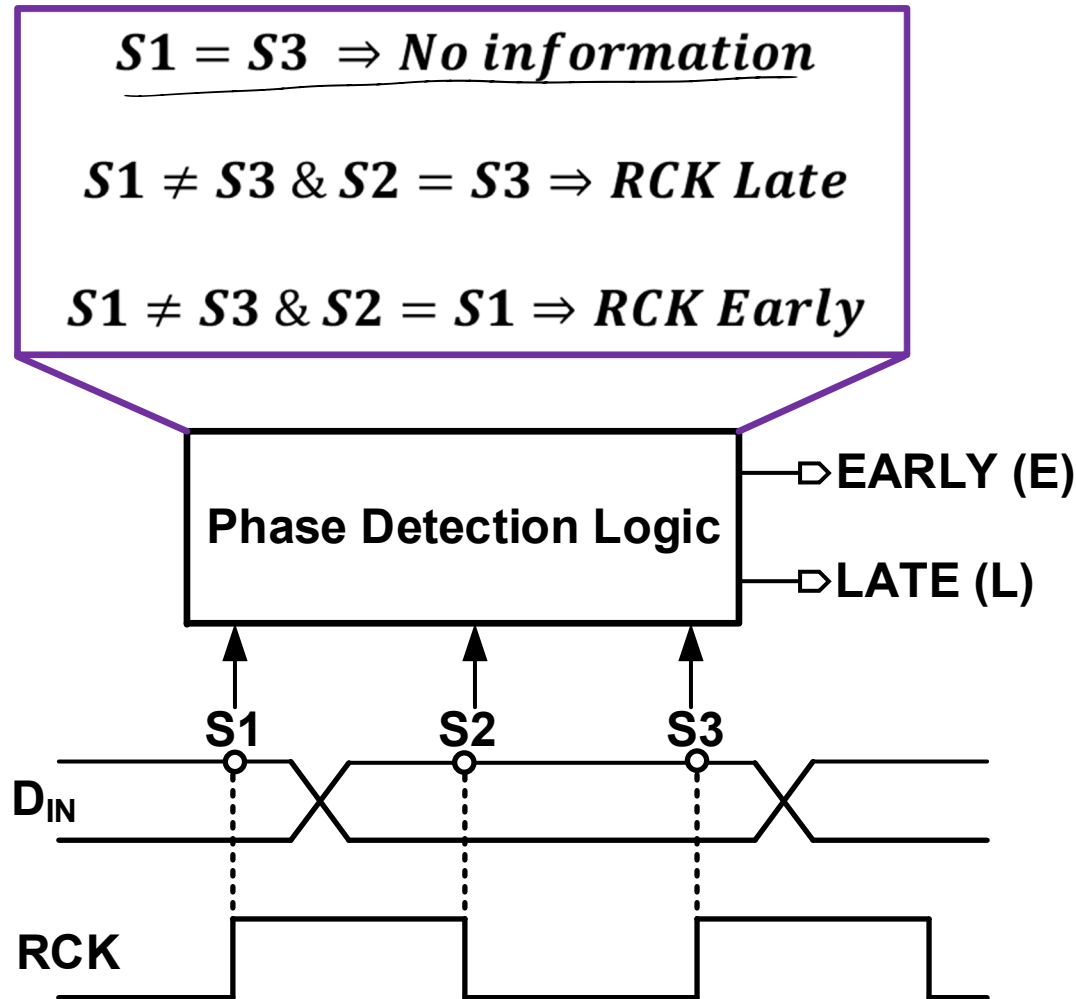
- D_E/D_R pulses not aligned in time ↑ causing VCO $f \uparrow$
 - "Tri-wave" on V_C causes Data Dependent Jitter (DDJ)
- See [2] for modified Hogge PD to mitigate DDJ

Bang-Bang Phase Detector (I)

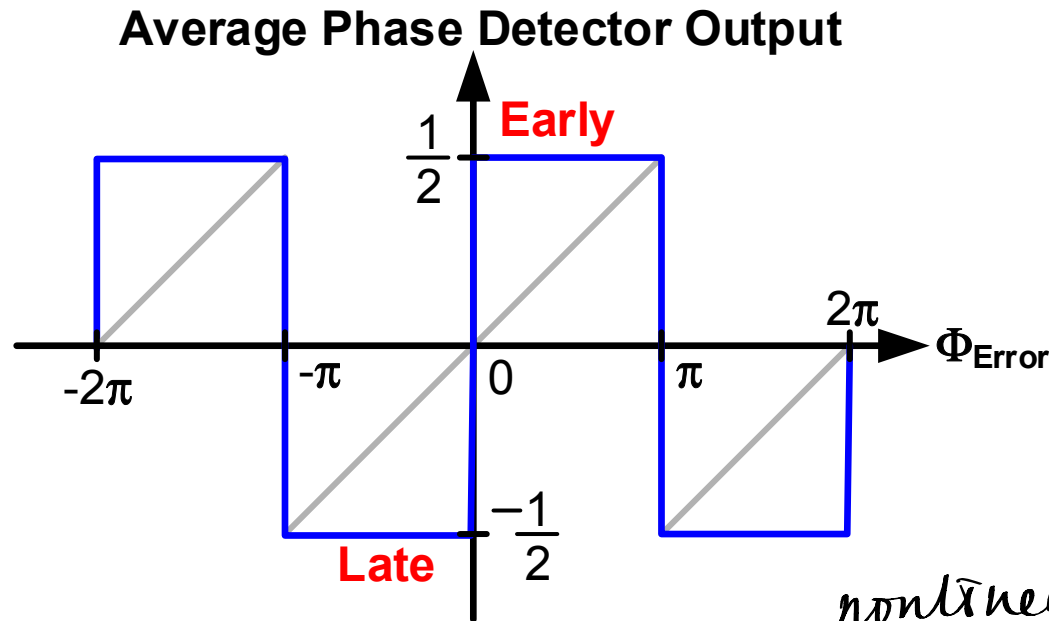
□ Can we detect phase error from sampled data?



Bang-Bang Phase Detector (II)^[3]



BBPD Characteristics: w/o Jitter

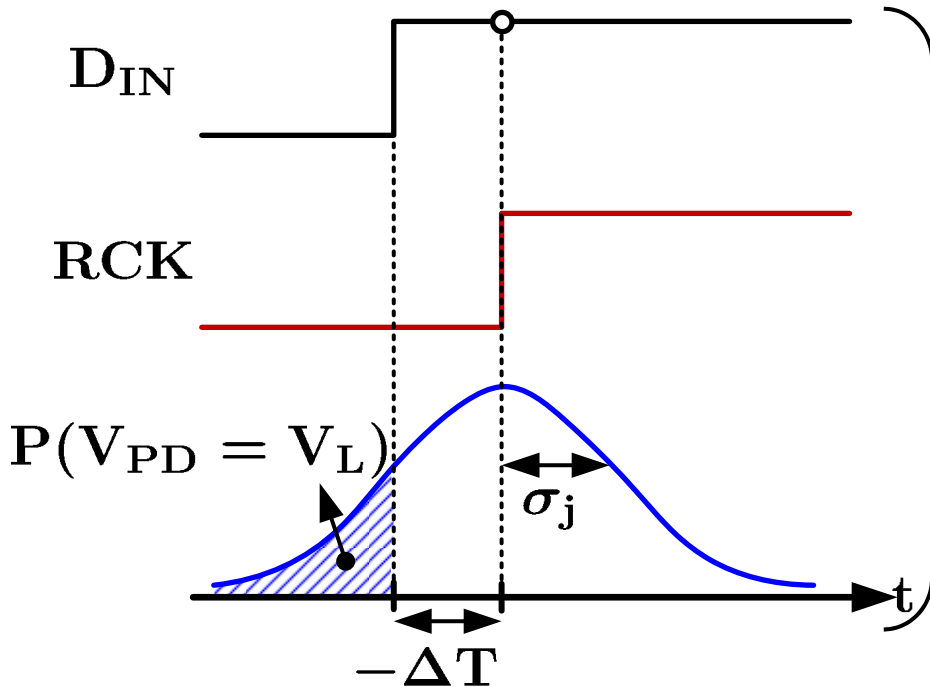


nonlinear

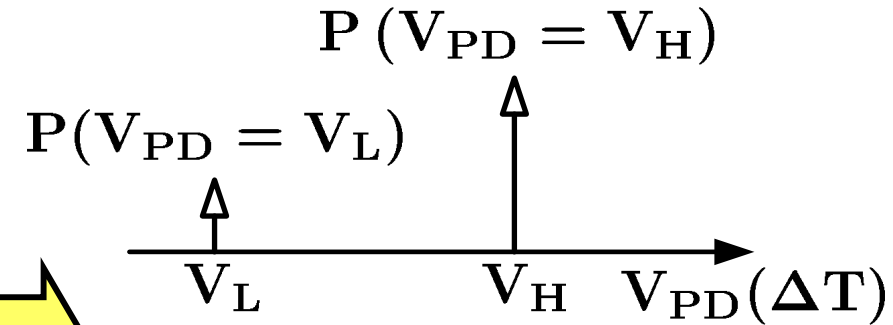
Useful range : $\pm\pi$

Gain $K_{PD}|_{\Delta T=0} = \infty$

BBPD Characteristics: w/Jitter (I)^[4]



Discrete PDF of BBPD output



$$P(V_{PD} = V_L) = \int_{-\infty}^{-\Delta T} p(x) dx$$

$$P(V_{PD} = V_H) = \int_{-\Delta T}^{\infty} p(x) dx$$

$$\overline{V_{PD}(\Delta T)} = V_L \cdot P(V_{PD} = V_L) + V_H \cdot P(V_{PD} = V_H)$$

BBPD Characteristics: w/ Jitter (II)

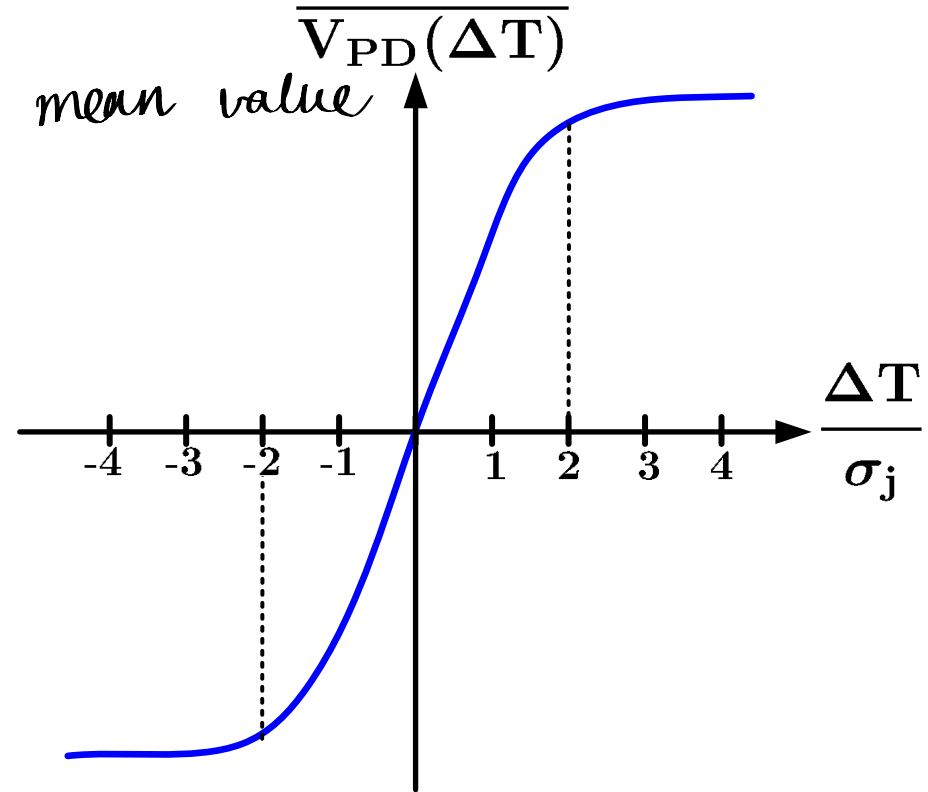
$$\overline{V_{PD}(\Delta T)} = 2V_o \cdot \text{erf} \left(\frac{\Delta T}{\sigma_j} \right)$$

$$\text{erf}(x) = \frac{1}{\sqrt{2\pi}} \int_0^x e^{-y^2} dy$$

$$\text{Gain } K_{PD}|_{\Delta T=0} = \frac{2V_o}{\sqrt{2\pi}\sigma_j}$$

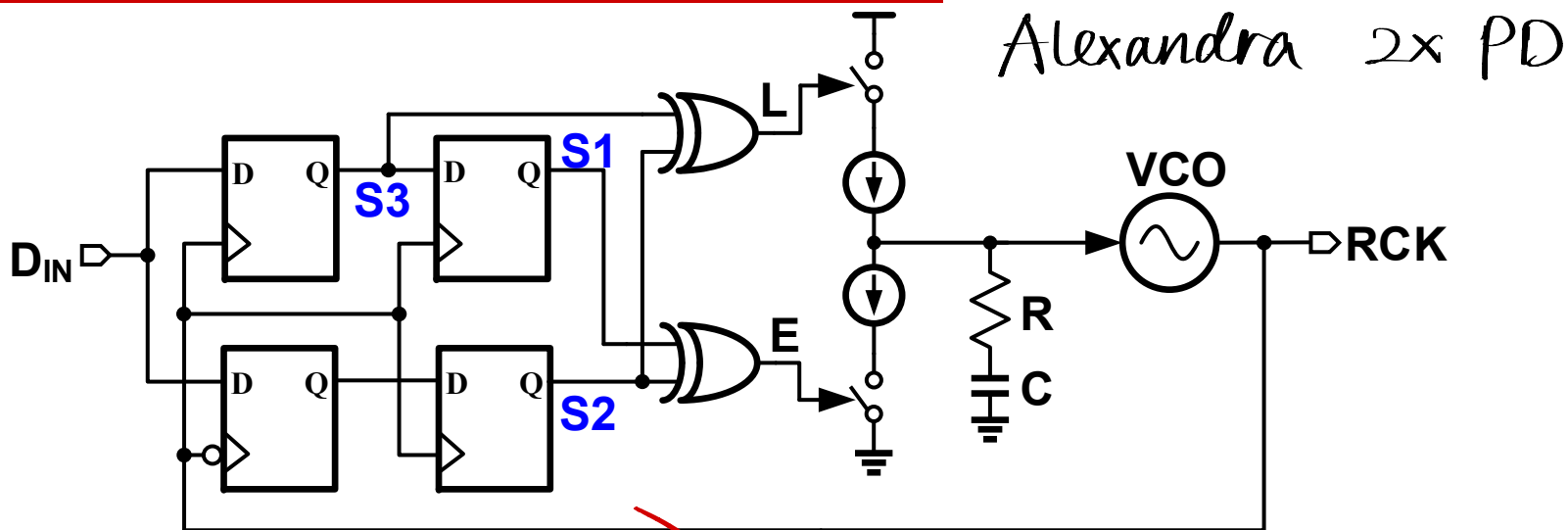
Linear range : $\pm 2\sigma_j$

linearized due to jitter.



$$K_{PD} = \frac{d\overline{V_{PD}(\Delta T)}}{d\Delta T} = \frac{2V_o}{\sqrt{2\pi}\sigma_j} e^{\left(-\frac{\Delta T}{\sqrt{2}\sigma_j}\right)^2}$$

Full-rate Bang-Bang CDR

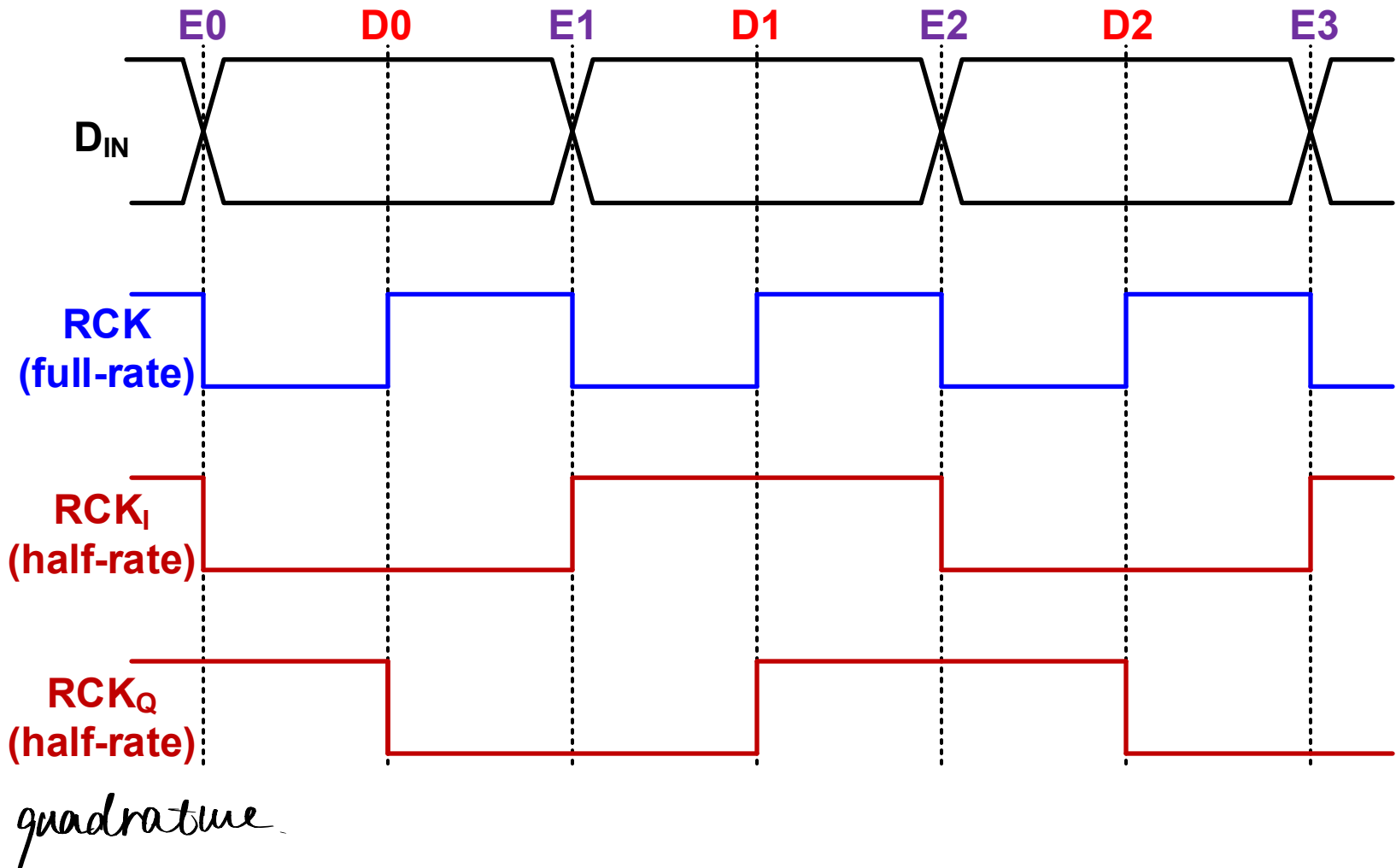


- ❑ Type-II response
- ❑ Near-zero static phase offset
- ❑ Insensitive to charge-pump non-idealities
- ❑ VCO & PD operate at full-rate ($F_{VCO} = F_{DIN}$)
 - Could become a speed bottleneck
 - Solution: Half-rate bang-bang CDR

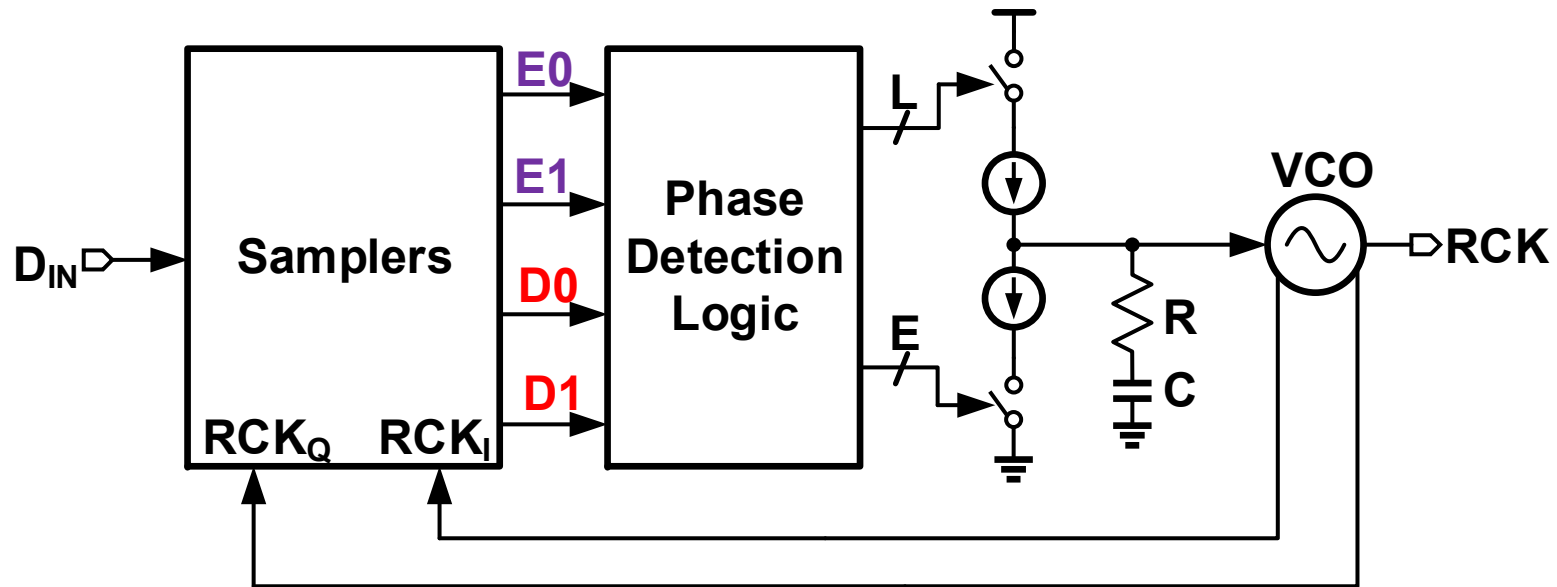
sub-rate CDR

because K_{PD} is large!
 might be a PDK limitation

Half-rate CDR Waveforms

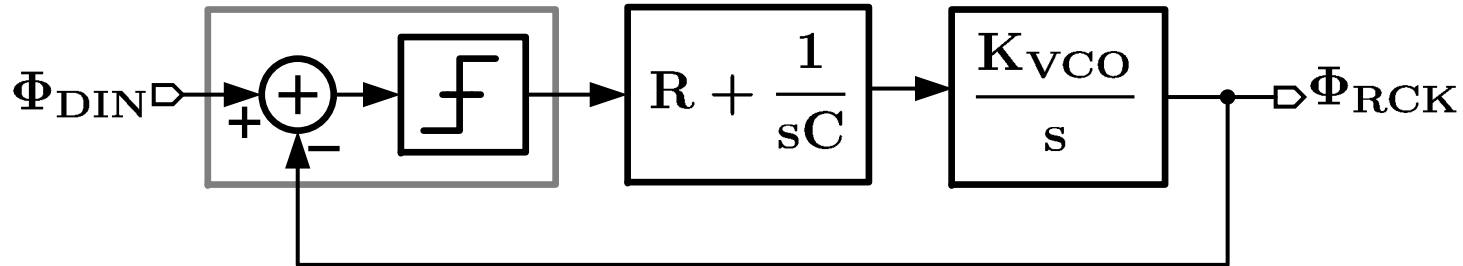


Half-Rate Bang-Bang CDR



- Topology same as full-rate architecture
 - Same phase detection logic as full-rate
- Requires quadrature VCO
- Lower loop update rate → higher loop latency

Choosing Loop Parameters^[5]



$$\text{Damping factor } \xi = \frac{\Delta\Phi_{RCK} \text{ due to prop. path}}{\Delta\Phi_{RCK} \text{ due to integ. path}} = \frac{2RC}{T_{UPDATE}} \gg 1$$

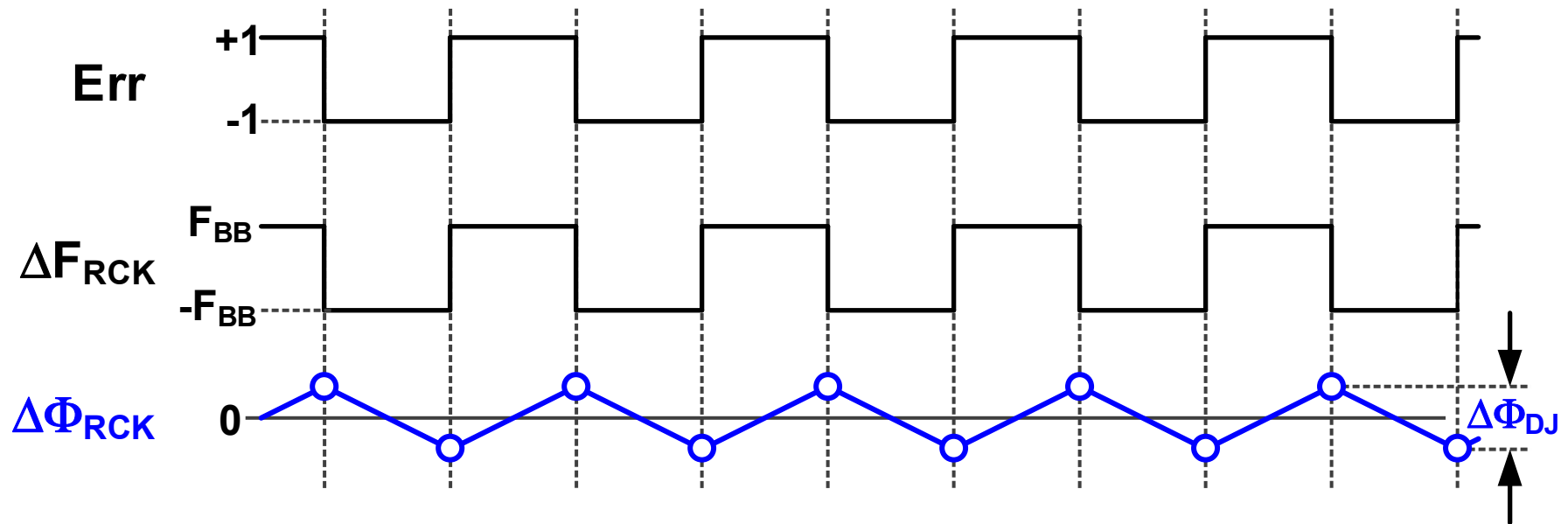
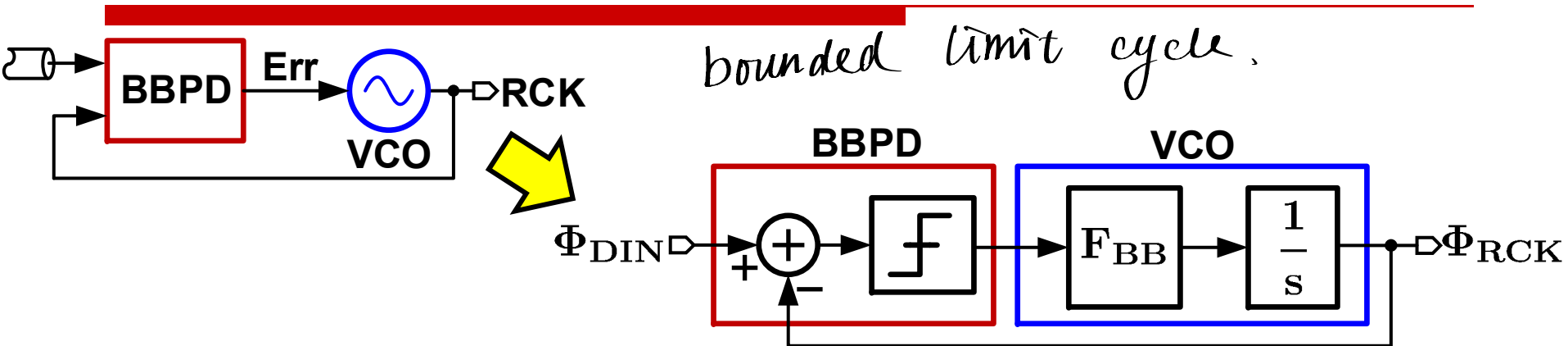
- BBPD makes the loop non-linear
 - Cannot use transfer function analysis
 - Loop gain is infinite $\rightarrow$ unstable in “linear sense”

- Ensure stability by choosing large damping factor
 - Relatively independent proportional and integral paths

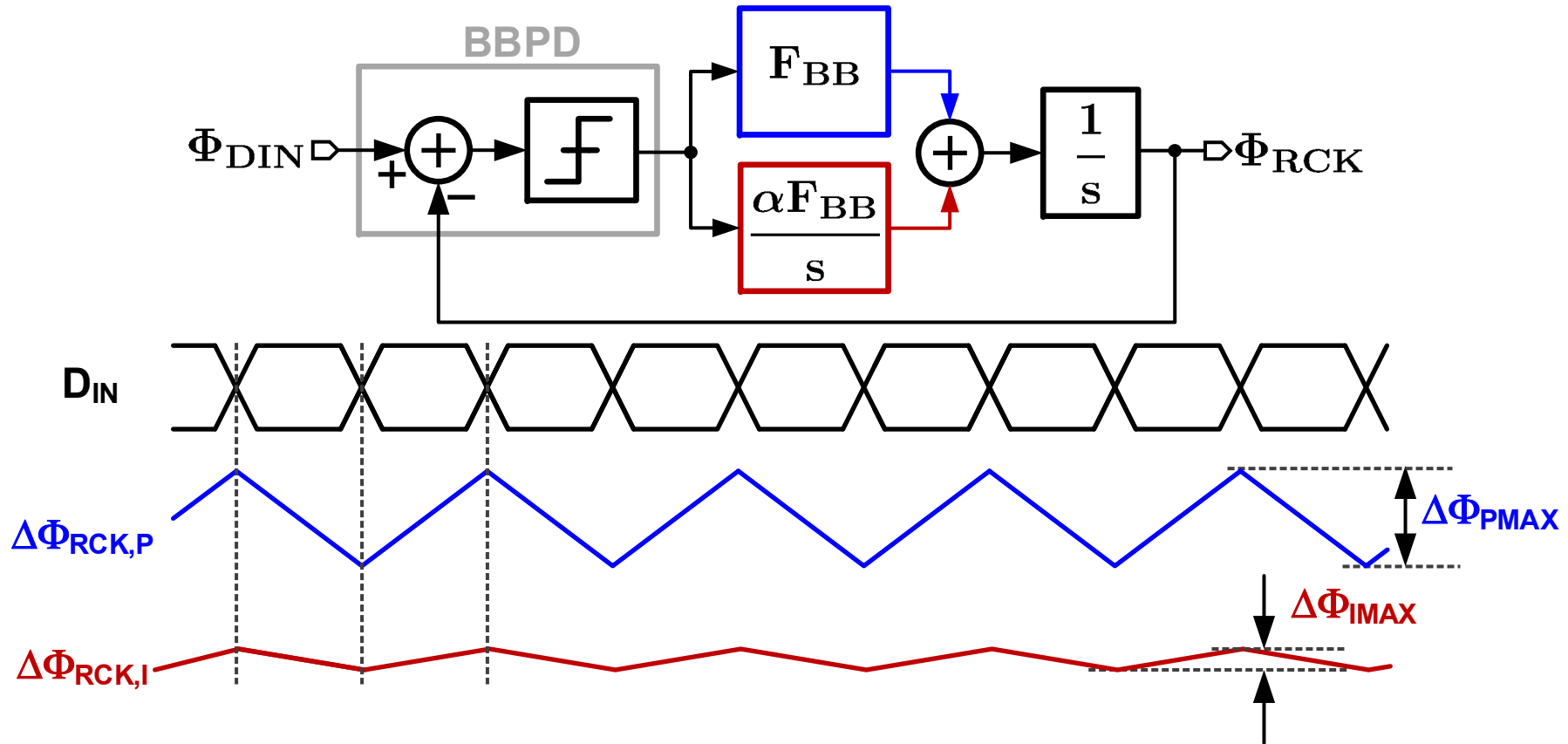
Bang-Bang CDR Drawbacks

- ❑ Coupled JTRAN and JTOL
 - ❑ Jitter peaking
 - ❑ Large loop filter area
- } Similar to linear CDR
-
- ❑ **JGEN caused by limit cycles** (*nonlinear behavior*)
 - ❑ **JTRAN dependence on input jitter**

Steady-State Limit Cycles (I)



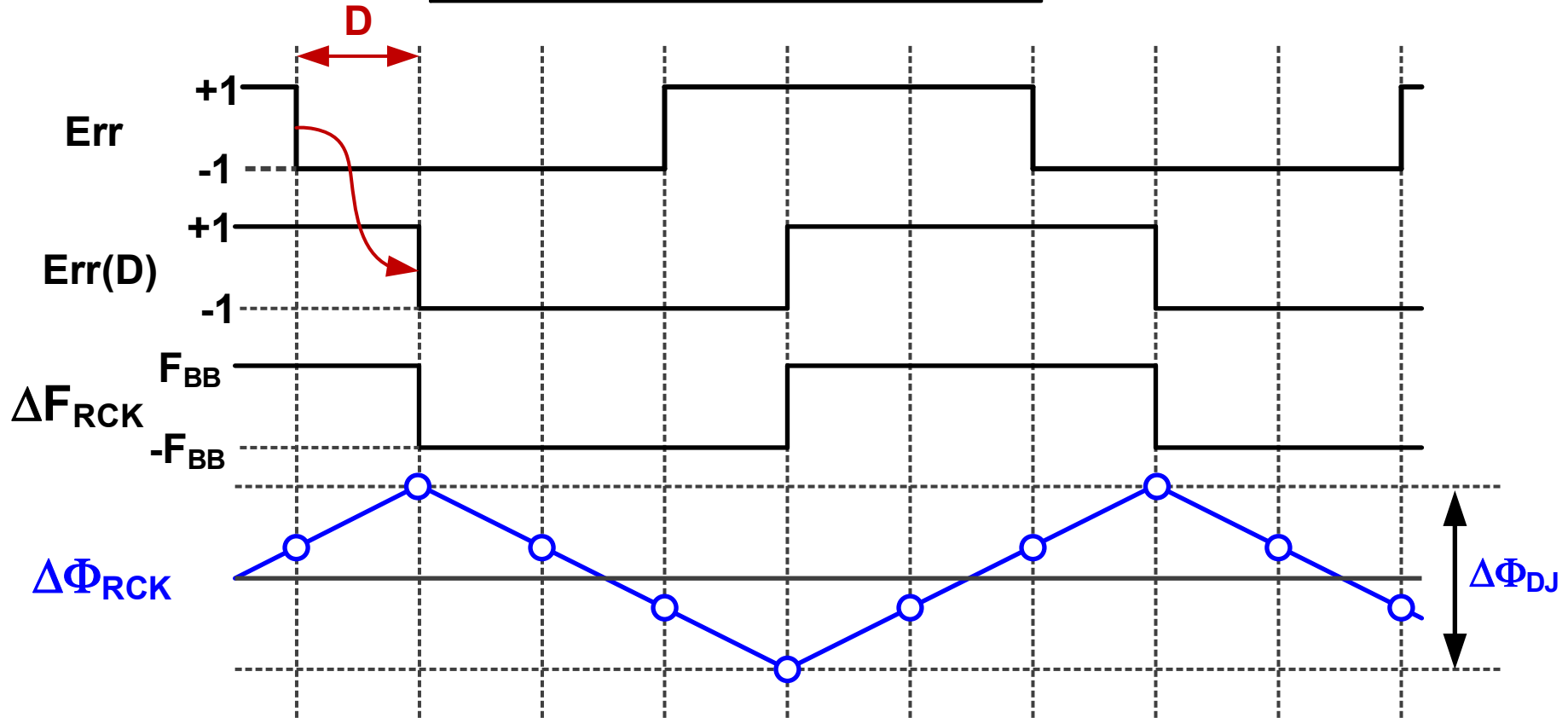
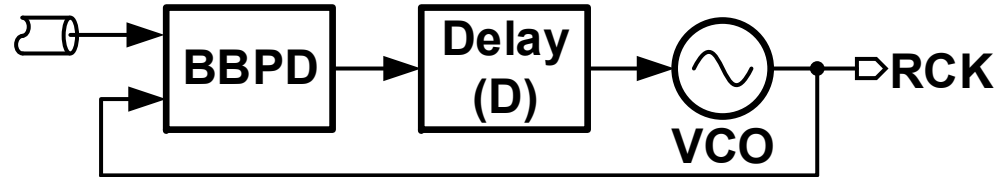
Steady-State Limit Cycles (II)



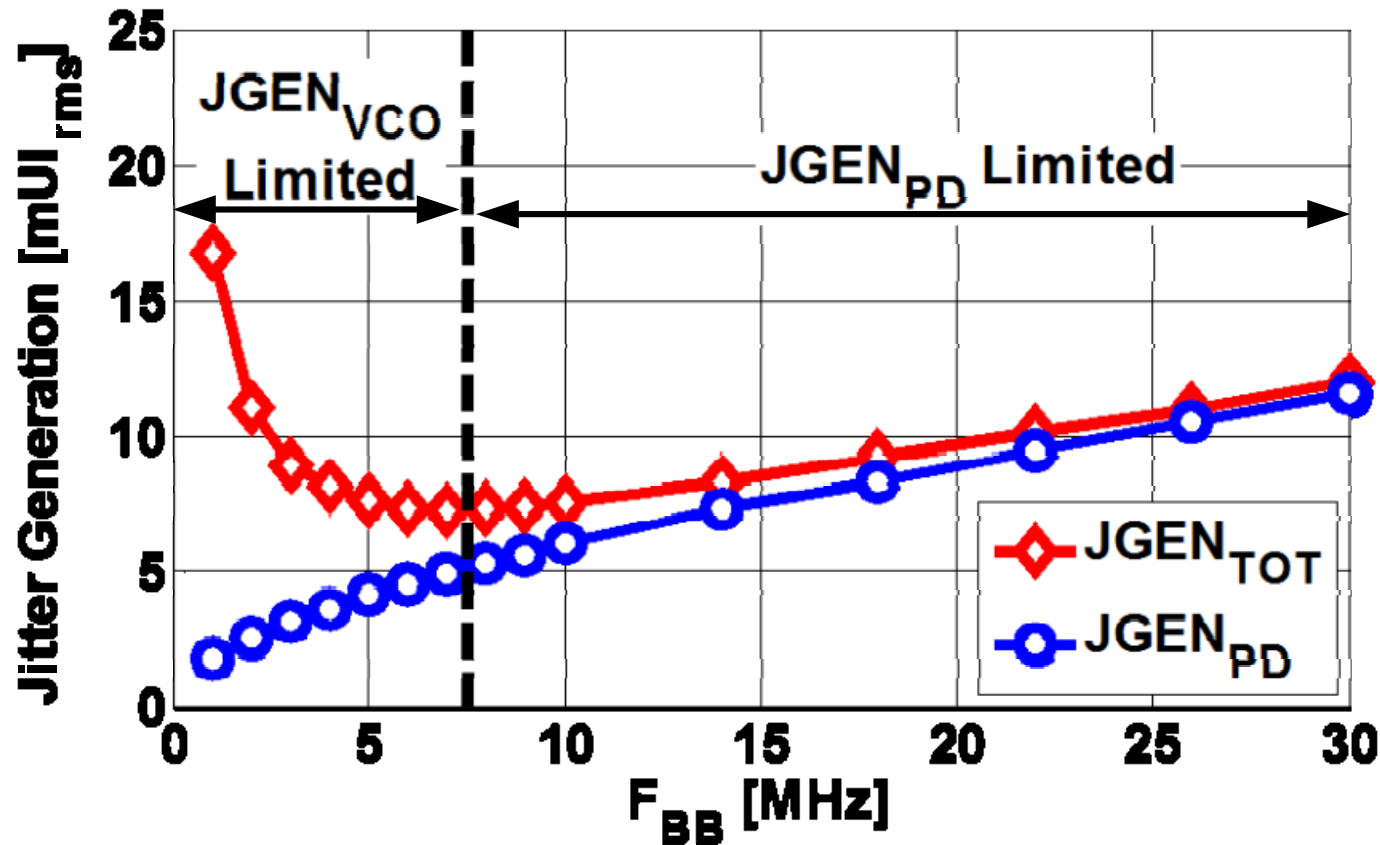
Proportional path dominates dithering jitter

I path suppress low f drifts

Loop Delay Increases JGEN



JGEN vs. Bang-Bang Step Size



- As bang-bang step size increases:
 - Contribution of PD noise to output noise increases
 - Contribution of VCO noise to output noise decreases

JTRAN Dependence on Jitter

$$\text{JTRAN BW} \approx K_{PD} \cdot K_P \cdot K_{VCO}$$

$$\text{Gain } K_{PD}|_{\Delta T=0} = \frac{2V_o}{\sqrt{2\pi}\sigma_j} \implies \text{JTRAN BW} \propto \frac{1}{\sigma_j}$$

- JTRAN is inversely proportional to input jitter
 - Difficult to predict *a priori*
- Set JTRAN for minimum input jitter condition
 - Makes it more susceptible to VCO phase noise

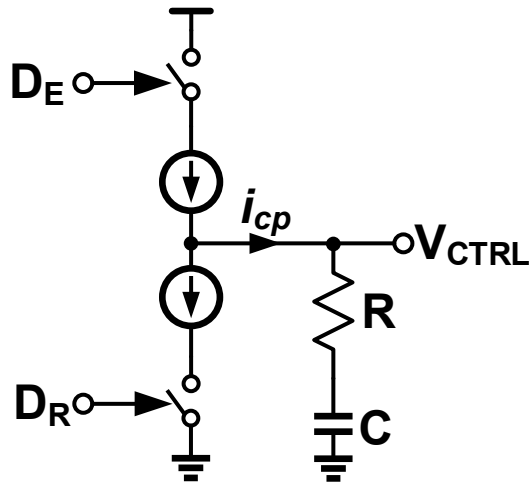
Tutorial Roadmap

- Performance metrics
 - **Basic architectures**
 - **Linear/Bang-bang**
 - **Digital**
 - **Hybrid**
 - Application-specific CDRs
 - Multi-lane chip-to-chip links
 - Repeaters for optical and active cables
 - Summary
-

Eliminating Loop Filter Capacitor

Just like digital PLL

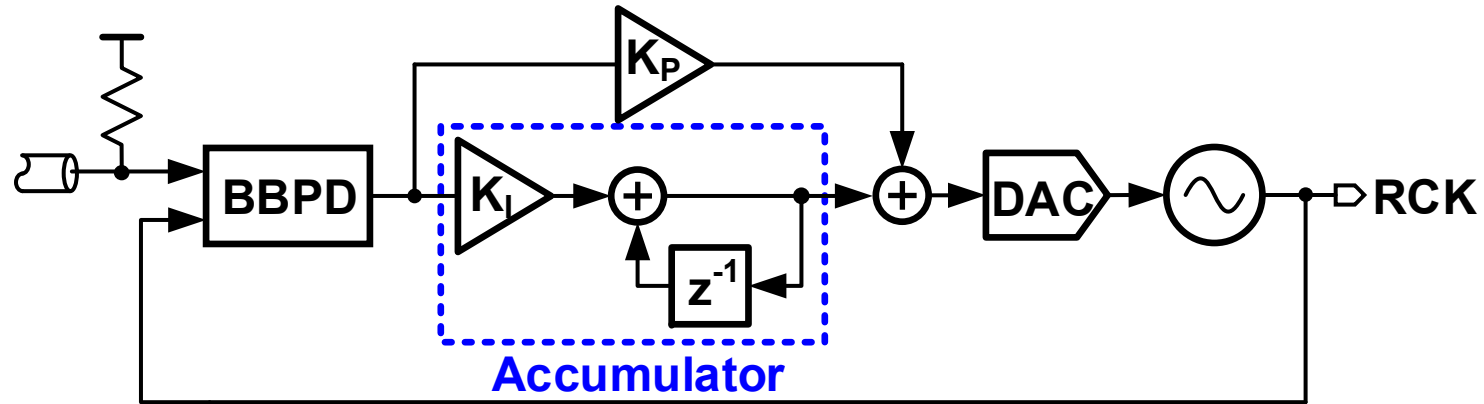
- Map CP + LF into digital domain directly



$$i_{cp}R + \frac{i_{cp}}{Cs} \Rightarrow i_{cp}R + \frac{i_{cp}T}{C} \frac{z^{-1}}{1-z^{-1}}$$

- Digital accumulator replaces loop filter capacitor
 - Large time constant with small area
 - Infinite DC gain → ideal Type-II behavior
 - PVT insensitive
 - Easy to reconfigure for loop dynamics control

Simple Digital CDR^[6]

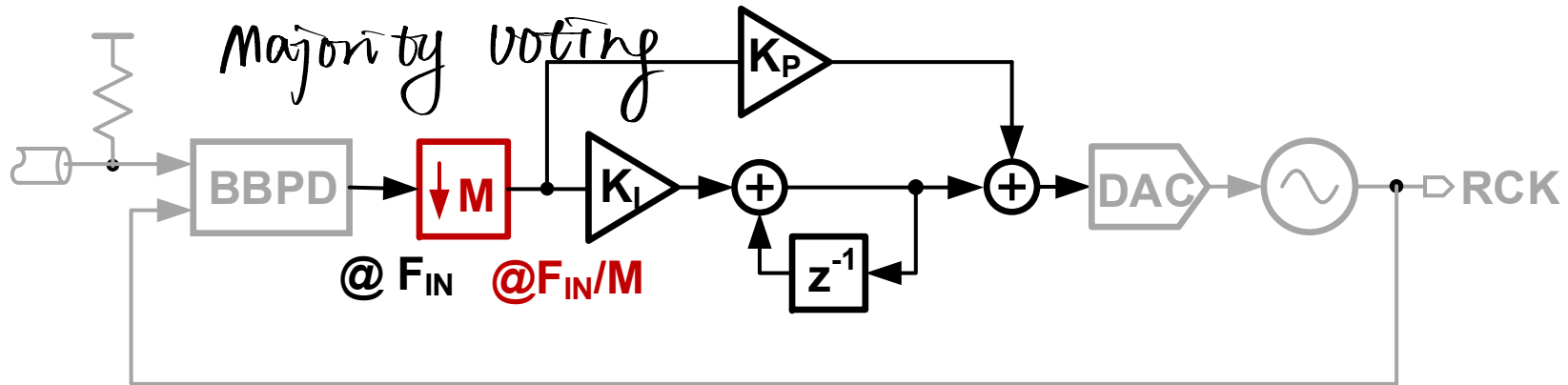


□ Loop filter must operate at data rate

□ **Need:**

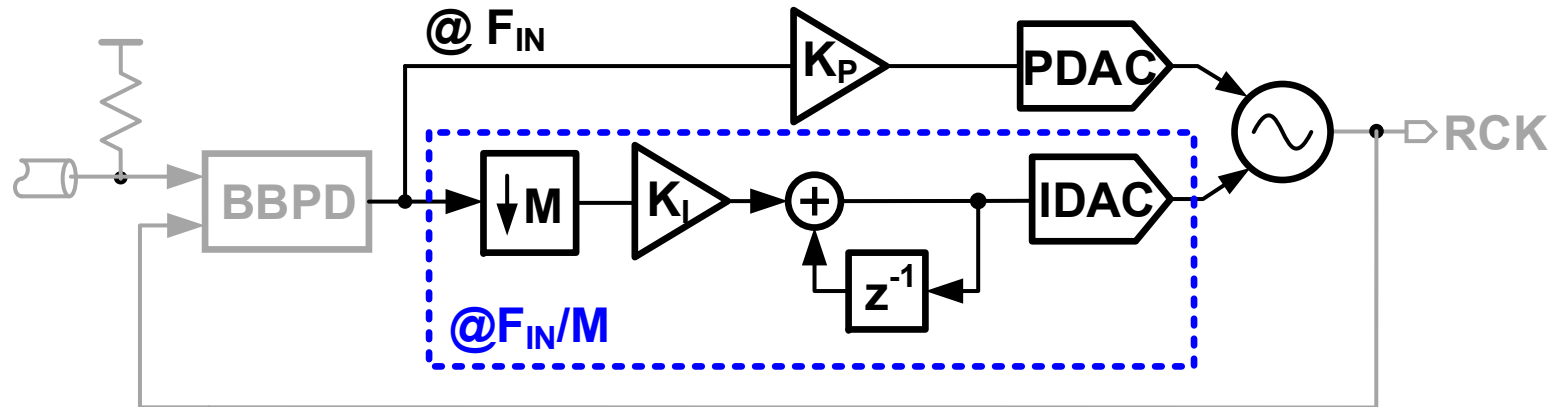
- Wide operand high-speed adders
- High speed/resolution **D**igital to **A**nalog **C**onverter (**DAC**)

Reducing Speed Requirements (I)



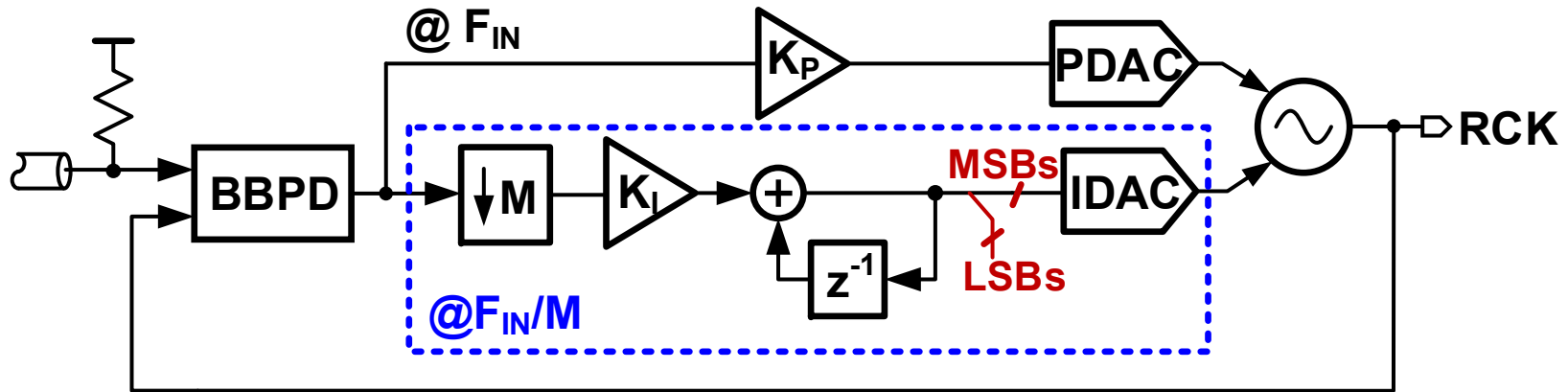
- ❑ Decimation eases speed requirements
- ❑ Lower update rate $\rightarrow$ increases loop latency
 - Loop latency increases dithering jitter
- ❑ Observation: Proportional path dominates jitter

Reducing Speed Requirements (II)



- Fast proportional path *no latency*
 - Minimize latency → reduce dithering jitter
- High resolution integral path
 - Minimize tracking jitter
- Minimal hardware penalty
 - Needs only 2-level high-speed PDAC

Practical Digital CDR

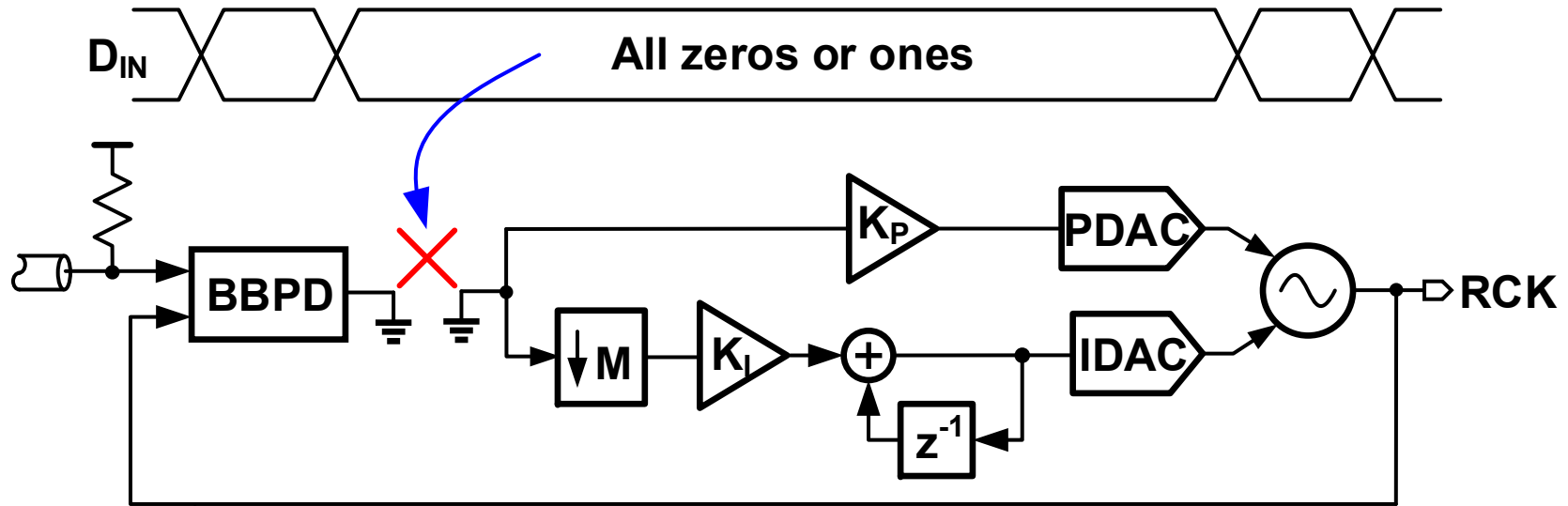


- ❑ Loop delay increases integral path dithering jitter
 - Reduce gain by dropping lower LSBs
- ❑ IDAC implemented using $\Delta\Sigma$ techniques
- ❑ Proportional and integral controls summed in VCO
- ❑ Area efficient $\rightarrow$ can be fully integrated

Digital CDR Drawbacks

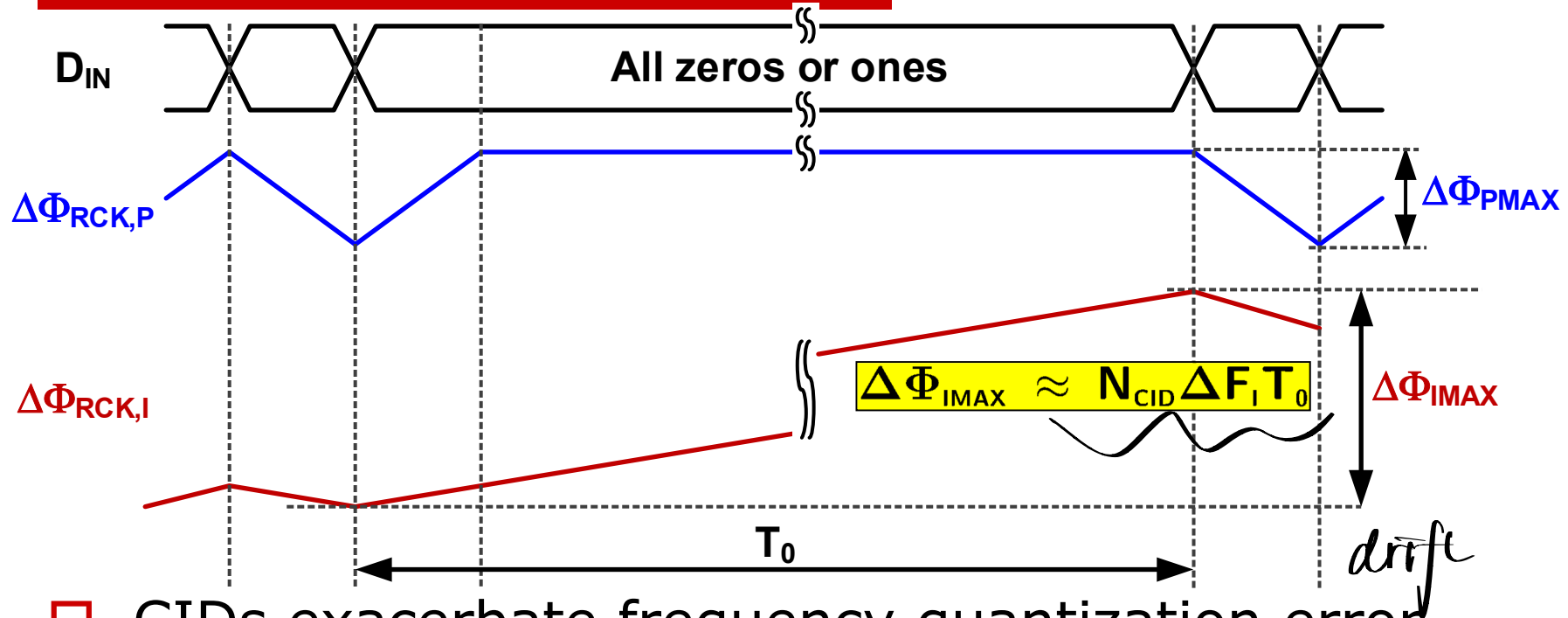
- ❑ Coupled JTRAN and JTOL
 - ❑ Jitter peaking
 - ❑ ~~Large loop filter area~~
- } Similar to linear CDR
-
- ❑ JGEN vs bang-bang step size tradeoff
 - ❑ JTRAN dependence on input jitter
- } Similar to BB CDR
-
- ❑ **Sensitive to Consecutive Identical Digits (CIDs)**

Impact of CIDs^[7]



- ❑ BBPD output is zero for the duration of CIDs
 - CDR operates in open loop
- ❑ All benefits of feedback are lost
 - Noise, leakage, PVT sensitivity, ...

JGEN due to CIDs



- CIDs exacerbate frequency quantization error
 - Output phase drifts at a rate proportional to freq. error
- Impact on JGEN depends on:
 - Number of CIDs (N_{CID})
 - Integral path frequency quantization error (ΔF_I)

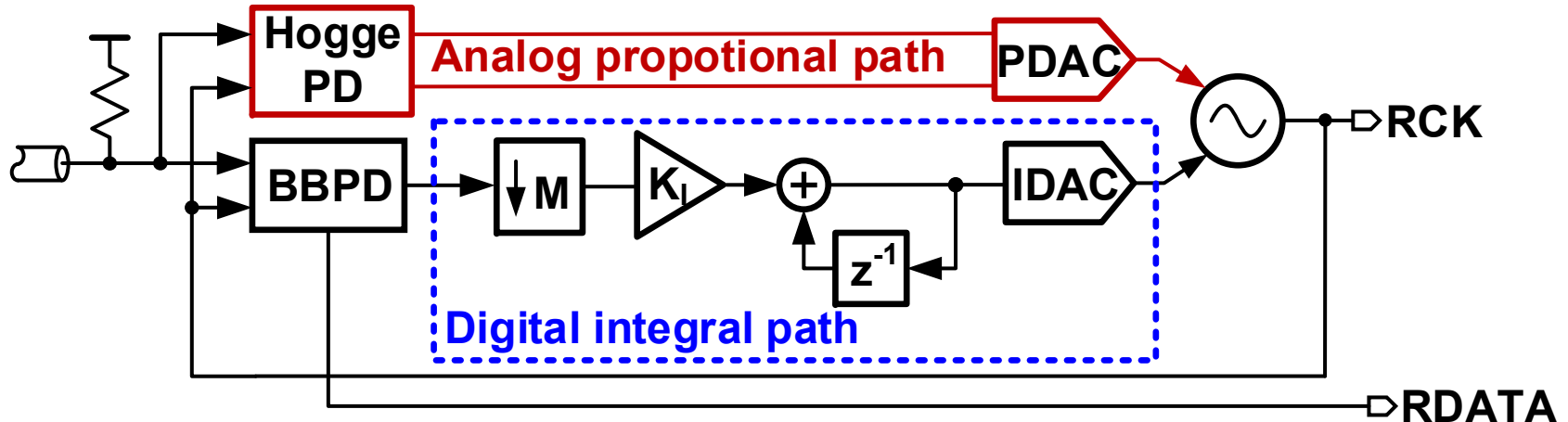
Analog vs. Digital CDRs

- Analog CDR using linear PD
 - Well-controlled loop dynamics
 - PD non-idealities degrade timing margin/BER
 - Large loop filter capacitor

- Digital CDR using bang-bang PD
 - Non-linear loop dynamics (JTRAN depends on jitter)
 - Bang-bang PD maximizes timing margin
 - No large capacitor (small area)

- Can we combine the advantages?

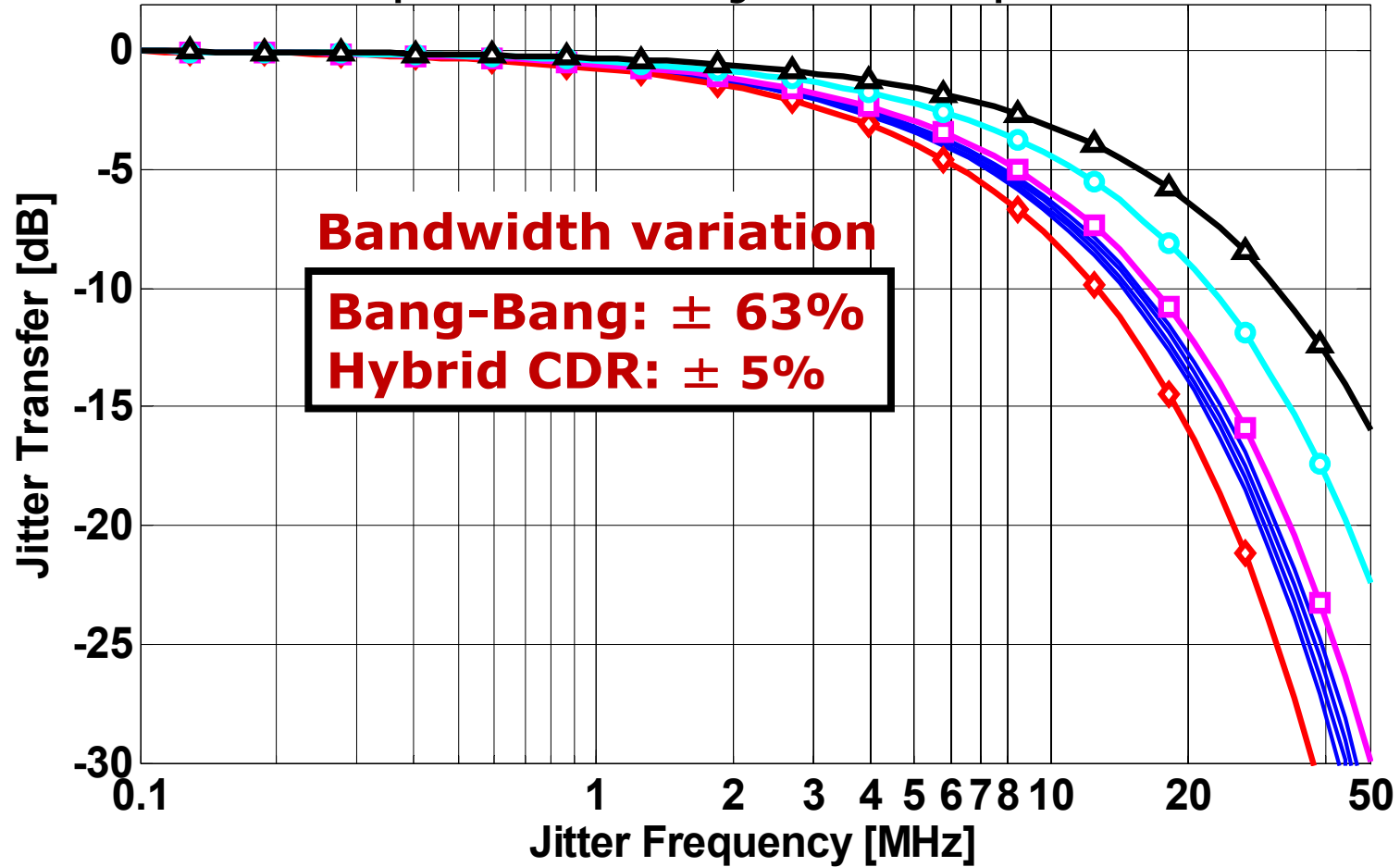
Hybrid Analog/Digital CDR^[8]



- Proportional path sets loop bandwidth (JTRAN)
 - Fixed gain leads to linear loop dynamics
 - Eliminates phase quantization error
- Digital integral path sets steady state
 - Makes it insensitive to linear PD phase offset
 - Accumulator filters BBPD quantization error
 - Causes ripple on the proportional path

Hybrid CDR JTRAN Characteristics

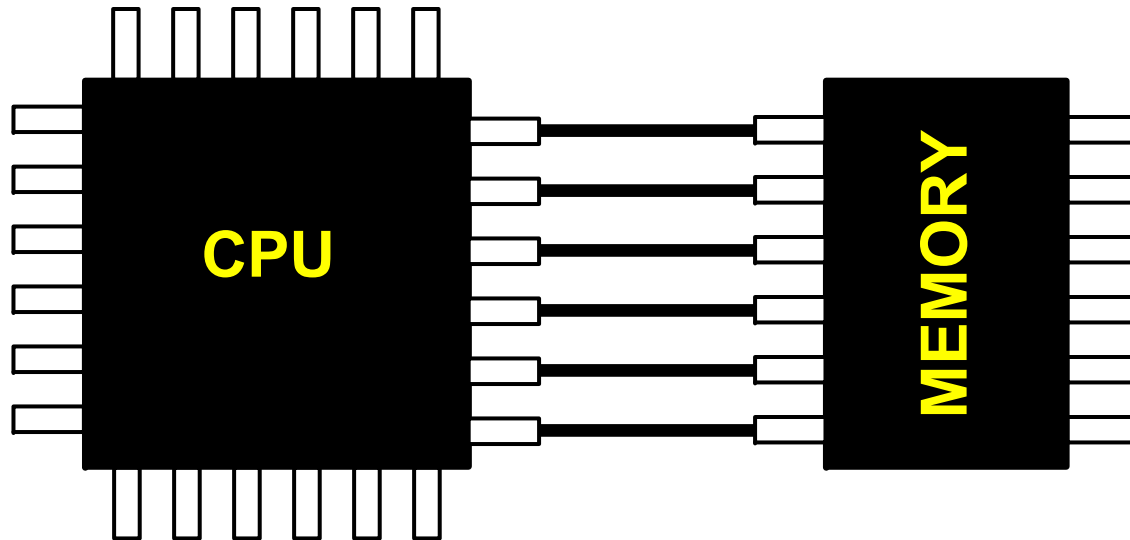
- JTRAN independent of jitter amplitude



Tutorial Roadmap

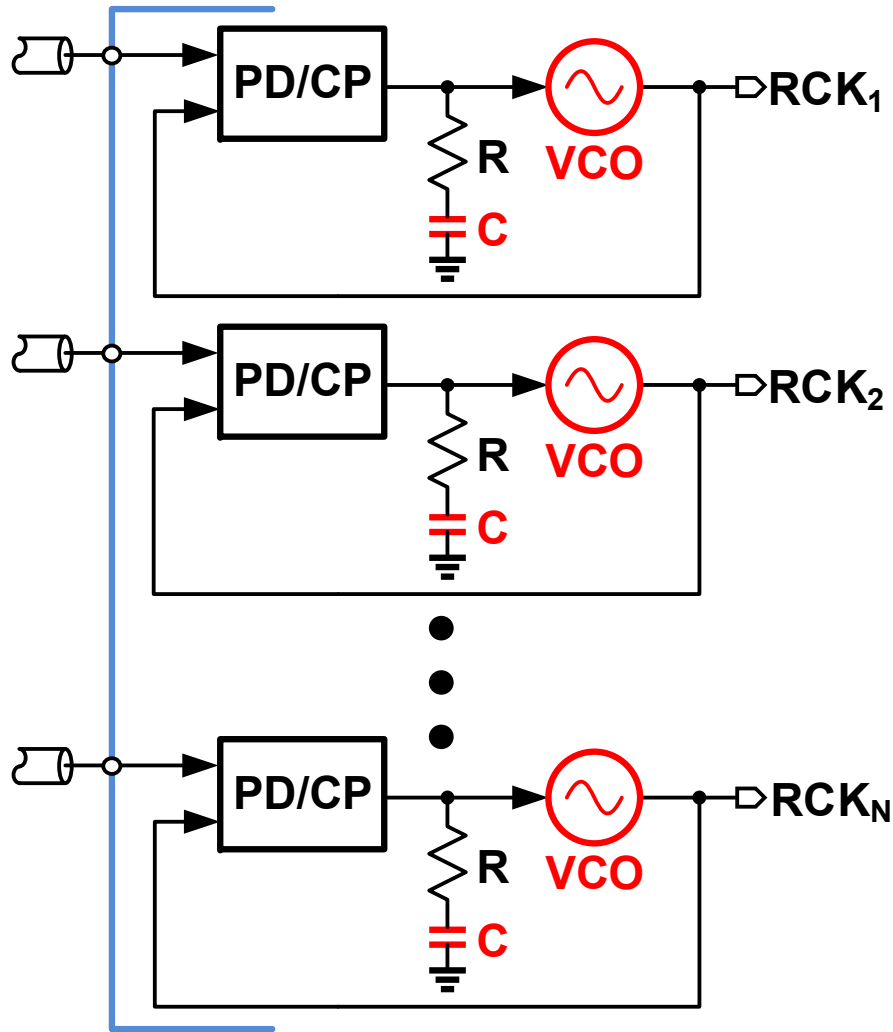
- Performance metrics
 - Basic architectures
 - Linear/Bang-bang
 - Digital
 - Hybrid
 - **Application-specific CDRs**
 - **Multi-lane chip-to-chip links**
 - **Repeaters for optical links and active cables**
 - Summary
-

Multi-lane Chip-to-Chip Links



- ❑ Source synchronous clocking is common
clock forwarding
- ❑ BUT many standards mandate embedded clocking
 - Examples: PCIe, XAUI, SATA, etc.

Multi-Lane CDR Challenges

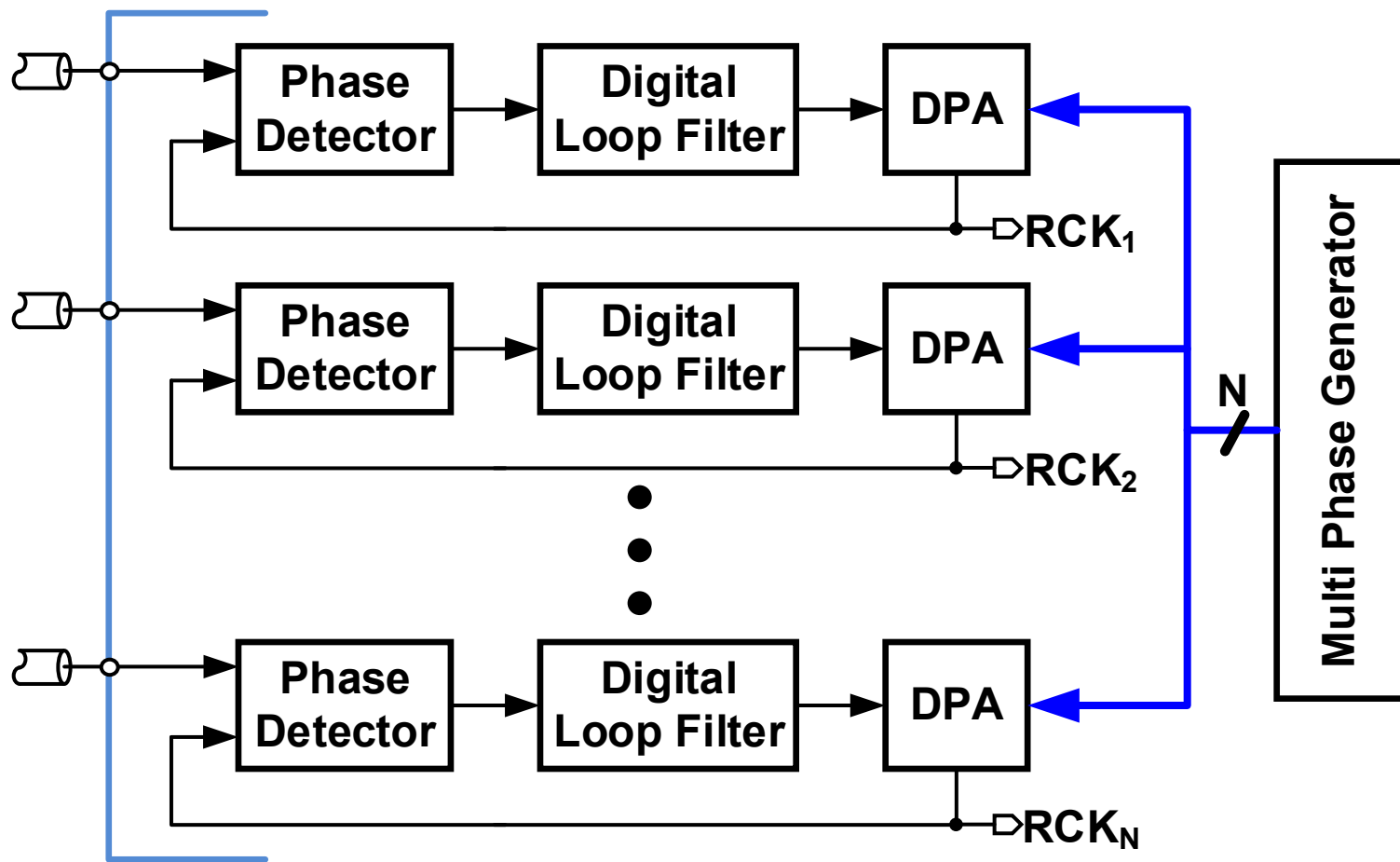


- Many VCOs & capacitors
- Large area
- Harmonic locking

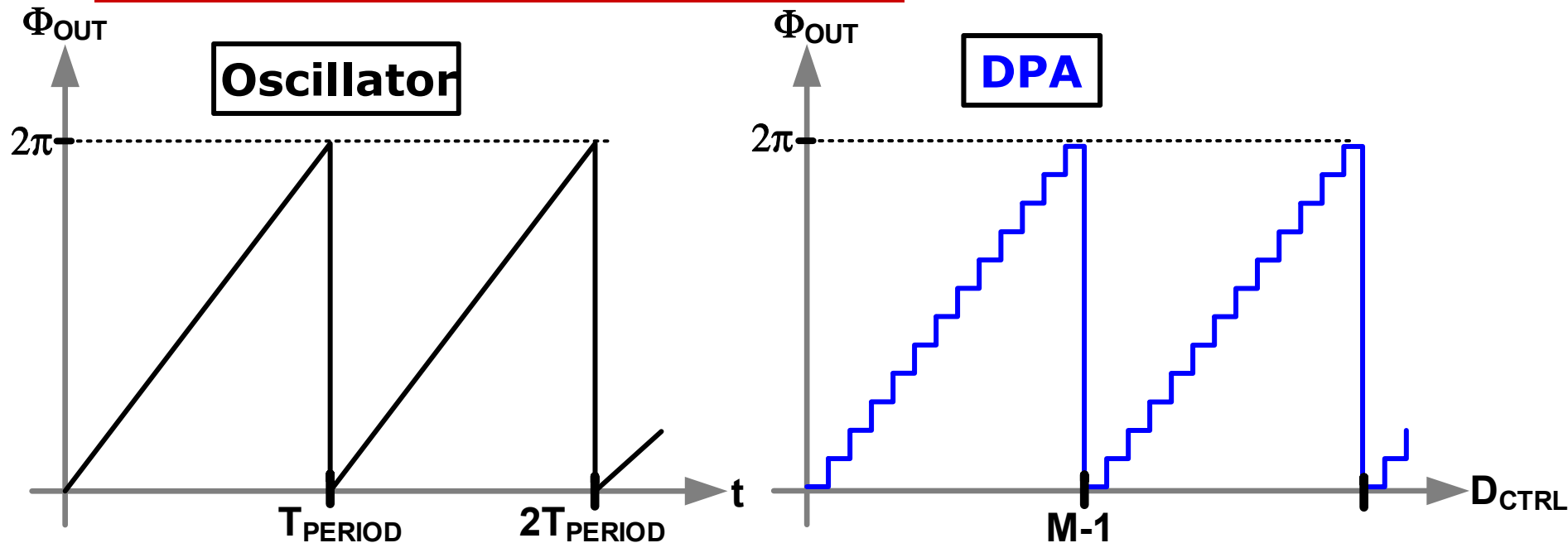
Multi-Lane CDR Solution

Digital Phase Accumulator (DPA)

8-10 phases

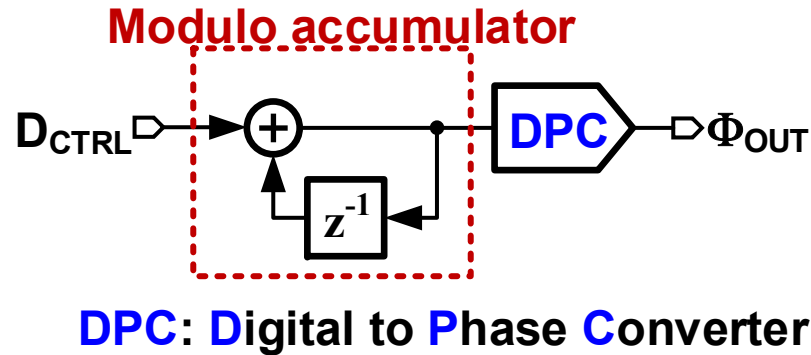


Digital Phase Accumulator (DPA)



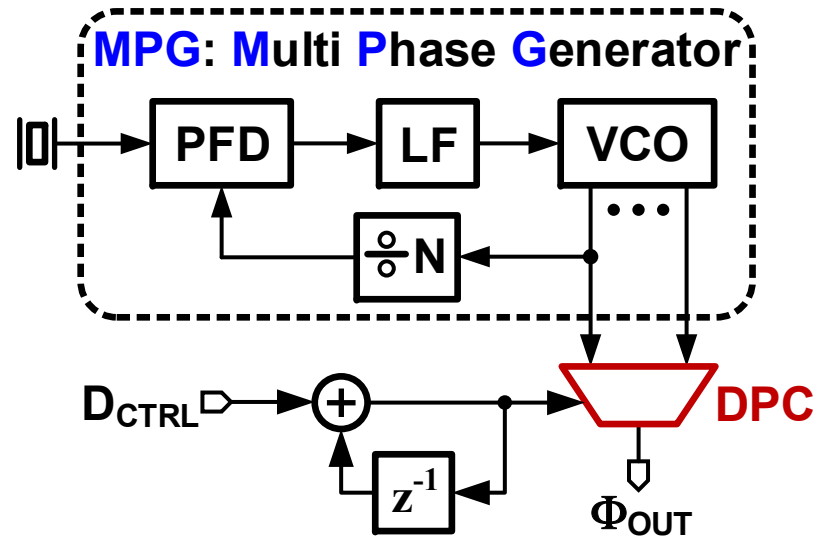
- ❑ Oscillators are phase accumulators
 - Modulo 2π accumulation
- ❑ Mimic oscillator operation
 - Explicit phase accumulation
 - Infinite phase shifting by modulo accumulation

Conceptual DPA Realization



- Digital accumulator mimics phase accumulation
 - Rate governed by clock frequency
 - Modulo arithmetic maps 2π phase to 0
- DPC generates output phase
 - DPC non-idealities directly appear at the output

Practical DPA Realization

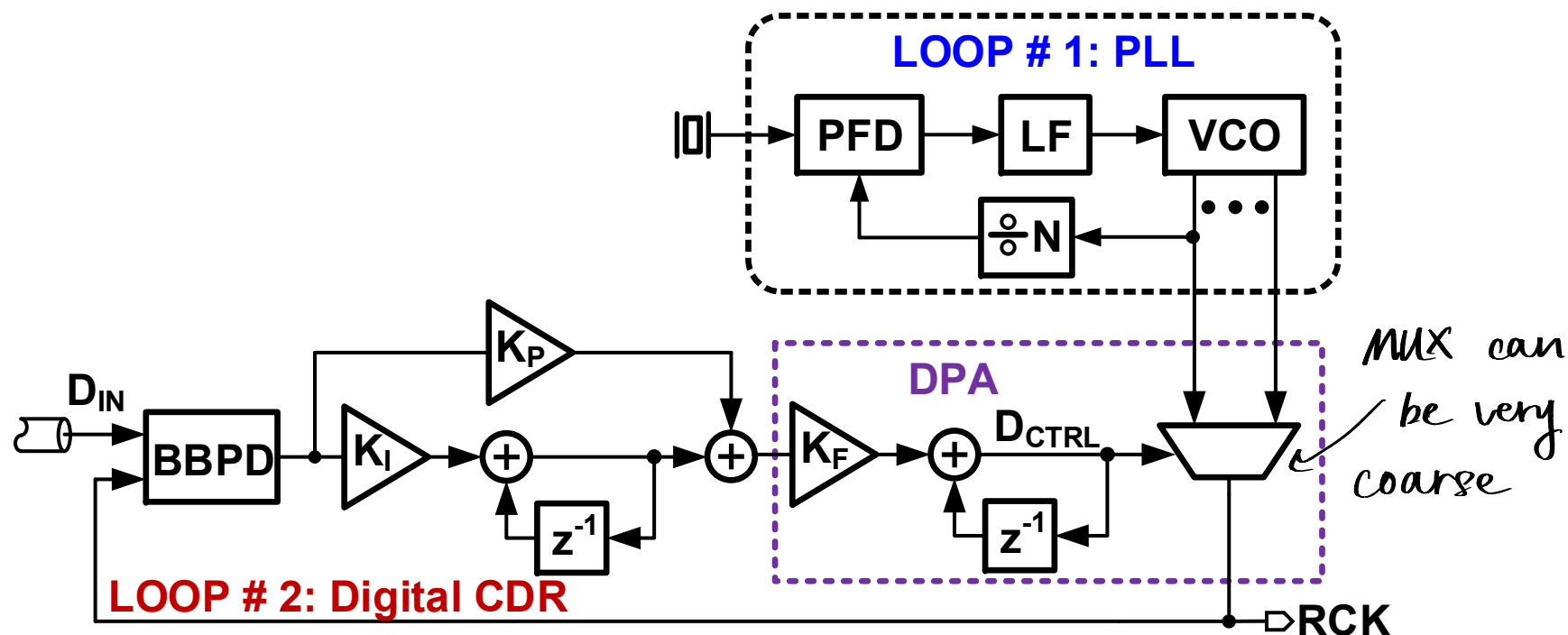


□ DPC implemented using a mux

$$\Phi_{OUT} = \Phi_{VCO} + D_{CTRL} \cdot K_{DPC}$$

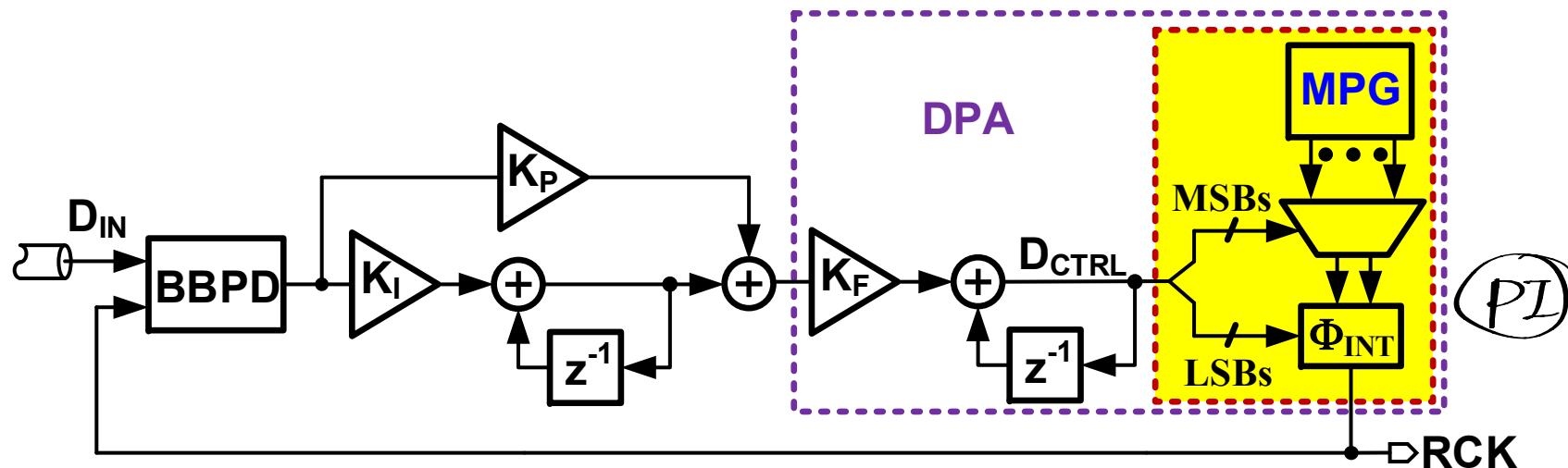
$$\text{For M-phase VCO : } K_{DPC} = \frac{2\pi}{M}$$

Dual-Loop CDR^[9]



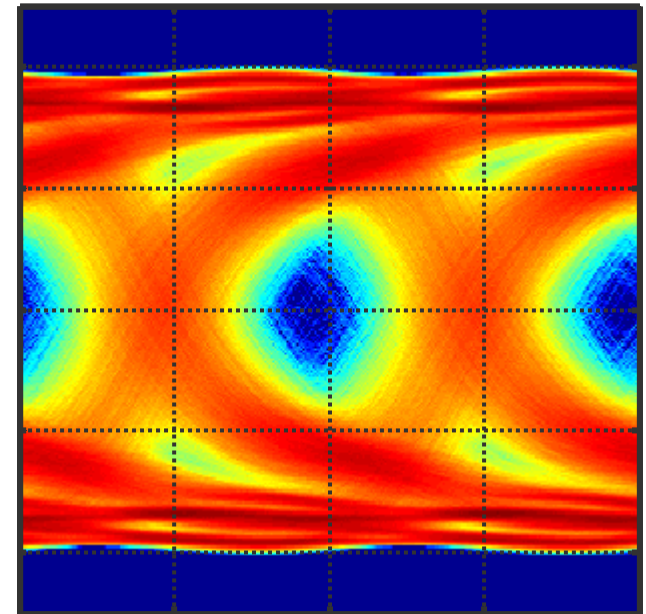
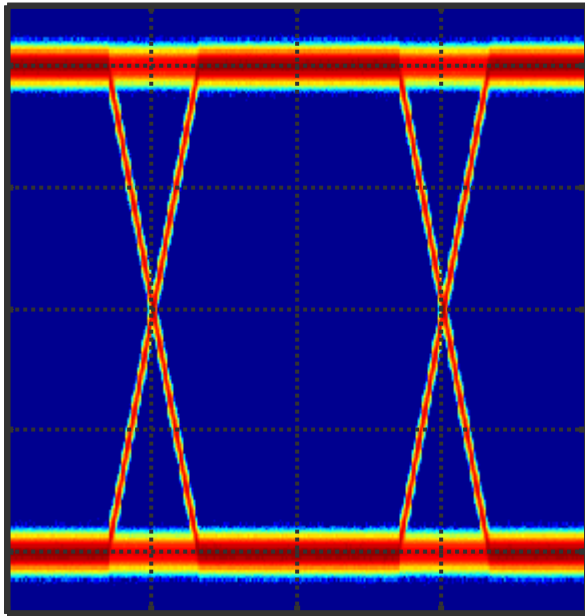
- ❑ Dual loop: Loop # 1: PLL(MPG), Loop # 2: CDR
- ❑ DPA replaces DCO in the CDR
- ❑ PLL guarantees frequency locking

Practical Dual-Loop CDR^[10]



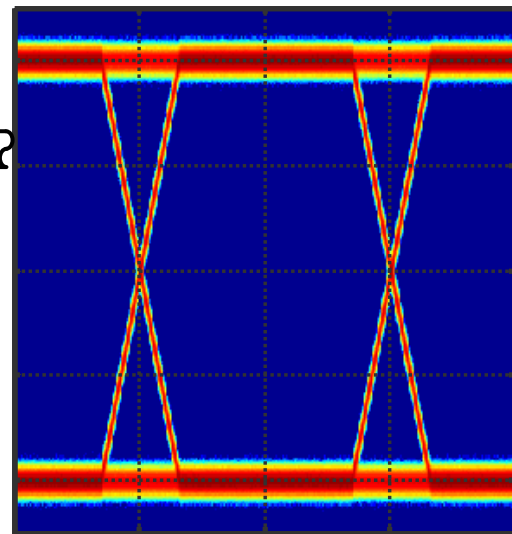
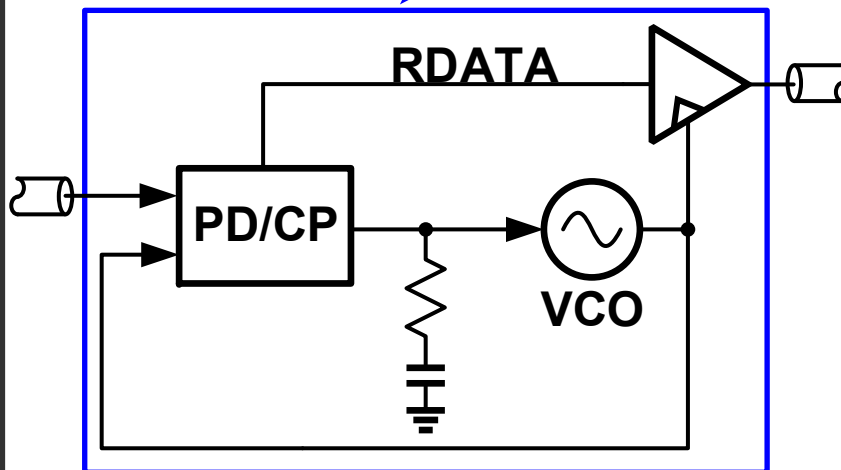
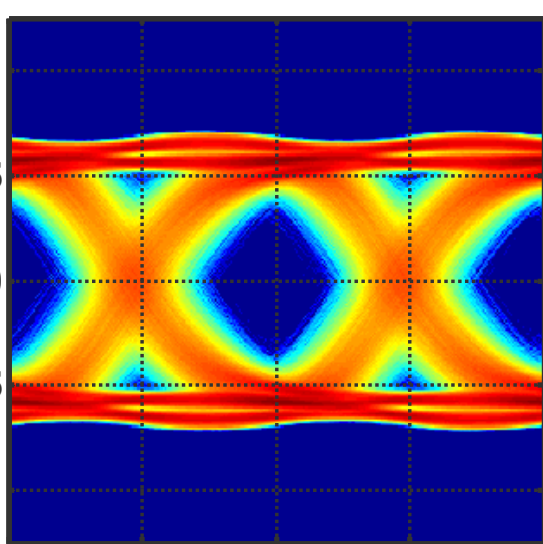
- ❑ Phase interpolator (Φ_{INT}) improves DPA resolution
 - Better JGEN
- ❑ PI resolution depends on many factors
 - Input rise time, input phase spacing, PI BW,...

Long-Haul Communication



- Difficult to achieve error-free operation

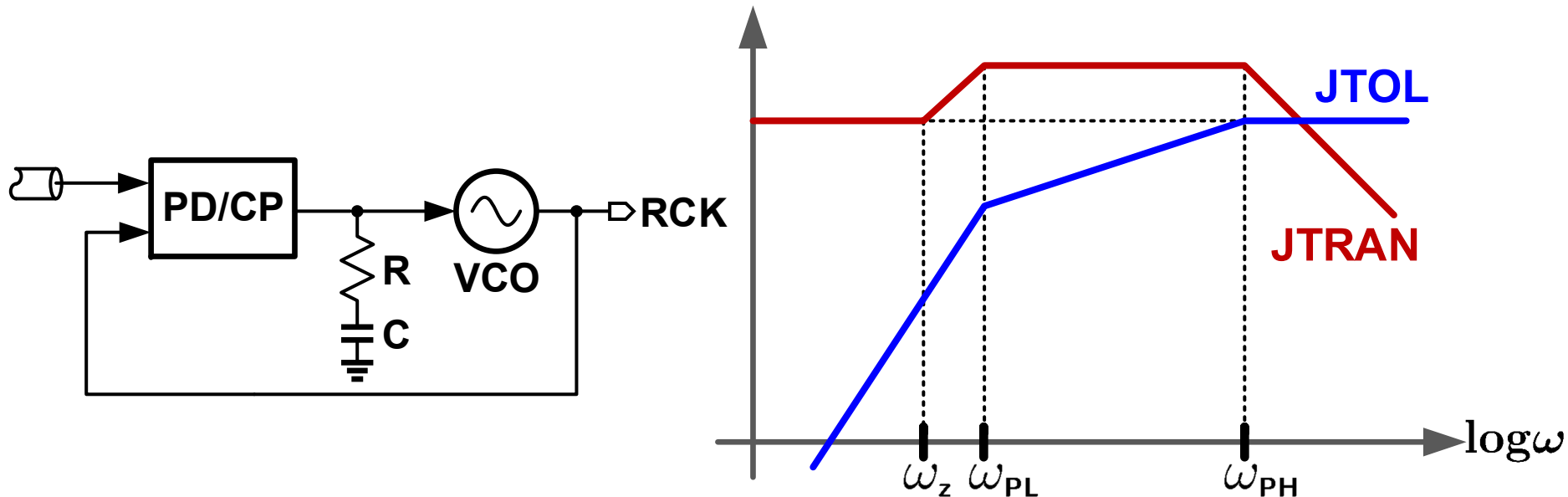
Active Repeaters *= receive, denoise, retransmit*



□ Repeater requirements

- Tolerate large input jitter (high JTOL)
- Filter input jitter (low JTRAN) w/ minimal peaking
- Re-transmit with low jitter (Low JGEN)

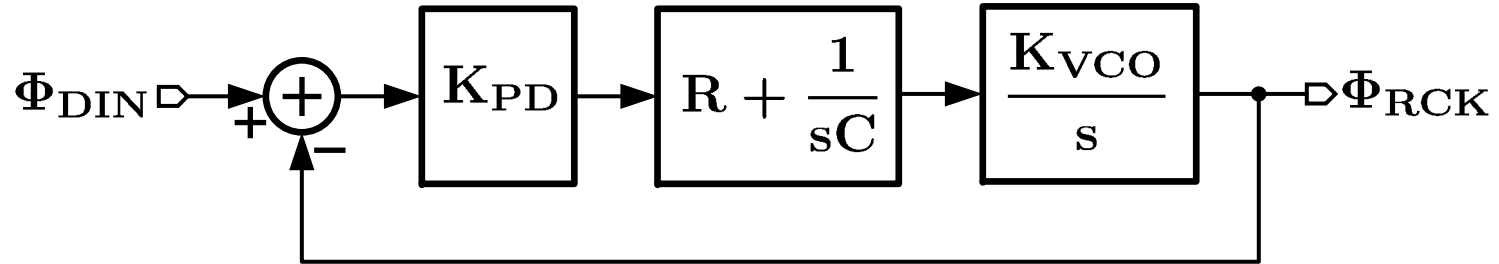
How About Conventional CDR?



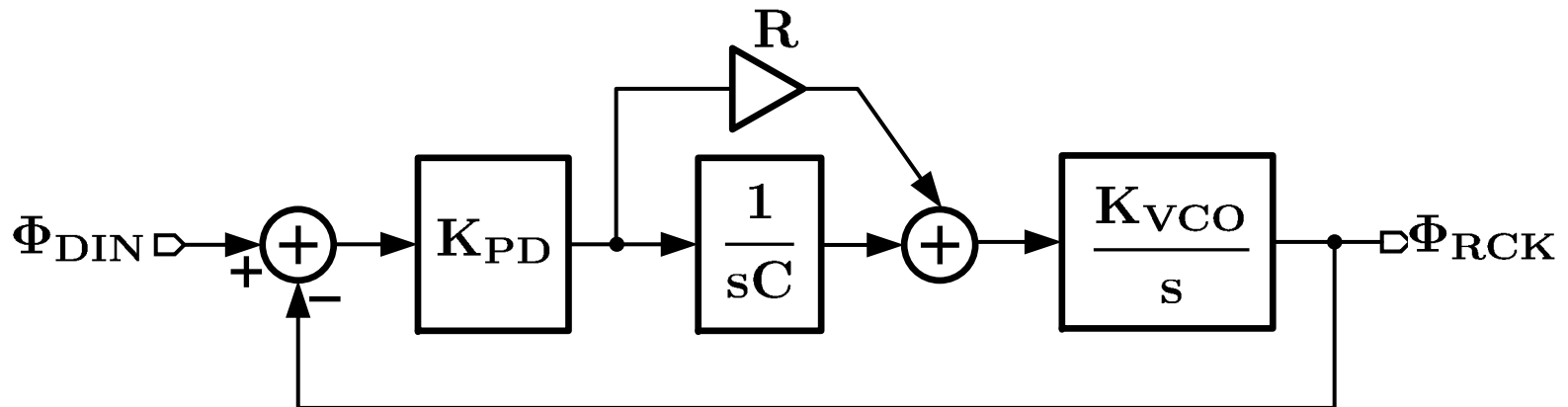
- ❑ Jitter peaking $\rightarrow$ large loop filter capacitor
small jitter peaking.
- ❑ High JTOL $\rightarrow$ high JTRAN *due to coupling*
 - Cannot adequately filter i/p jitter $\rightarrow$ degrades RCK jitter

How to Eliminate Jitter Peaking?

- **Main idea:** Remove zero in feed-forward path

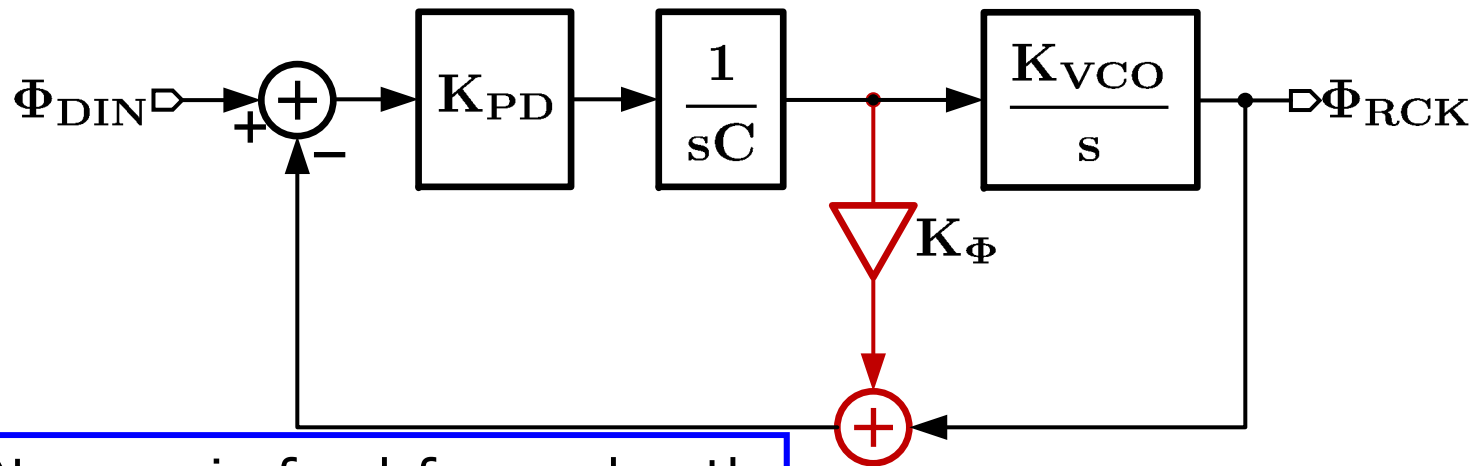
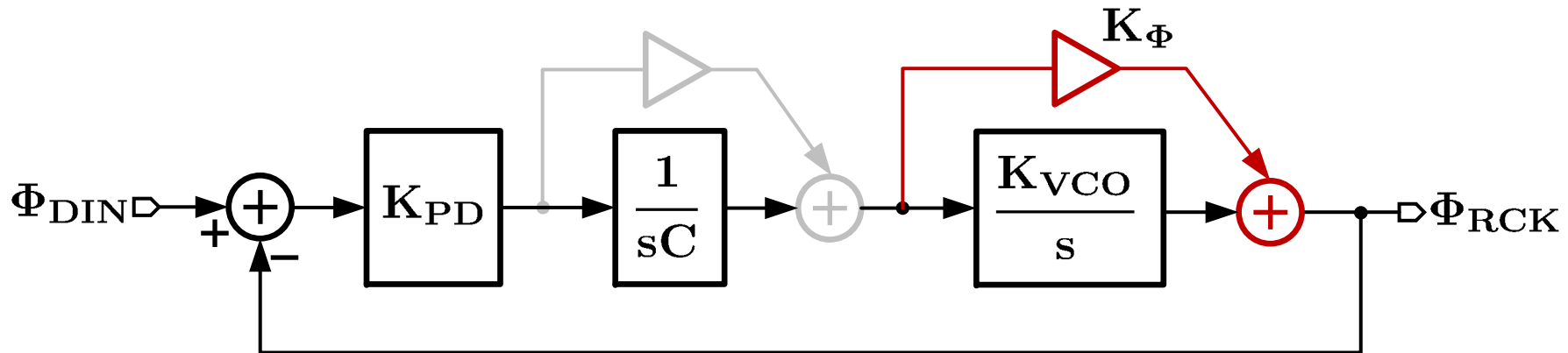


Feed-forward across **current** integrator



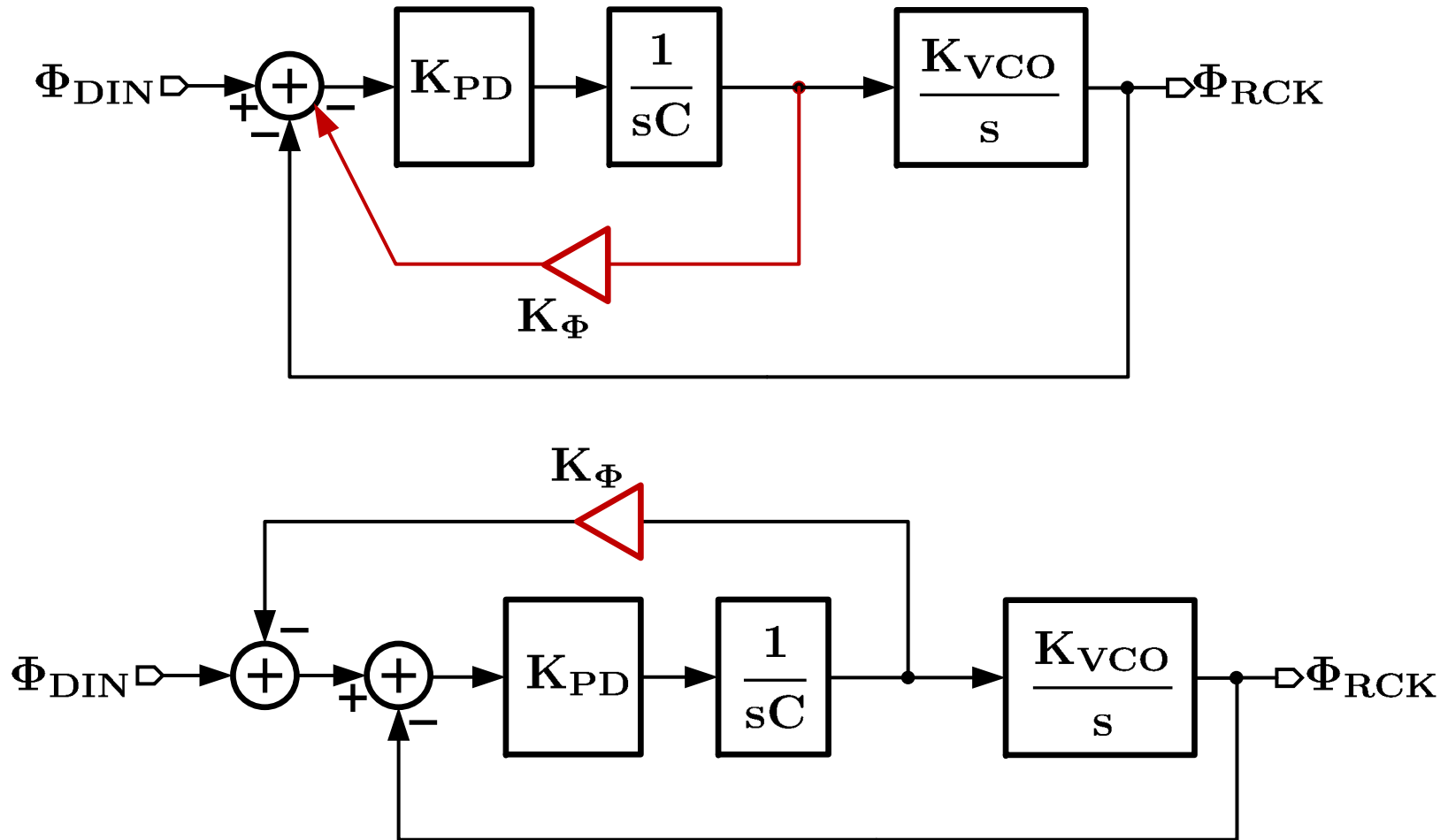
Eliminate Jitter Peaking (I)^[11]

Feed-forward across **phase** integrator (VCO)

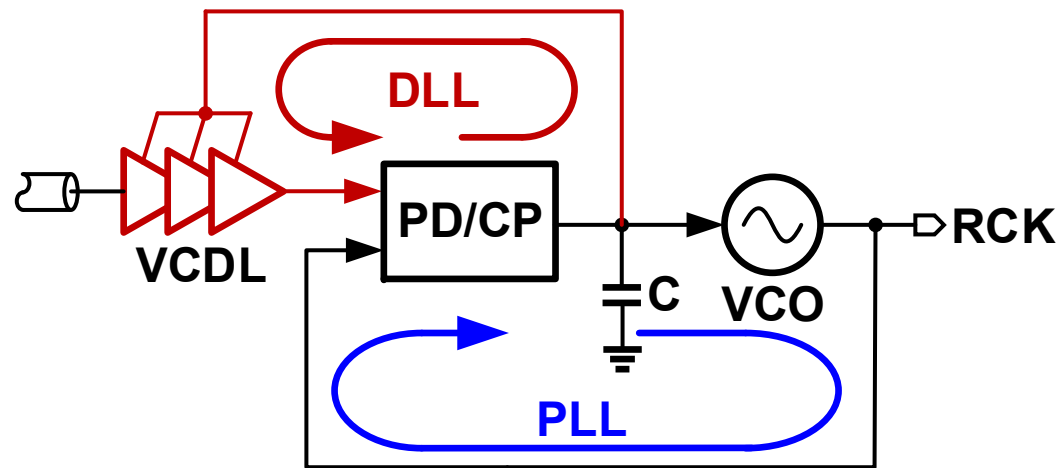
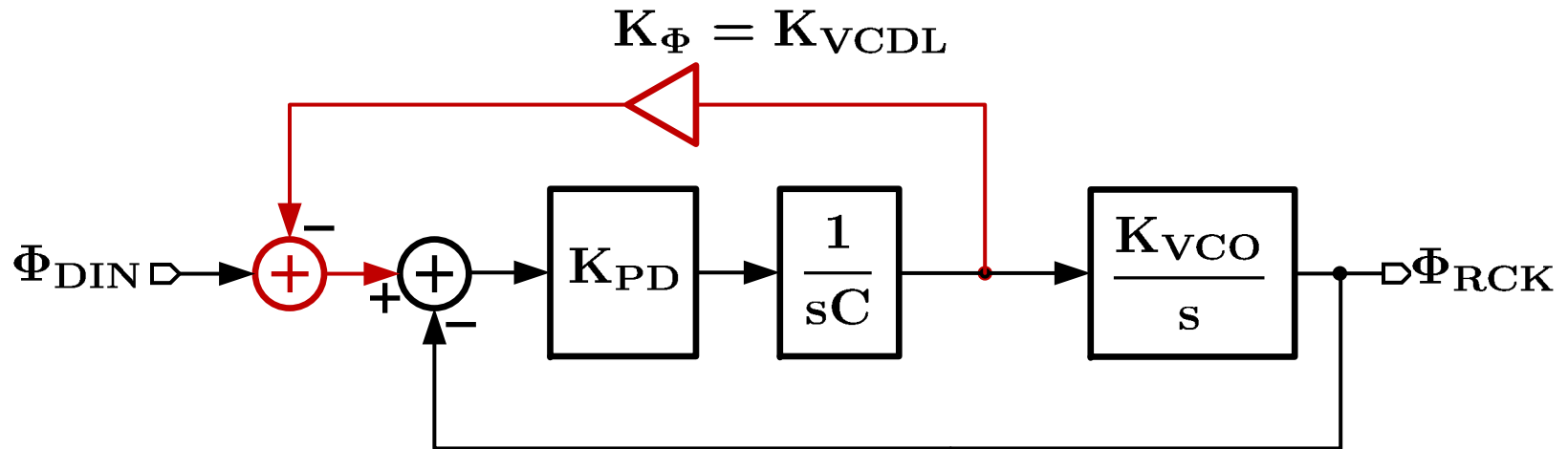


No zero in feed-forward path

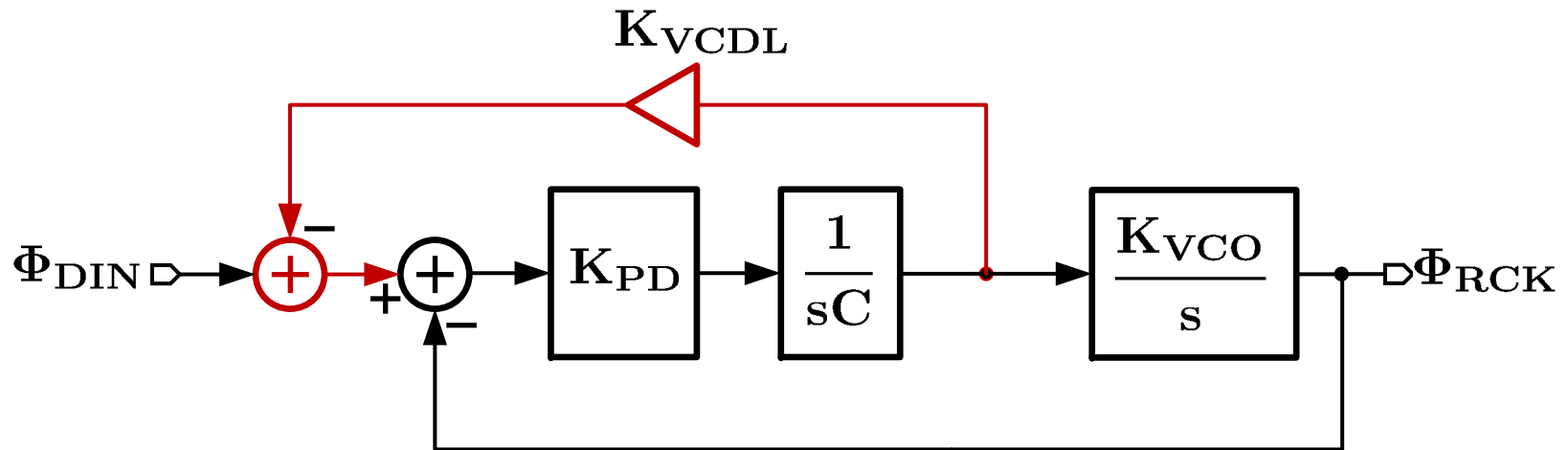
Eliminate Jitter Peaking (II)



Country	Year	Value
Algeria	2010	0.0000
Algeria	2011	0.0000
Algeria	2012	0.0000
Algeria	2013	0.0000
Algeria	2014	0.0000
Algeria	2015	0.0000
Algeria	2016	0.0000
Algeria	2017	0.0000
Algeria	2018	0.0000
Algeria	2019	0.0000
Algeria	2020	0.0000
Algeria	2021	0.0000
Algeria	2022	0.0000
Algeria	2023	0.0000
Algeria	2024	0.0000
Algeria	2025	0.0000
Algeria	2026	0.0000
Algeria	2027	0.0000
Algeria	2028	0.0000
Algeria	2029	0.0000
Algeria	2030	0.0000
Algeria	2031	0.0000
Algeria	2032	0.0000
Algeria	2033	0.0000
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Algeria	2035	0.0000
Algeria	2036	0.0000
Algeria	2037	0.0000
Algeria	2038	0.0000
Algeria	2039	0.0000
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Algeria	2044	0.0000
Algeria	2045	0.0000
Algeria	2046	0.0000
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Algeria	2082	0.0000
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Algeria	2084	0.0000
Algeria	2085	0.0000
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Algeria	2087	0.0000
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Algeria	2110	0.0000
Algeria	2111	0.0000
Algeria	2112	0.0000
Algeria	2113	0.0000
Algeria	2114	0.0000
Algeria	2115	0.0000
Algeria	2116	



D/PLL CDR Jitter Transfer (I)



$$\mathbf{LG}_{\text{PLL}}(s) = K_{\text{PD}} \cdot \frac{1}{sC} \cdot \frac{K_{\text{VCO}}}{s}; \quad \mathbf{LG}_{\text{DLL}}(s) = K_{\text{PD}} \cdot \frac{1}{sC} \cdot K_{\text{VCDL}}$$

$$\begin{aligned} \mathbf{H}_{\text{JTRAN}}(s) &= \frac{\Phi_{\text{RCK}}(s)}{\Phi_{\text{DIN}}(s)} = \frac{\mathbf{LG}_{\text{PLL}}(s)}{1 + \mathbf{LG}_{\text{PLL}}(s) + \mathbf{LG}_{\text{DLL}}(s)} \\ &= \frac{1}{1 + s \cdot \frac{K_{\text{VCDL}}}{K_{\text{VCO}}} + s^2 \cdot \frac{C}{K_{\text{VCO}}K_{\text{PD}}}} \end{aligned}$$

no zero, no peaking if overdamped.

D/PLL CDR Jitter Transfer (II)

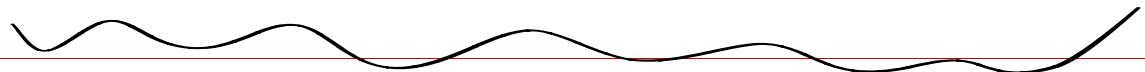
$$\mathbf{H}_{\text{JTRAN}}(\mathbf{s}) = \frac{1}{1 + \mathbf{s} \cdot \frac{\mathbf{K}_{\text{VCDL}}}{\mathbf{K}_{\text{VCO}}} + \mathbf{s}^2 \cdot \frac{\mathbf{C}}{\mathbf{K}_{\text{VCO}}\mathbf{K}_{\text{PD}}}} \equiv \frac{1}{(1 + \mathbf{s}/\omega_{\text{PL}})(1 + \mathbf{s}/\omega_{\text{PH}})}$$

$$\Rightarrow \omega_{\text{PL}} \approx \mathbf{K}_{\text{VCO}}/\mathbf{K}_{\text{VCDL}}; \omega_{\text{PH}} \approx \mathbf{K}_{\text{VCDL}} \cdot \mathbf{K}_{\text{PD}}/\mathbf{C}$$

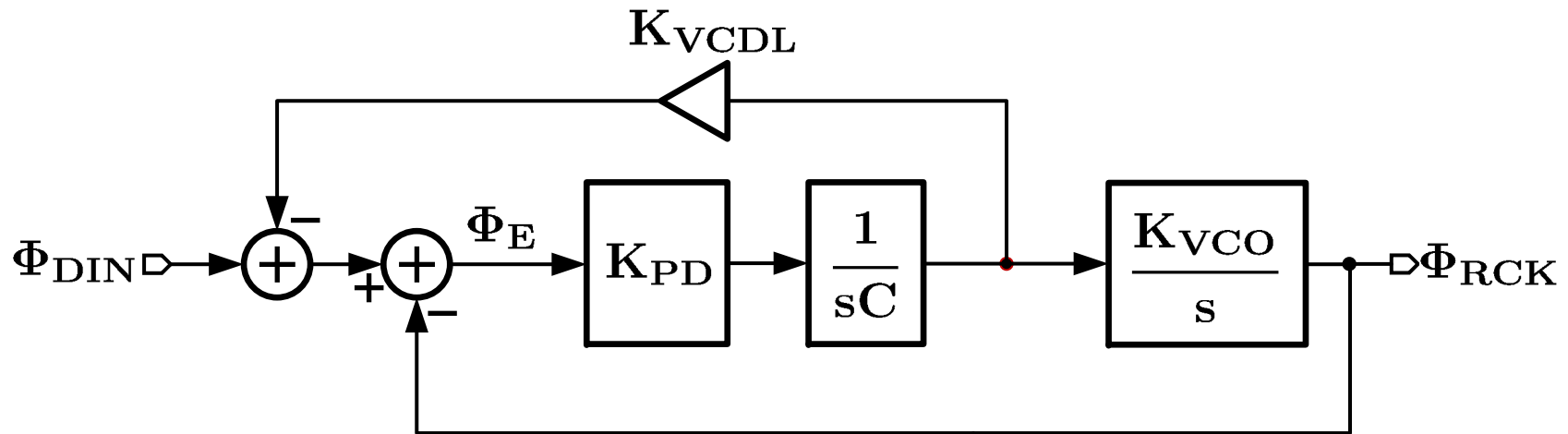
$$\mathbf{JTRAN\ BW} = \omega_{-3\text{dB}} \approx \omega_{\text{PL}} = \mathbf{K}_{\text{VCO}}/\mathbf{K}_{\text{VCDL}}$$

□ No jitter peaking if damping factor > 0.707

□ JTRAN BW = lower of the 2 pole frequencies



D/PLL CDR Jitter Tolerance (I)



$$\begin{aligned}
 H_{JTRACK}(s) &= \frac{\Phi_E(s)}{\Phi_{DIN}(s)} = \frac{1}{1 + LG_1(s) + LG_2(s)} \\
 &= \frac{s^2 \cdot \frac{C}{K_{VCO}K_{PD}}}{1 + s \cdot \frac{K_{VCDL}}{K_{VCO}} + s^2 \cdot \frac{C}{K_{VCO}K_{PD}}} \\
 &\equiv \frac{s^2 / \omega_{PL}\omega_{PH}}{(1 + s/\omega_{PL})(1 + s/\omega_{PH})}
 \end{aligned}$$

D/PLL CDR Jitter Tolerance (II)

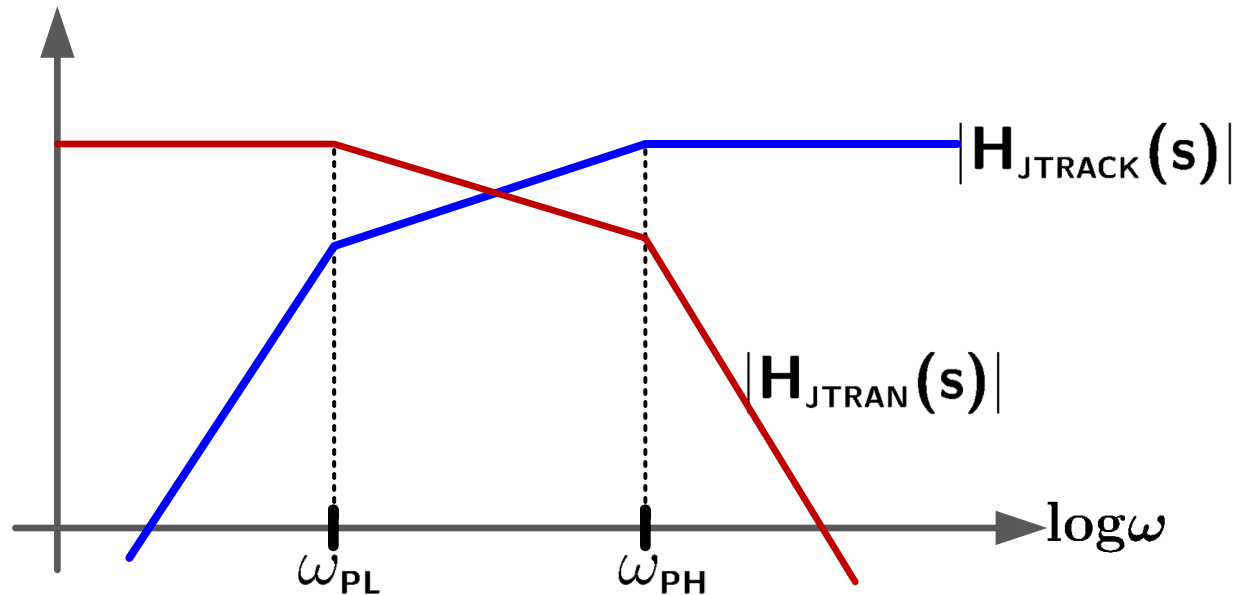
$$\mathbf{H}_{\text{JTRACK}}(s) = \frac{s^2 \cdot \frac{C}{K_{\text{VCO}}K_{\text{PD}}}}{1 + s \cdot \frac{K_{\text{VCDL}}}{K_{\text{VCO}}} + s^2 \cdot \frac{C}{K_{\text{VCO}}K_{\text{PD}}}} \equiv \frac{s^2 / \omega_{\text{PL}}\omega_{\text{PH}}}{(1 + s/\omega_{\text{PL}})(1 + s/\omega_{\text{PH}})}$$
$$\implies \omega_{\text{PL}} \approx K_{\text{VCO}}/K_{\text{VCDL}}; \omega_{\text{PH}} \approx K_{\text{VCDL}} \cdot K_{\text{PD}}/C$$

$$\text{JTRACK/JTOL Corner} = \omega_{-3\text{dB}} \approx \omega_{\text{PH}} \approx K_{\text{VCDL}} \cdot K_{\text{PD}}/C$$

□ JTOL corner = **higher** of the 2 pole frequencies



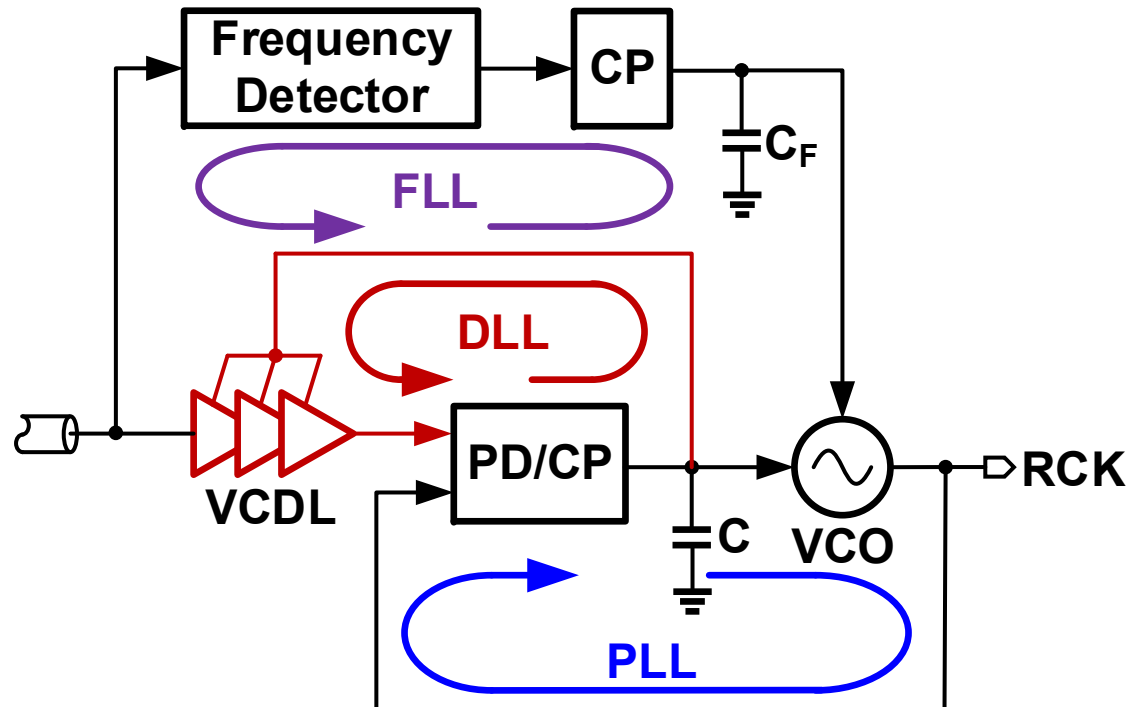
D/PLL CDR JTRAN vs. JTOL



□ Decoupled JTRAN and JTOL

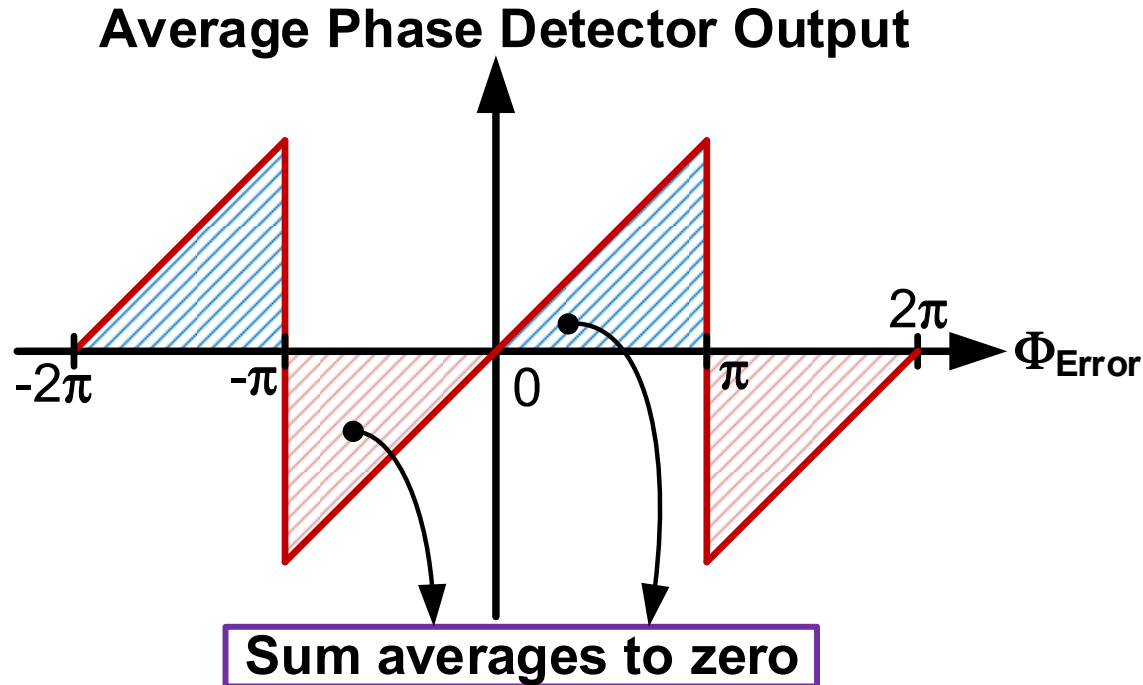
- ω_{PL} sets JTRAN BW
- ω_{PH} sets JTOL corner frequency

A Practical D/PLL CDR



- Additional frequency-locking loop (FLL)
- Challenges:
 - Frequency detection of random data
 - Interaction between FLL and PLL

Limited Capture Range

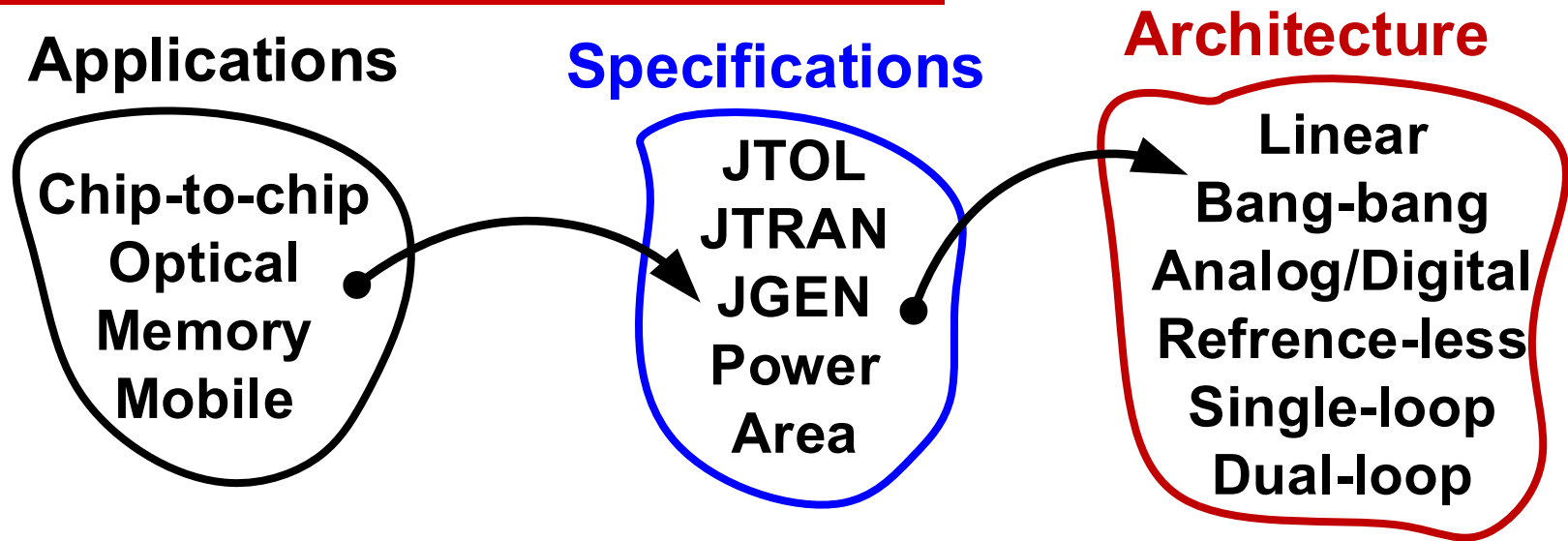


- Symmetric PD transfer characteristic
- Avg. PD output becomes zero w/ frequency error
 - PD cannot detect large frequency error

Frequency Detectors

- Rotational frequency detector^[12-14]
- Quadri-correlator frequency detector^[15-16]
- Stochastic reference clock generator^[17]
- Miscellaneous FDs
 - Strobed linear PD^[18]
 - Counting BBPD outputs ^[19]

Summary



- Understanding of CDR specifications
 - What are the important jitter metrics for a given app.?
- Understanding of CDR architectures
 - What is the best architecture for a given app.?
- CDR design techniques
 - How to choose the loop parameters?

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